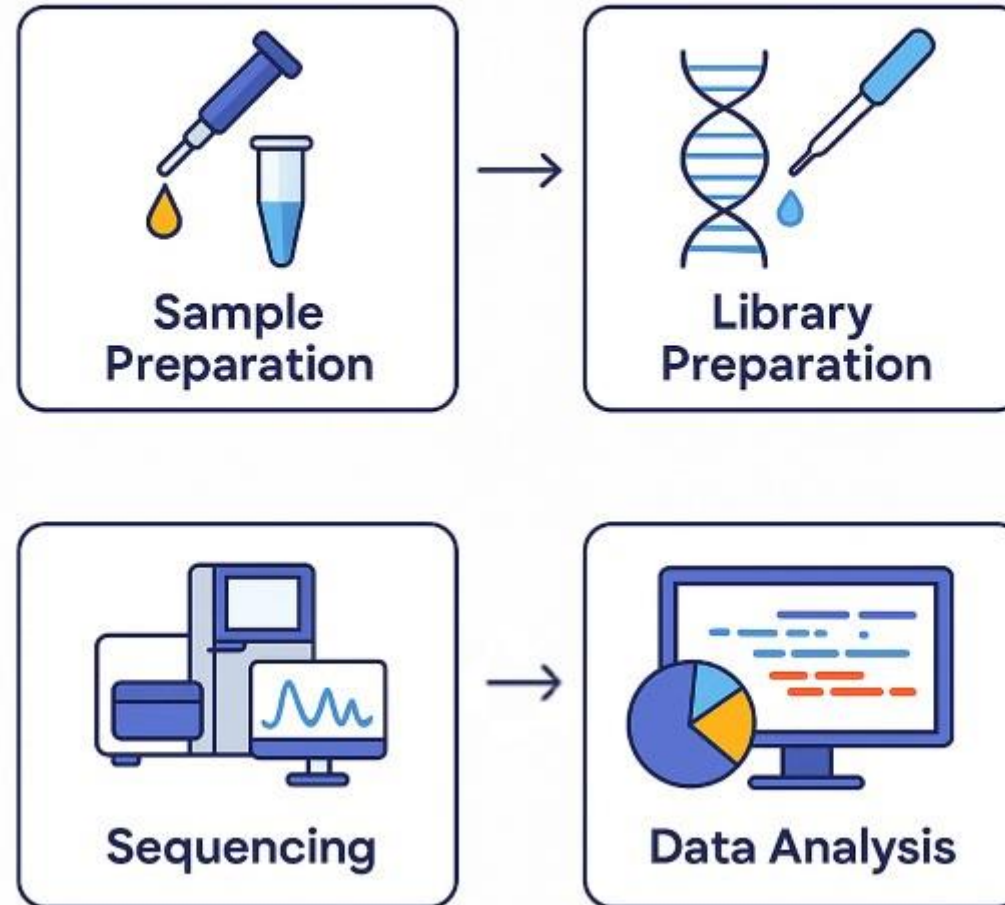


Key Steps in NGS



Lecture 01: Next-Generation Sequencing
Princess Rodriguez, PhD

Learning Objectives

01

Understand General Approaches of NGS

02

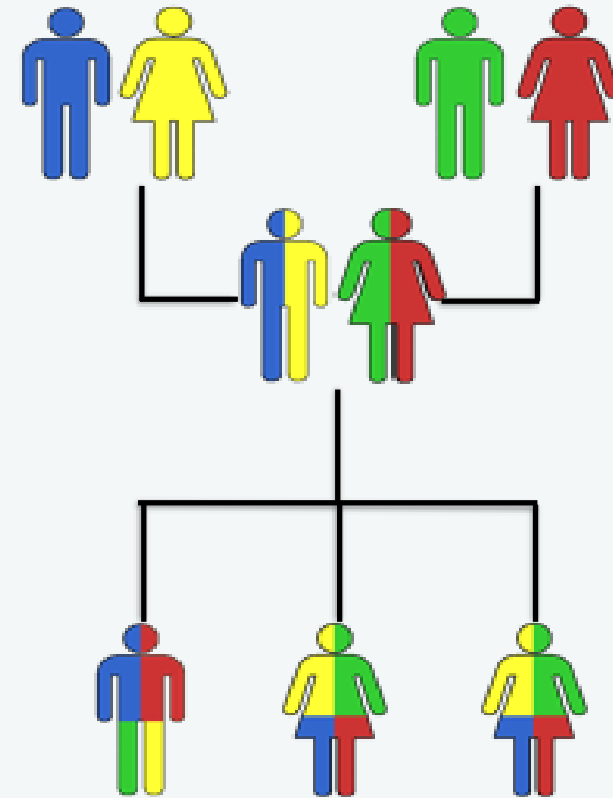
Relevance of NGS to R programming with biological data

03

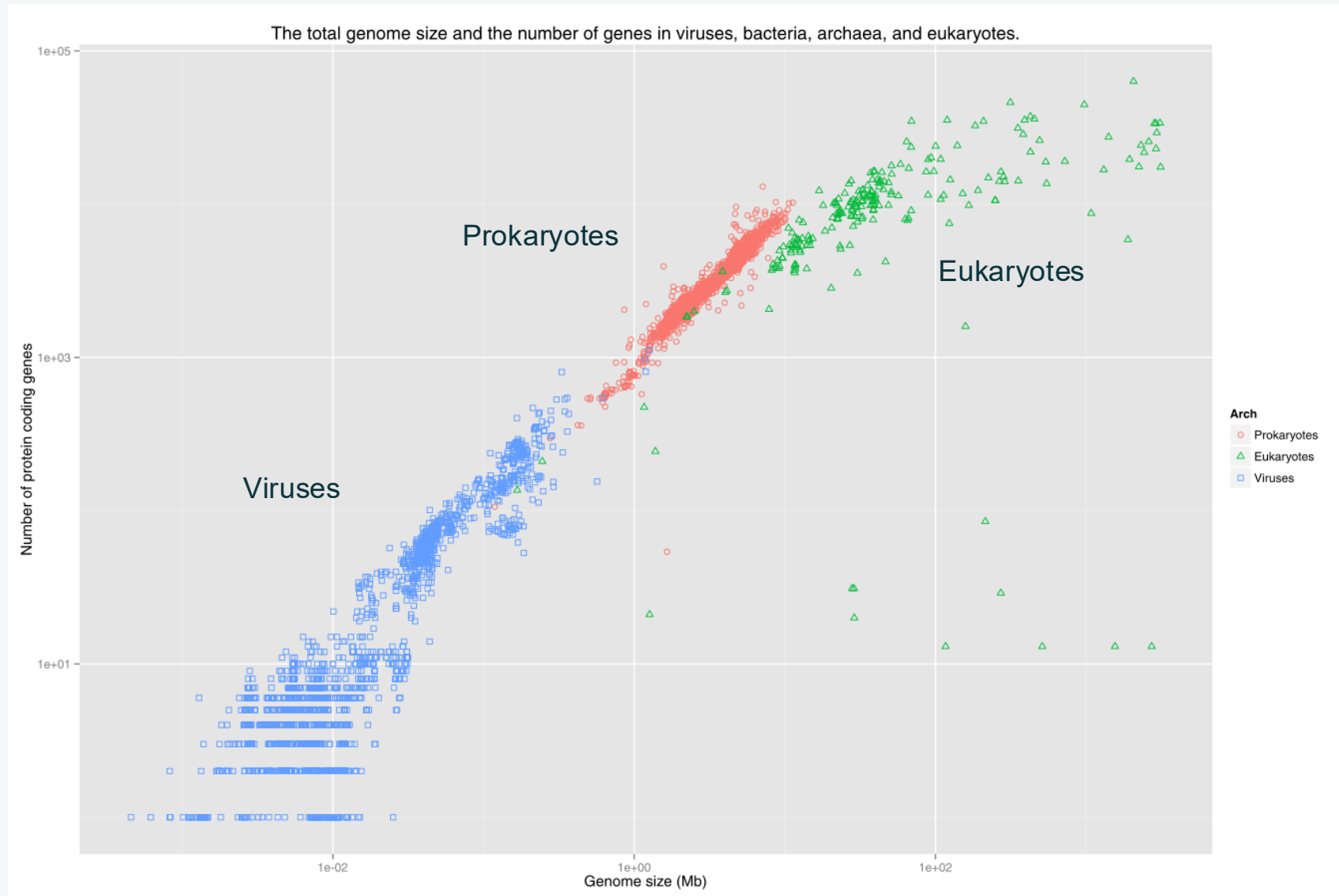
Build your vocabulary

What is a Genome?

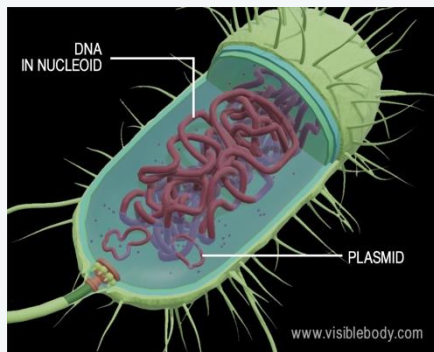
The total genetic material of an organism by which individual traits are encoded, controlled, and ultimately passed on to future generations



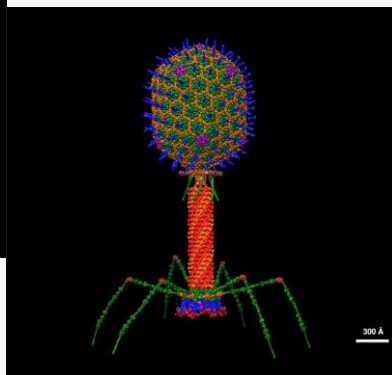
Genomes come in many different sizes



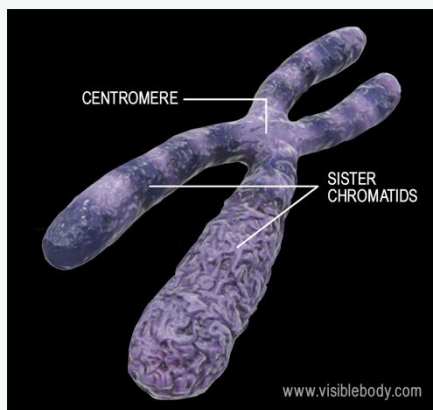
Genomes comes in different shapes



prokaryote



bacteriophage



eukaryote

Primarily DNA, but can be RNA in the case of some viruses

Some genomes are circular, others linear

Can be organized into discrete units (chromosomes) or freestanding molecules (plasmids)

Challenges of DNA

DNA has only four letters

They are strung together with NO obvious punctuation

There are no signals to say “a gene starts here”

AGGTTAGATTATGCCCCGAGGGCGCCCCAGCCGAAATTTTAAATGCAGGTTTAAATAGTTTAGAGC
CTGTGGGCTTCCATGGCTTGGTTCTGCTGTTCTTCACTGGGGACTTGGGGGACCCTGGGAGCTTG
TGATGGGGCCTGTCTCCACCTCTGTAAATCCAAGGAGTCAGATGACAAATCTGTCAATTTTCGGGCG
ACACACTCCCCTGAGGAAAGGGCCTTGCAGGAGGGCAGAGCAGCTTGCTGGGCATGGCAGGGAGT
GGAGAAGGGCAGGGGGCGCAGAGCAGGAGCAGCTTCCTGCCTCTGGGTGGGGACAGTGATCCCCA
CTGGGGACTGGCAAAGCCCCATGCTCTCTGTTCACCCTGGATGGGTGGCACCTGGGGGCAGGCAT
GGGGCCTGCAGGAGCCCCTGTGTGCCAGCCCTCCCCTGCCAGCATCCCATCTCCCAGGAGGCCCC
CAGGGCAGGTAAGTGCCAGGTCCCCCCTCAGCTCACCGTTGTCCTTCCCCTTGACGAACGCCTCC
CACTCCCGGAACCACTGCATGCTGATGCAGTAGATGACGCCCGGCGACTCCTCGGCCTGGAAGGC
CTTGTTCAACTGGCAGGCGGGCAGGGACGGGAGGAGACAGAGGGCAGGTCAGTCTCTATTTACCT
TCAGCAAGGATTTCCCAAATGCCCCCGCCCCAGTCCCTCACCCAAGAGTCTTACAAAACACCAG
ATACTCAAGGTGAAAAATCAAACCCCCAGACAAGCTCAAACAAAATGAACACACCCATCACTCAG
AGAGGACGGTGGTAACATTTTGTGGGTCTCTTCAATAACTGTTTGAACCAACTGAGACCACAAGG
GAGATTCTACTTTTTTGAGAAGGAATCTCACTCTGTCACCCAGGCTGGAGTGCAGTGGCGCGATCT
CGGCTTACTGCAACCTCTGCCTCCCAGGTTCAAGCGATTCTCCACCTCAGCCTCTTGAGTAGCT
GGGATTACAGGTGTGTGCCACCACCTGGCTACTTTTTTGTGTTTTTTAGTAGAGACGGGGTTTCGCC
ATGTTGGCCAGGCTGGTCTTGAACCTCCTGACCTCAGGTGATCCGCCCACCTCGACCTCTCAAAG
TGCTGGGATAACAGGCATGAACCACTGCGCCCGGCCTGGGAGATGCTAATTTTCTCCGGTTGAAT
AGAATGTGCCTATCTGCTCAGAGAGGCAGCTCTCCTTCTGACAGGAGCATTTTTCTTTTTTCGAGAT
GGGGGGTGGTCTCACTCTGTGCCAGGCGGGAGTGCAGCGGCGCAATCATGGCTCACTGCAGCCT
TGACCTCCTGGGCTCAAGTGATCCTCCACCTCAGCCTCCTGAGTAGGTTGGACCACAGGTGCAT
ACCACTAGGCCCAGCCCTGACAGTCTCTTTTTTCGTTTGTGTTCTGAGACAGGGTCTCACTCTATT
GCCCAGGCTGCGGTGCAGTGGCATGATCACGGCTCACTGCAGCCTCAACCTCCCAGGCTTAGGTG
ATCCTCCCAACTCACTCAGCCCTCCAGGTAGCGGGGACTACAGGTACACATCACCATGCCTGGCT
AATTTTTTGTATTGTTTGTAGAGATGGGGTTTCGCCATGTTGGCCAAGTTGGTCTTGAACCTCCTGG

Challenges of DNA

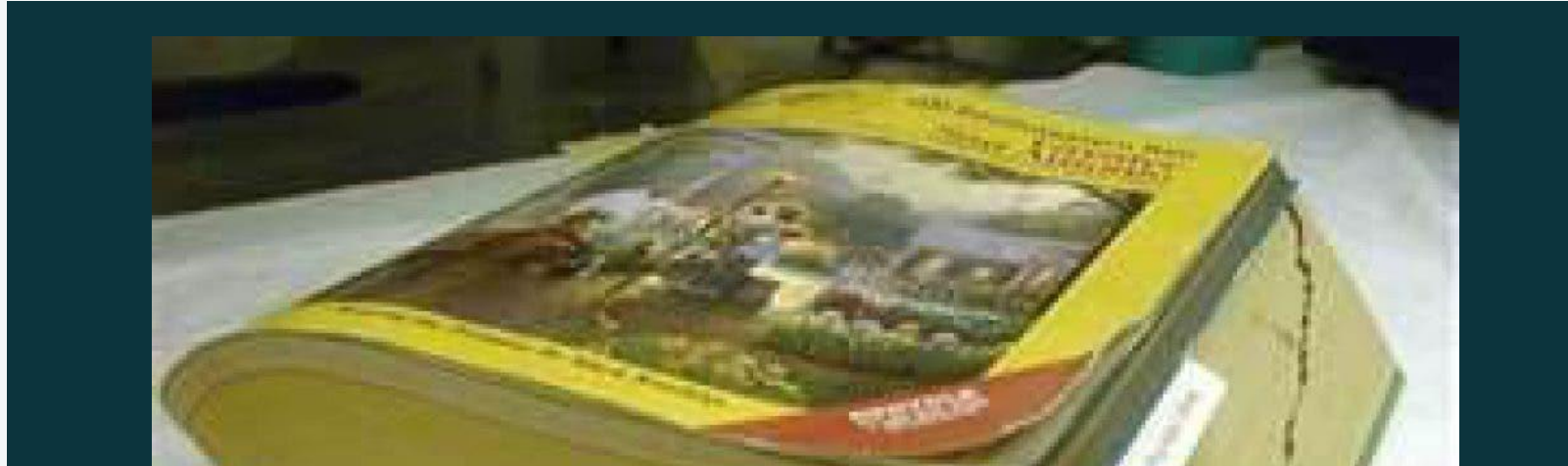
DNA has only four letters

They are strung together with
NO obvious punctuation

There are no signals to say
“a gene starts here”

Its large size

How large?

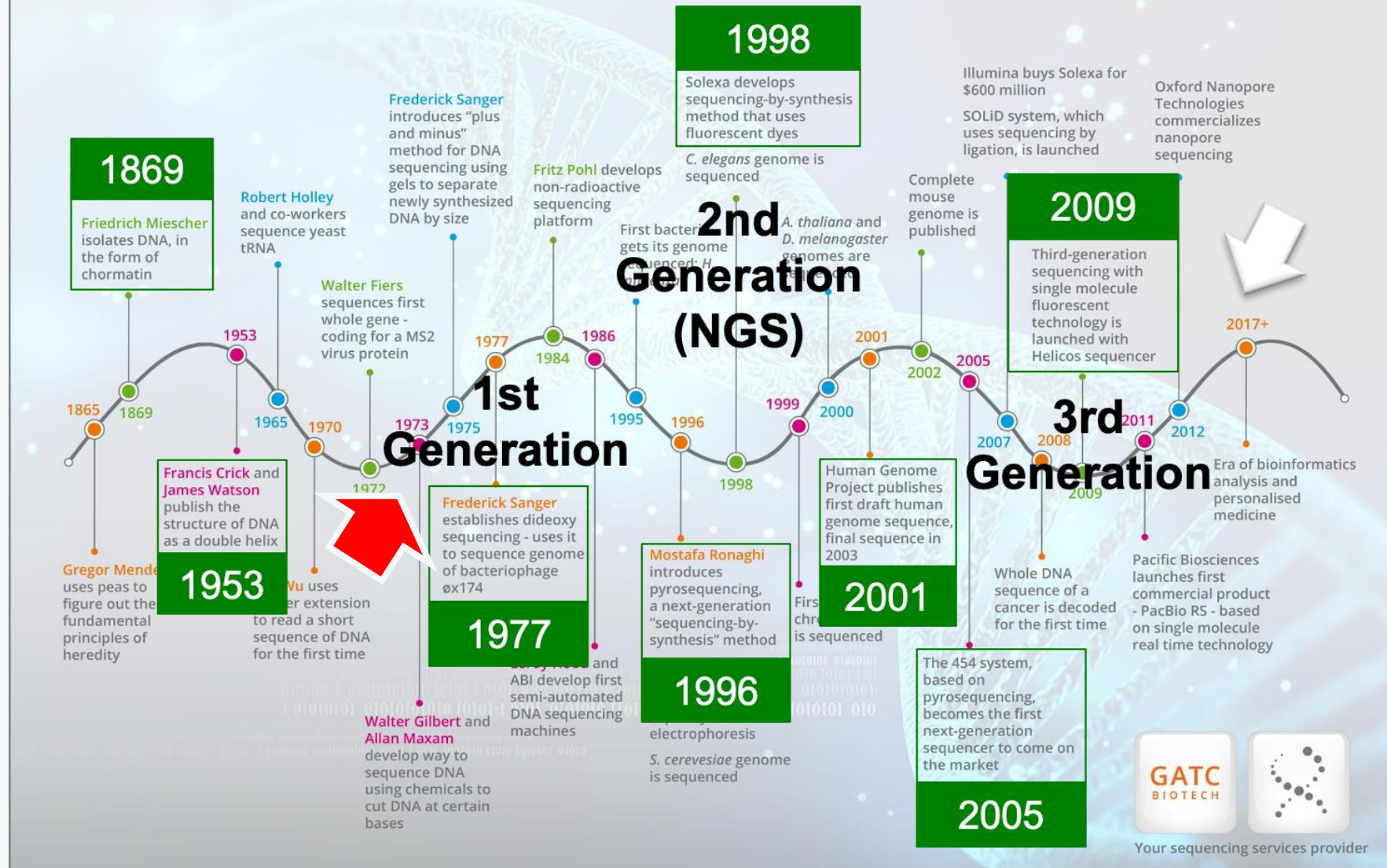


How did we sequence the human genome?



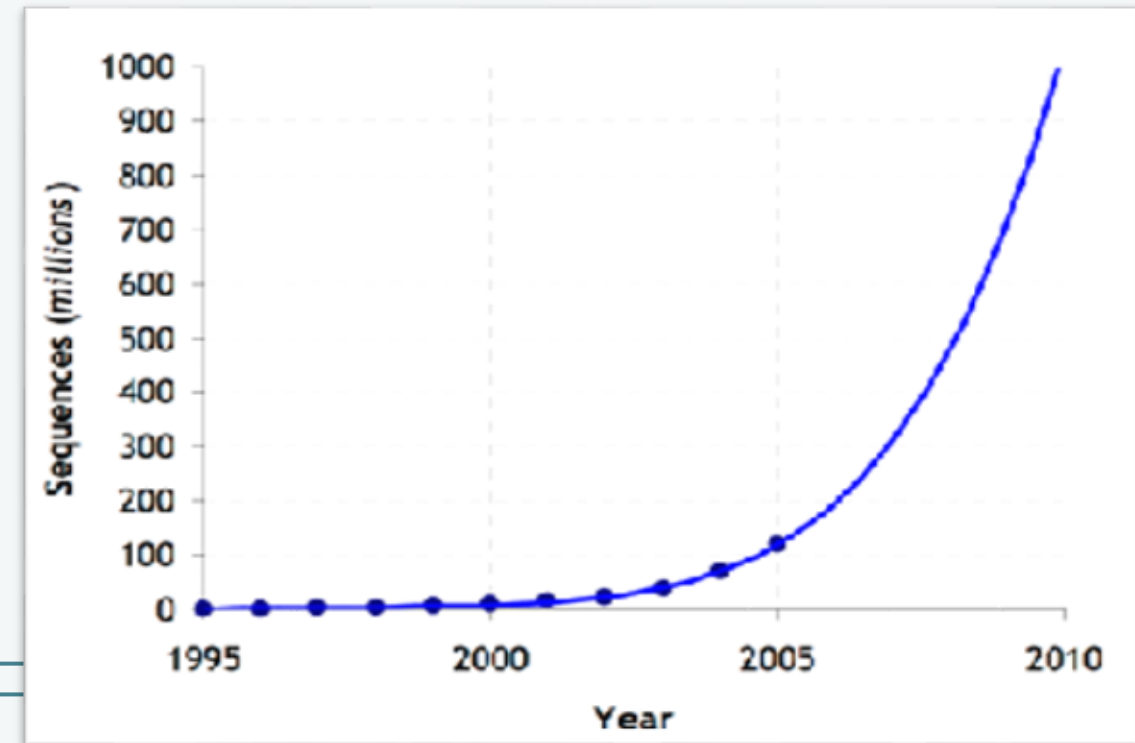
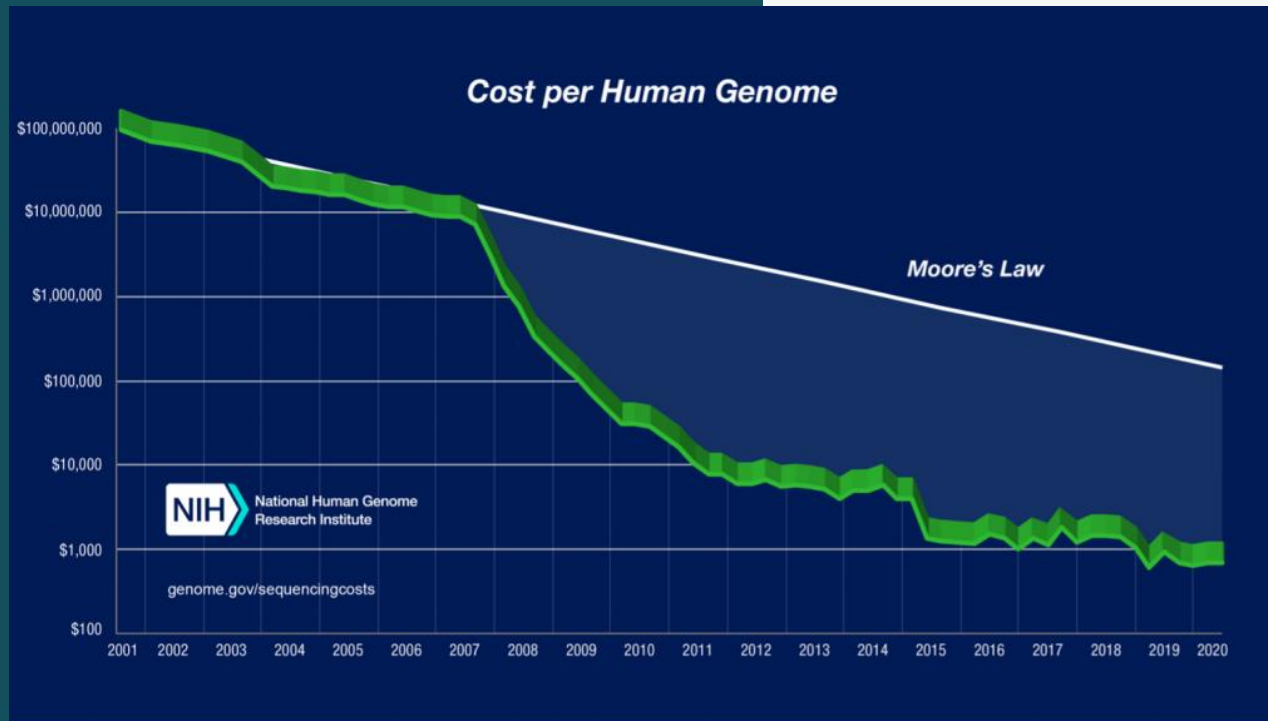
If compiled in books, it would fill an estimated 200 volumes the size of a Manhattan telephone book (at ~1000 pages each) and reading it would require 26 years working around the clock.

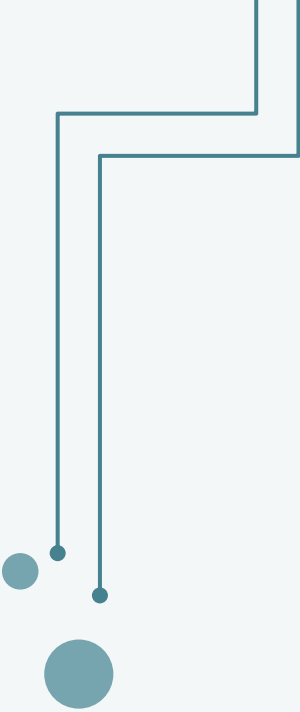
A BRIEF HISTORY OF DNA SEQUENCING



Where did bioinformatics come from?

- Bioinformatics arose as molecular biology began to be transformed by the emergence of molecular sequence and structural data



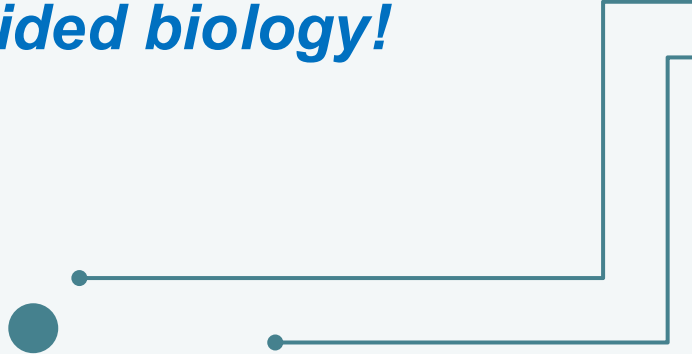


What is Bioinformatics?

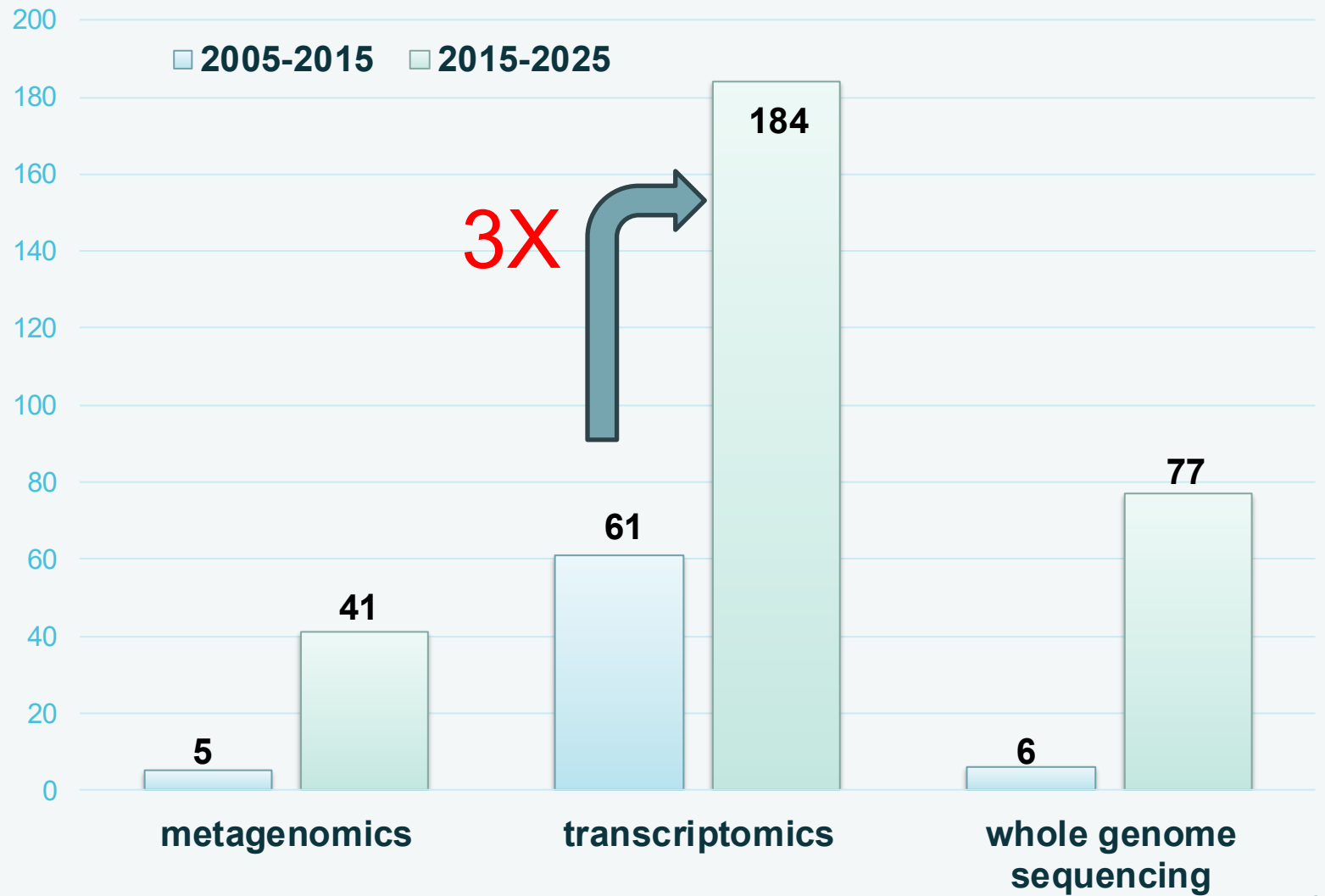
The use of biological databases and computational programs to analyze proteins, genes, and the complete collection of DNA that comprises an organism (the genome)

... Bioinformatics is a hybrid of biology and computer science

... Bioinformatics is computer-aided biology!



The number of publications from UVM with Omics has increased





How do we deal with it?

We accept that the
needs are changing



Molecular

1980s:

- RFLP
- PCR

1990s – 2000s:

- microarrays
- RNAi

2010 and beyond....

- RNA-Seq
- Single-cell
- Spatial

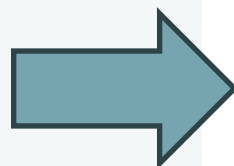
Computational

- Word processing

- BLAST
- Databases

- **R programming**
- **Linux/UNIX**
- **Python**

In the lab, someone does this....



... this might be the next step

Stain Deconvolution

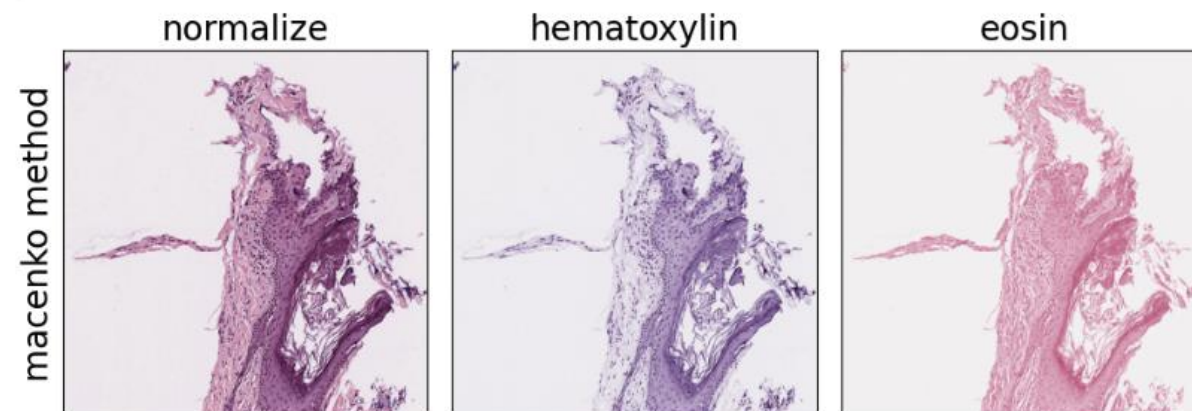
```
In [6]: fig, axarr = plt.subplots(nrows=2, ncols=3, figsize=(10, 7.5))

for i, method in enumerate(["macenko", "vahadane"]):
    for j, target in enumerate(["normalize", "hematoxylin", "eosin"]):
        # initialize stain normalization object
        normalizer = StainNormalizationHE(target=target, stain_estimation_method=method)
        # apply on example image
        im = normalizer.F(source_region)

        # plot results
        ax = axarr[i, j]
        ax.imshow(im)
        if j == 0:
            ax.set_ylabel(f"{method} method", fontsize=fontsize)
        if i == 0:
            ax.set_title(target, fontsize=fontsize)

for a in axarr.ravel():
    a.set_xticks([])
    a.set_yticks([])

plt.tight_layout()
plt.show()
```



PathML

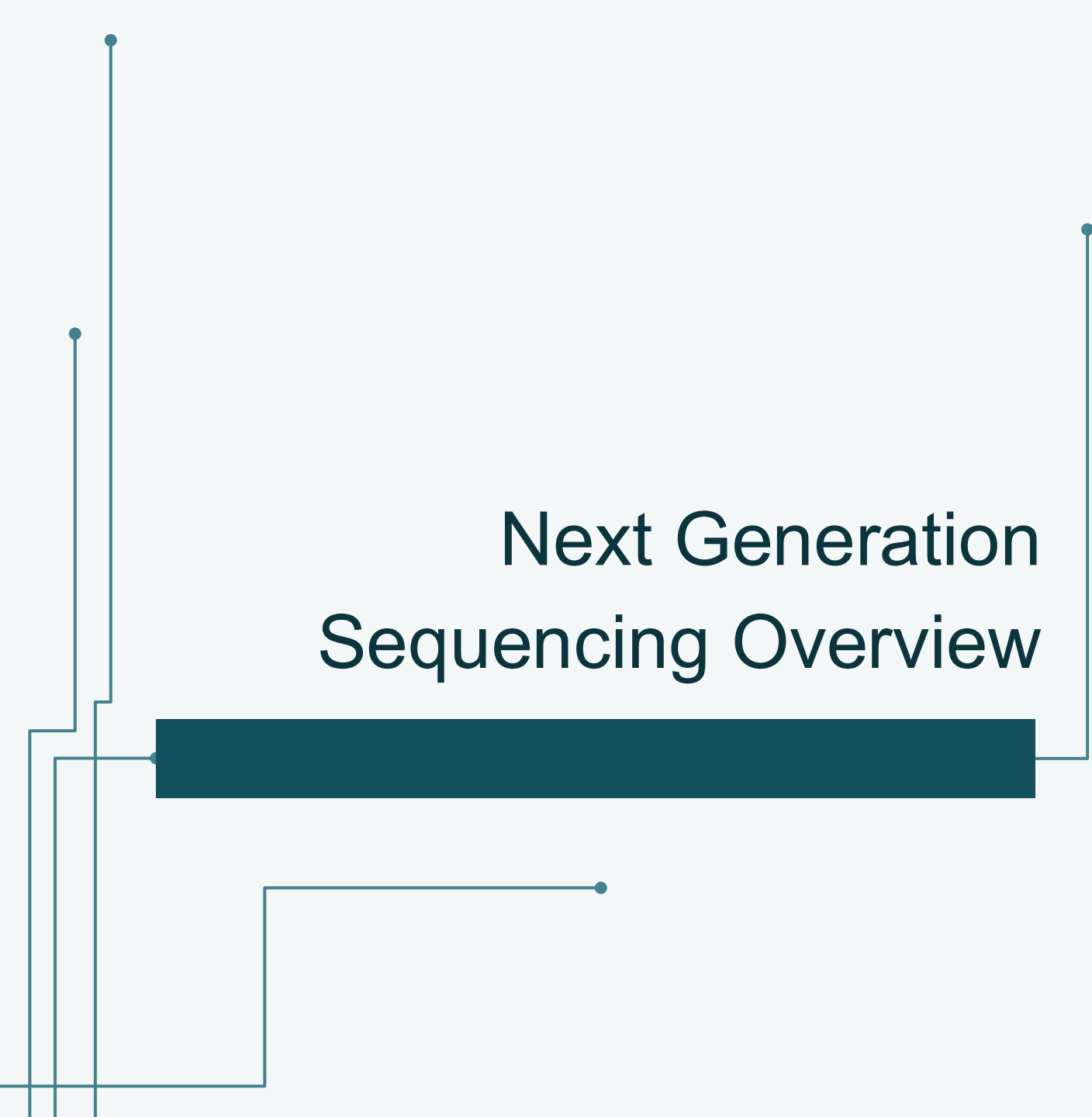


Tools for Computational Pathology

[View On GitHub](#)

[View On PyPI](#)

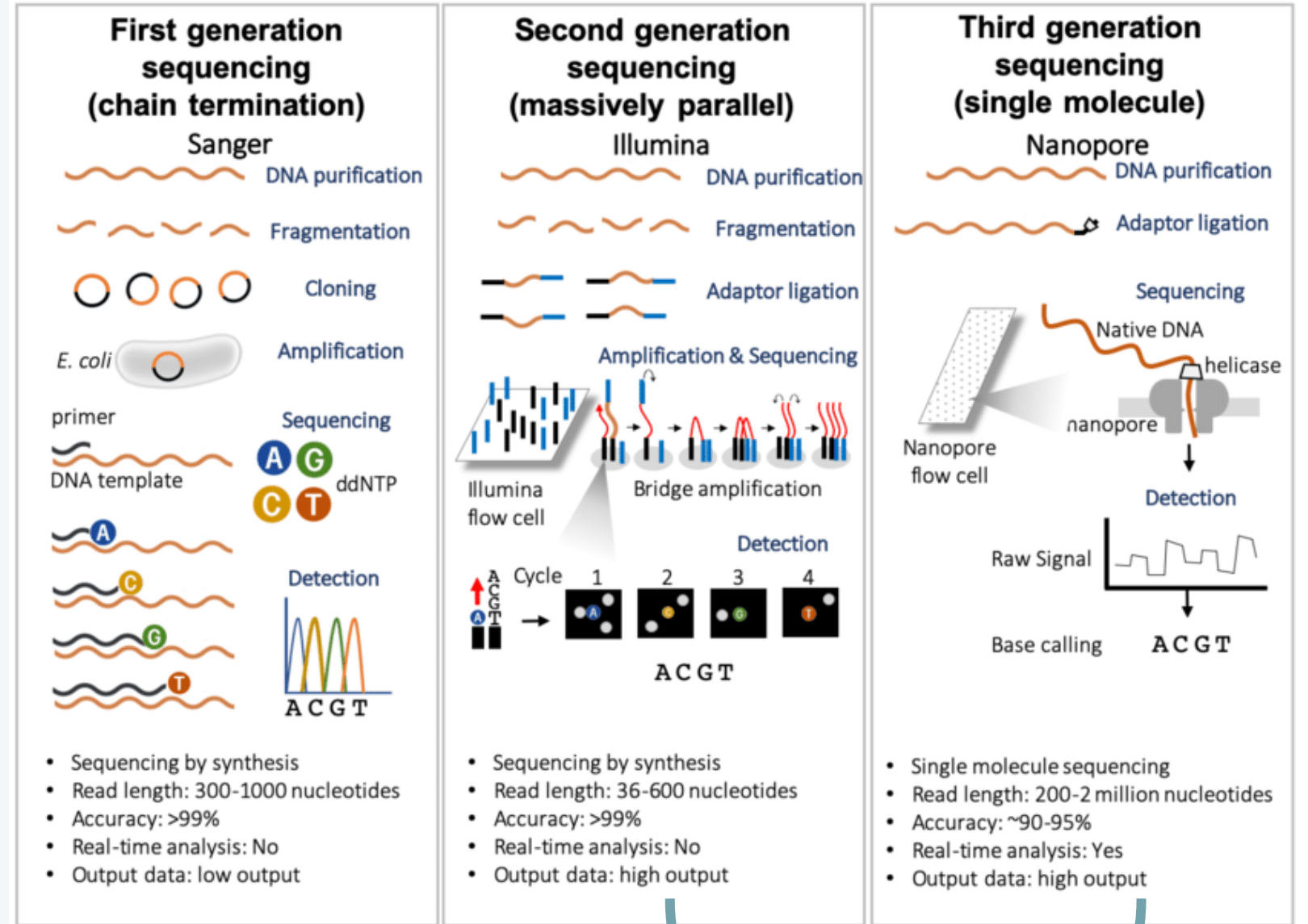
[View Docs](#)



Next Generation Sequencing Overview

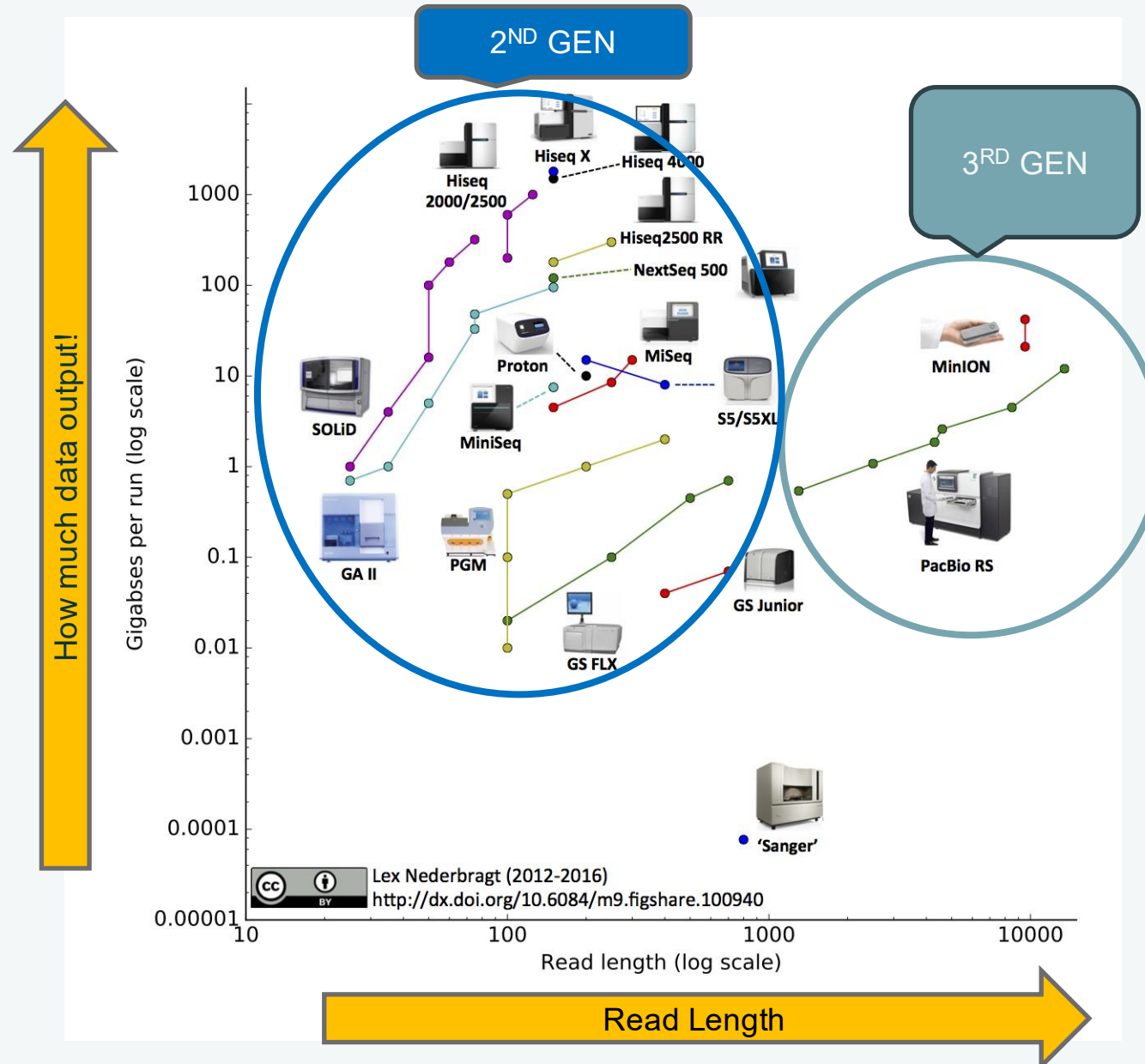
What is NGS?

- Is a high-throughput DNA sequencing methodology that makes use of parallelization to process up to half a million sequences concurrently
- Often taken to mean 2G technologies



Next Generation Sequencing (NGS)

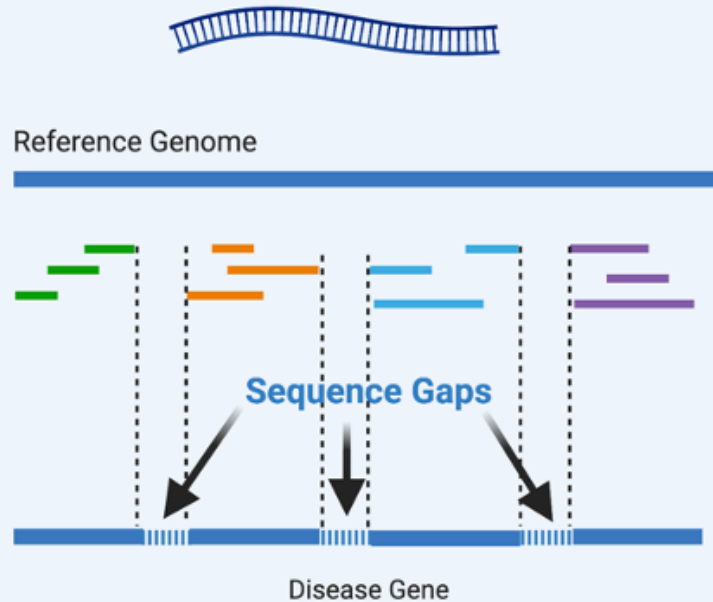
Diversity of sequencing technologies



- Roche (454) / ION
- **Illumina (Solexa)**
- Helicose
- ABI Solid
- Polonator
- PacBio
- **Nanopore**
- Element Biosciences

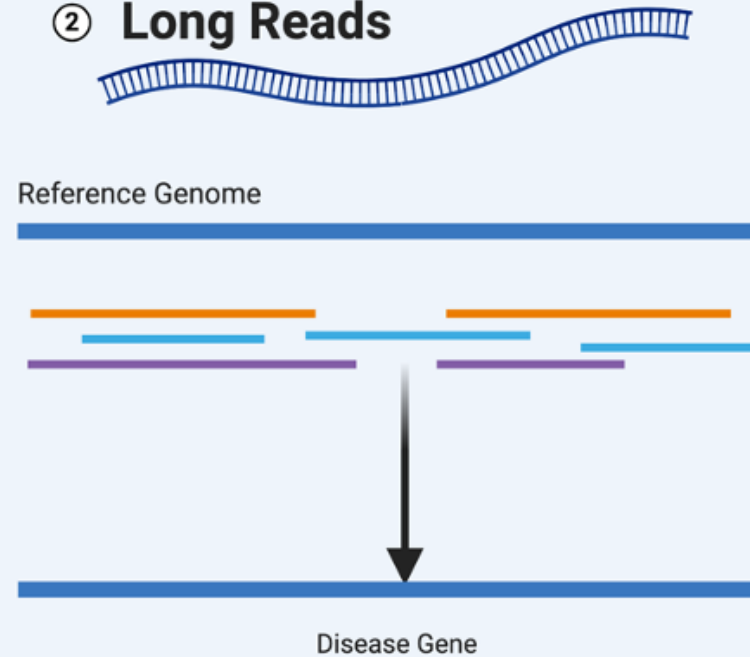
A *read* is an inferred *sequence* of base pairs corresponding to part of a DNA fragment

① **Short Reads**



Missing sequence data leads to gaps
in genome coverage and limits variant detection

② **Long Reads**



Long reads map uniquely and
span large variants providing
comprehensive variant detection

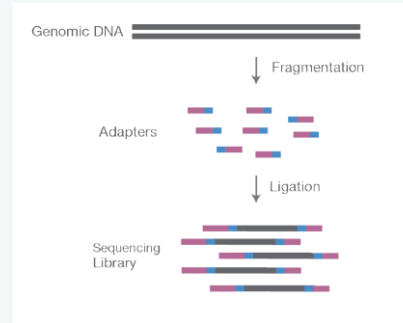
Next-generation sequencing (= 2nd generation)

- unifying characteristics of the different NGS platforms:
- **short reads (50-250 bp) and short fragments (250-1000 bp)**
 - Includes HiSeq, **NextSeq**, NovaSeq platforms
 - Run Time varies but can be up to > 48hrs
 - Maximum Output is ~25 to 400 million reads per lane

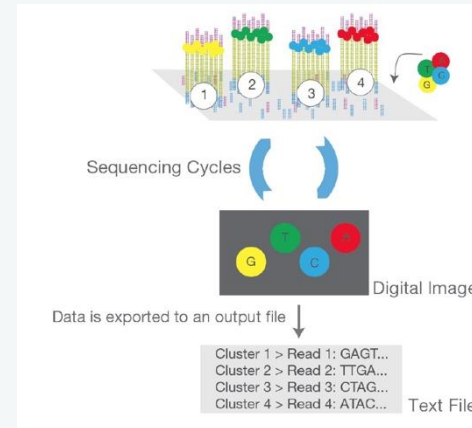


General approach for 2nd Gen NGS

1. Generate **library**



2. Sequence the library



3. Analyze the sequence



Figure 1. Dawadi et al.

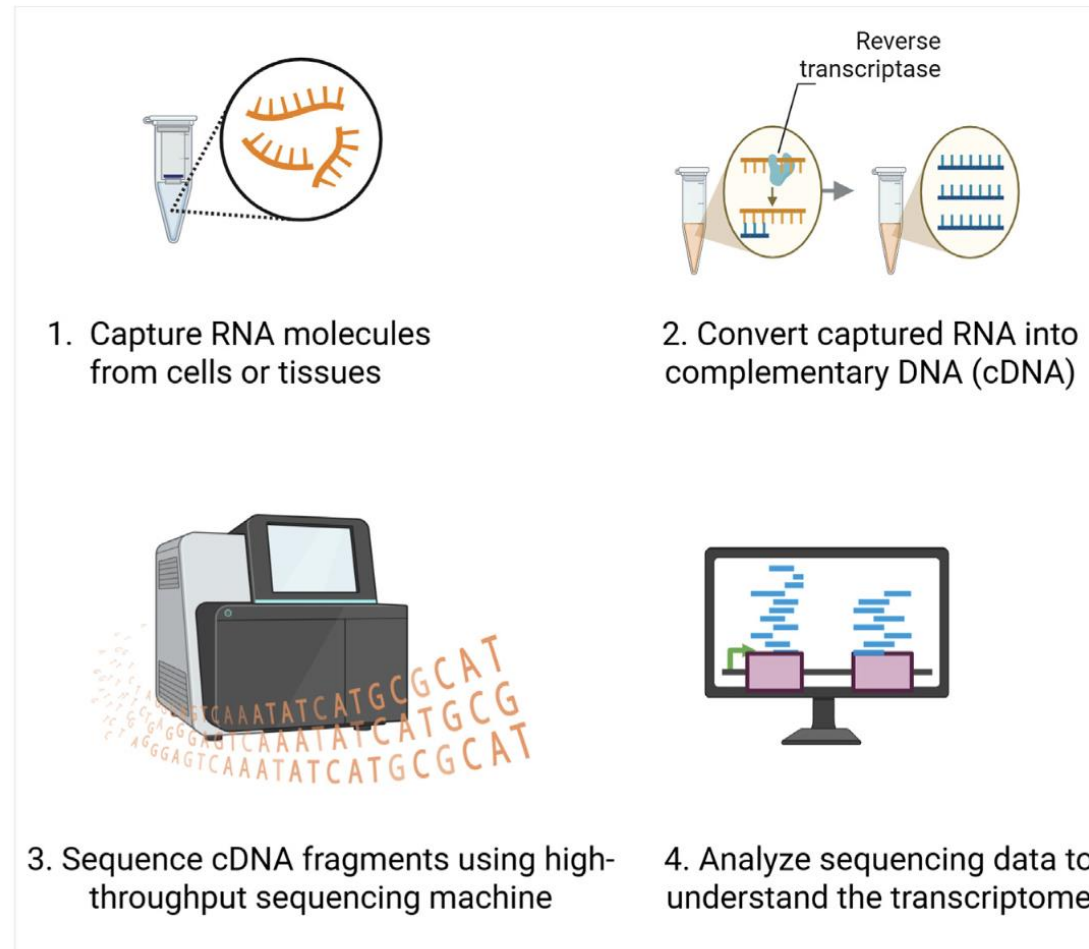
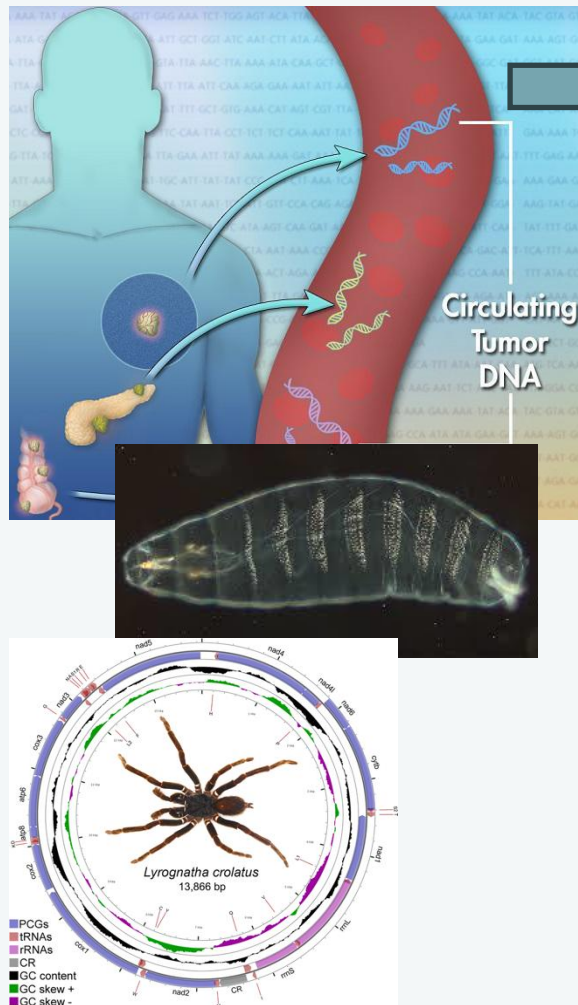
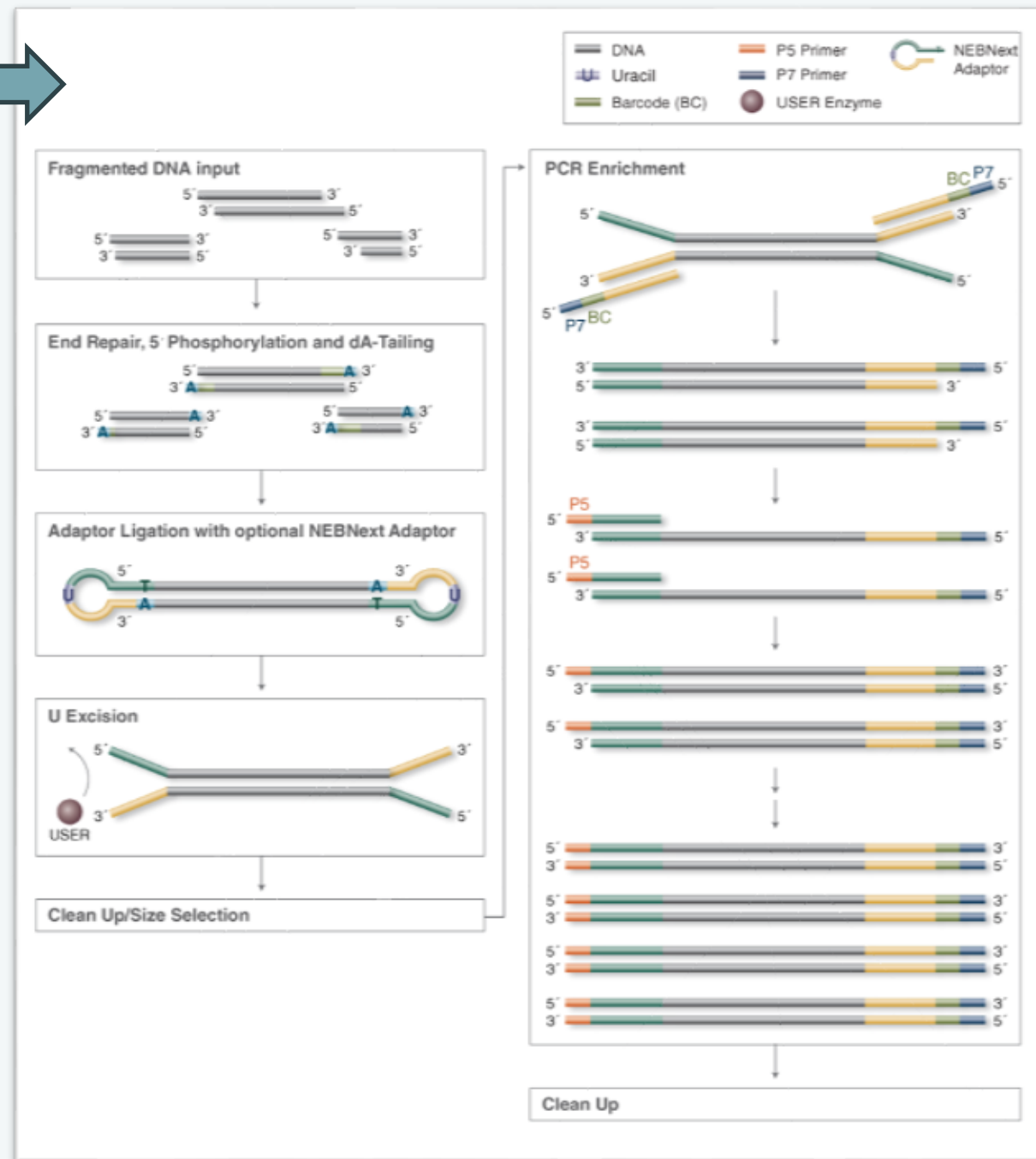


FIGURE 1

Overview of RNA sequencing workflow. RNA molecules are first extracted from cells or tissues, then converted into complementary DNA (cDNA) using reverse transcriptase. The resulting cDNA fragments are sequenced using high-throughput sequencing platforms, and the data are computationally analyzed.

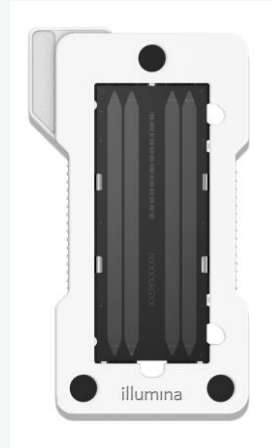


NGS library preparation



Libraries are then loaded onto a flow cell

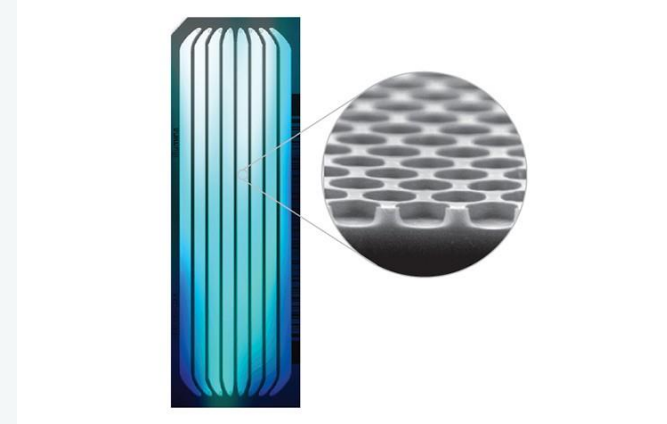
- The flow cell is a glass slide with lanes
- Libraries are loaded into a flow cell and the fragments are hybridized to the flow cell surface



NextSeq500

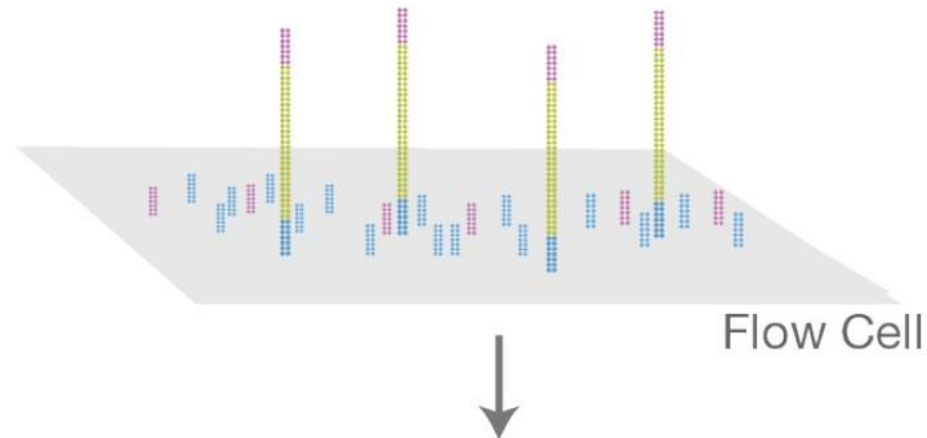


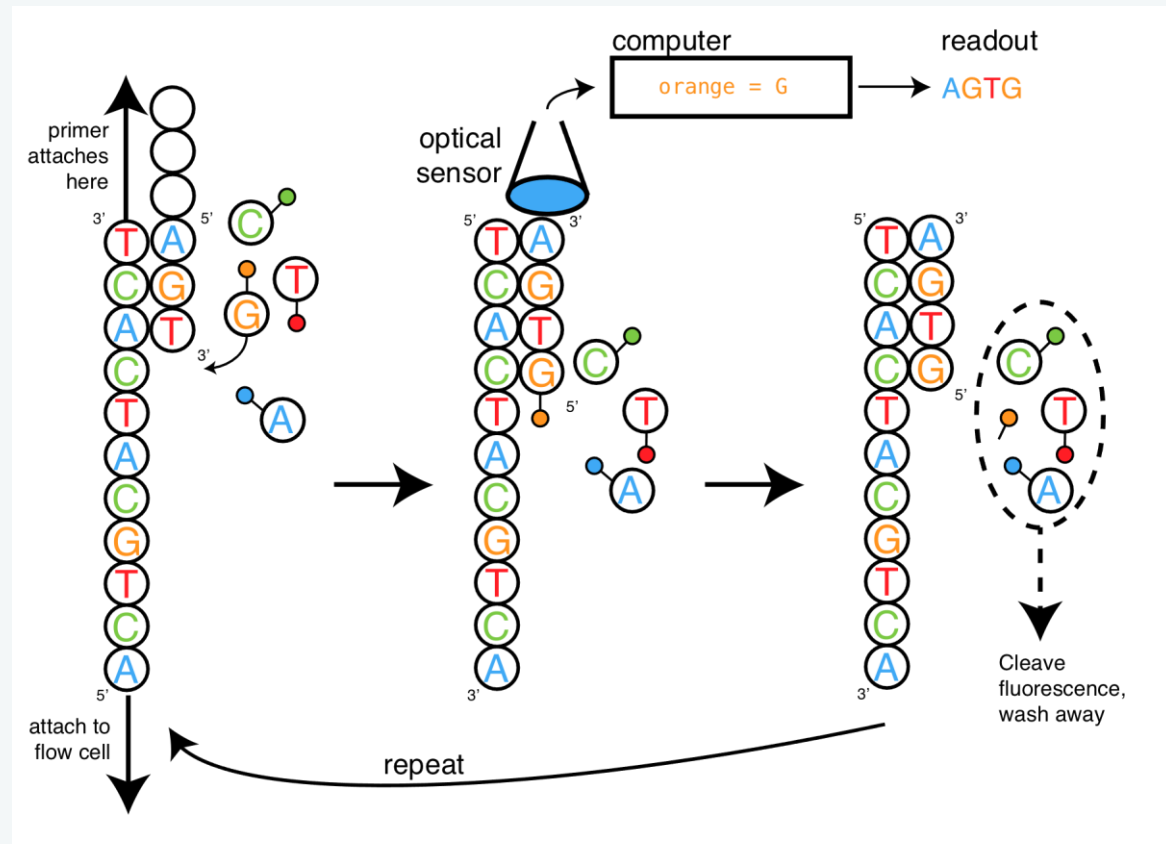
HiSeq2500



HiSeq3000/4000

B. Cluster Amplification





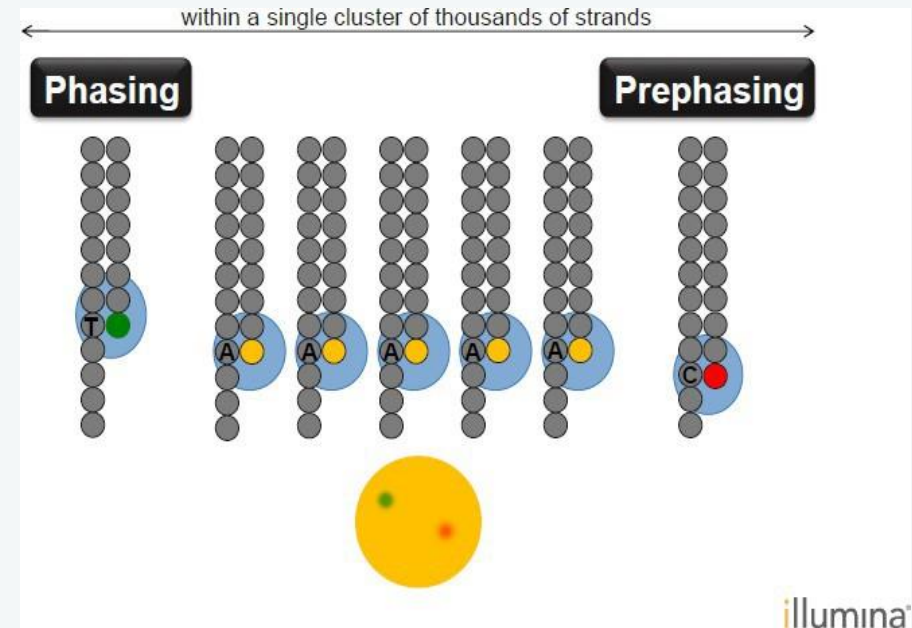
- Then sequencing begins!
 - Have fluorescent-labeled nucleotides
 - Each dNTP has a corresponding color
 - During each **cycle**, a labeled dNTP is added to the growing chain
 - An image is then taken
 - Then the fluorescent dye is cleaved to allow for the next nucleotide to be incorporated in the next **cycle**

Problem

- Sequencing by synthesis progressively becomes out-of-sync therefore, signal quality deteriorates as the read-length grows
- At each cycle, accumulates noise

Solution

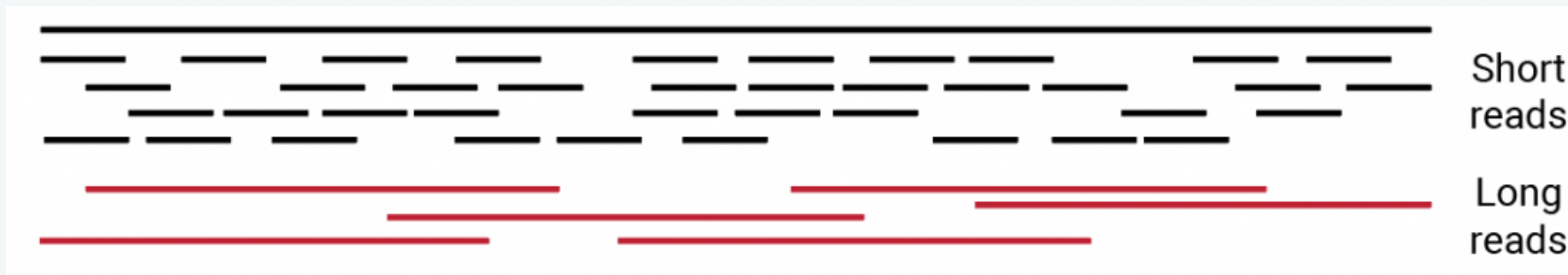
- Sequence only (up to 300bp)



The human genome is 3 billion base pairs i.e. this is a lot of reads generated !!

Third Generation Sequencing

- Typical characteristics:
 - No amplification step
 - **Long (1,000bp+) sequence reads**
 - Real-time monitoring of nucleotide incorporation
 - Reads the nucleotide sequence at the **single molecule** level
 - Often associated with nanopore technology



Oxford Nanopore Technology (ONT) sequencing

- Sequencers can be pocket-sized, USB connectable, cost \$150 to \$1,000
- Generates incredibly long reads: read length are typically between 10,000bp & as long as 1,000,000 bp!



Dr. Kathleen Rubins is an
microbiologist and NASA
astronaut

First person to sequence in space

“Analysis of anything, by
anyone, anywhere”



<https://vimeo.com/297106166>

<https://www.nasa.gov/astronauts/biographies/kathleen-rubins/biography>

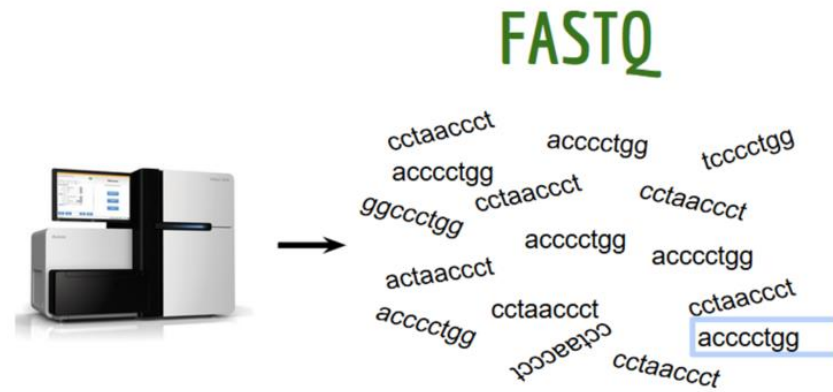
How does it work?

- A strand of DNA is passed through a nanopore
- The magnitude of current is changed as the bases G, A, T and C pass through the pore in different combinations



The resulting output

The FASTQ format.



A “standard” format for storing and defining sequences from next-generation sequencing technologies.

Sequence ID	@SEQ_ID
Sequence	GATTTGGGGTTCAAAGCAGTATCGATCAAATAGTAAATCCATTTGTTCAACTCACAGTTT
<separator>	+
Quality scores	! ' ' * ((((* * * +)) % % % + +) (% % % %) . 1 * * * - + * ' ')) * * 5 5 C C F > > > > C C C C C C C C 6 5

http://en.wikipedia.org/wiki/FASTQ_format

Figure 2. Dawadi et al.

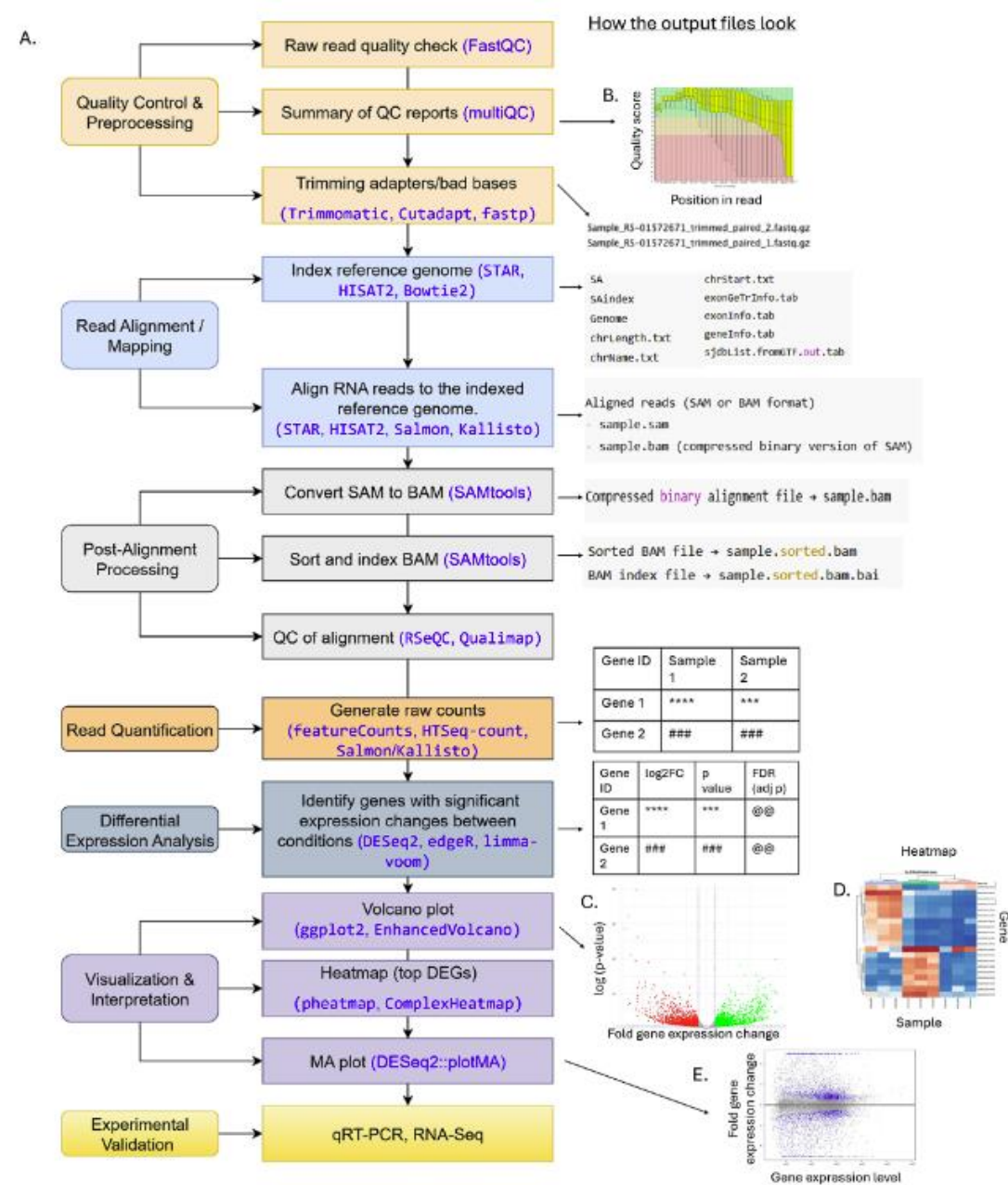
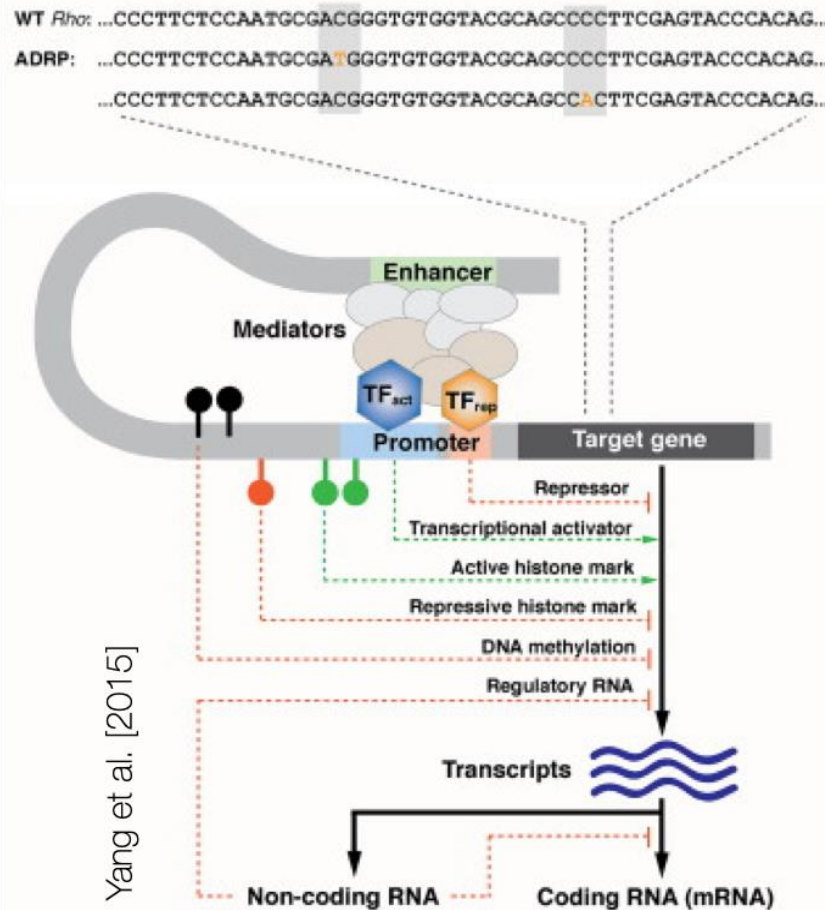


FIGURE 2
Flowchart illustrating an RNA-Seq analysis pipeline, from raw FASTQ files to functional interpretation. (A) Steps and tools. Each step includes key steps shown on the left and commonly used tools (in blue) and corresponding outputs (files, tables, or figures) displayed on the right next to the analysis. (B) Example of low-quality data using FastQC. The red and blue line is the median and mean of quality score at a given position of a read. The yellow box represents the inter-quartile range (25%–75%), and the upper and lower whiskers represent the 10% and 90% points. The higher the quality score is the better the base call. The background of the graph (green, orange, and red) indicates very good, reasonable, and poor quality calls, respectively. (C) Volcano plot. (D) Heatmap. (E) MA plot.

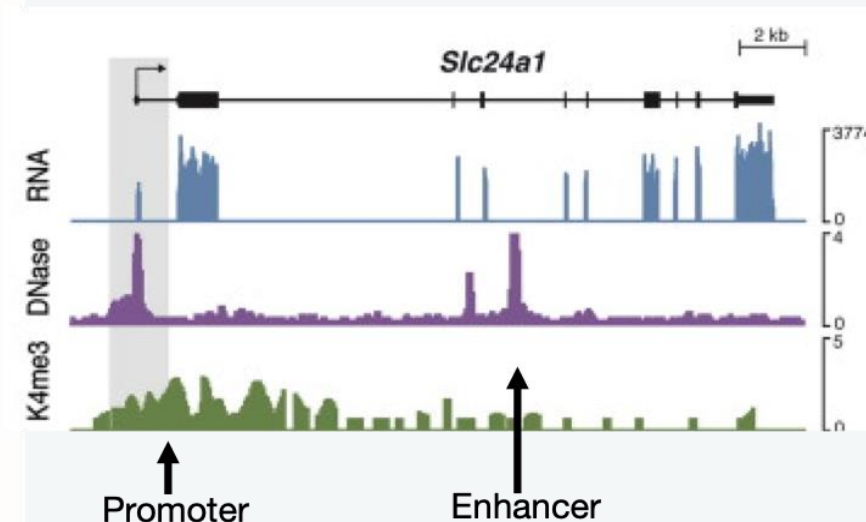
NGS can be used for (A) qualitative and (B) quantitative approaches

(A) “Reading” the actual sequence

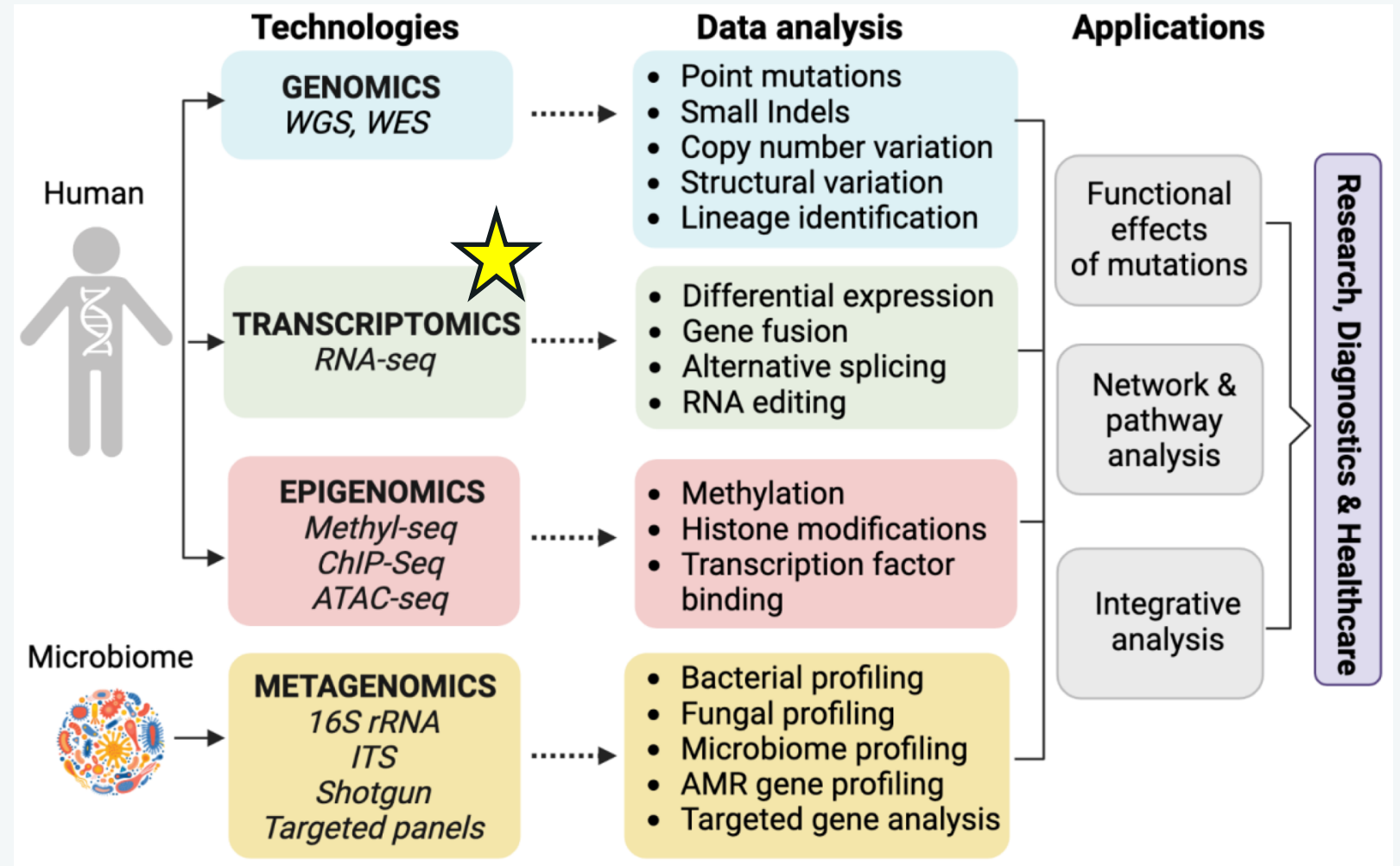


(B) Characterizing DNA regions with certain properties

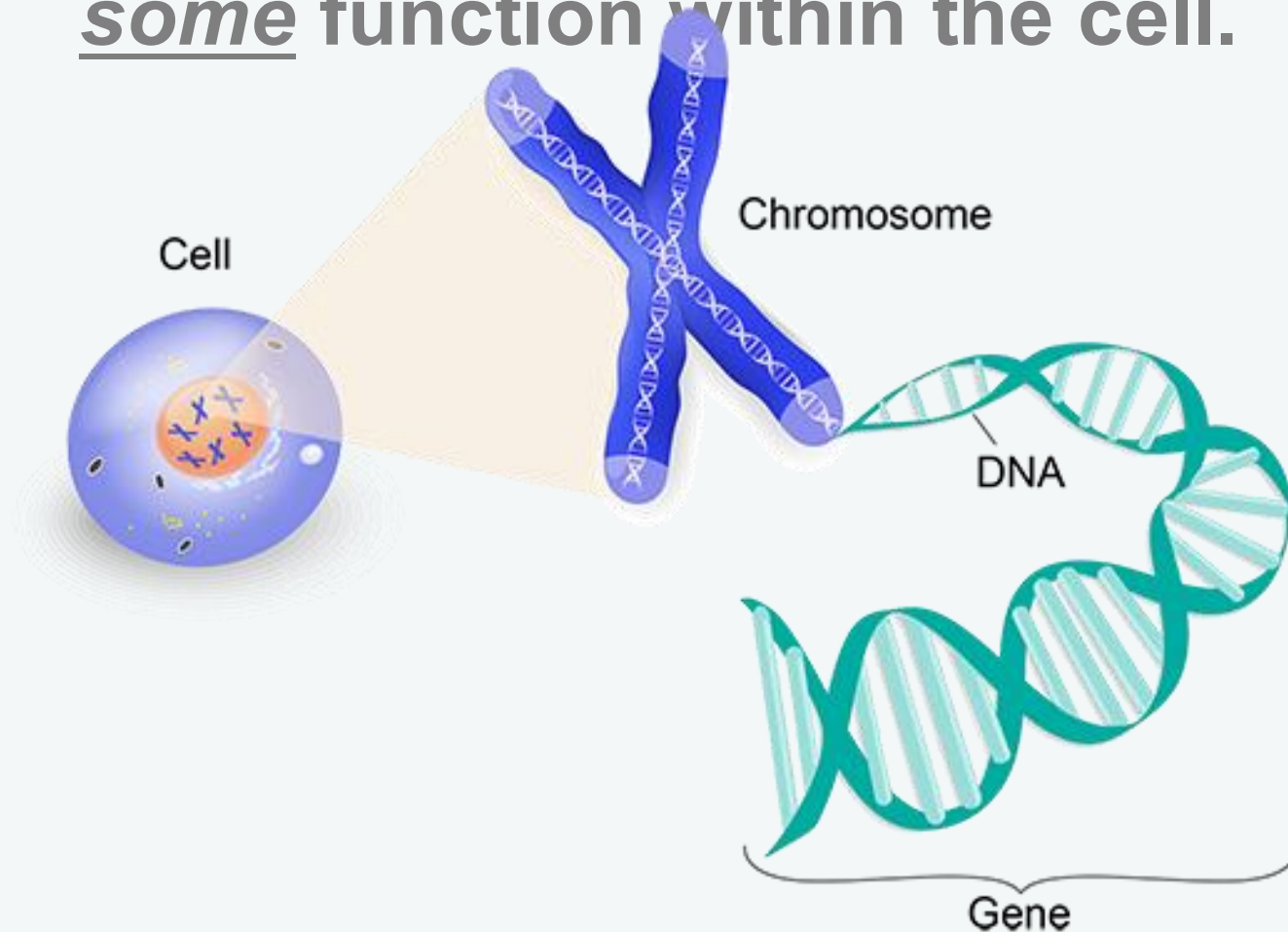
that are first biochemically enriched in a sample of interest – their DNA sequence is only needed to identify the locus of origin, the information of interest is based on the abundance (= enrichment) of seq. reads



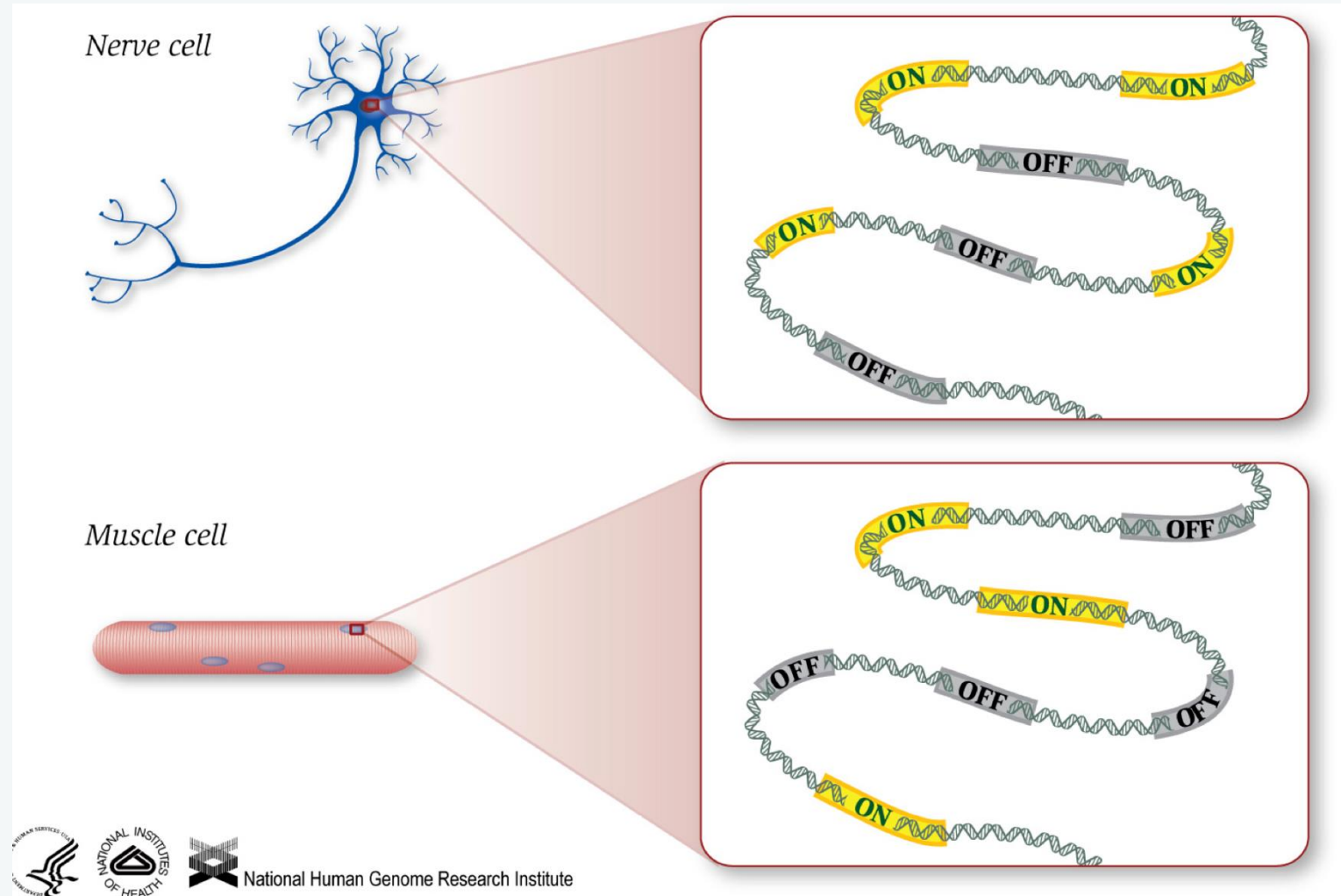
Common experimental designs



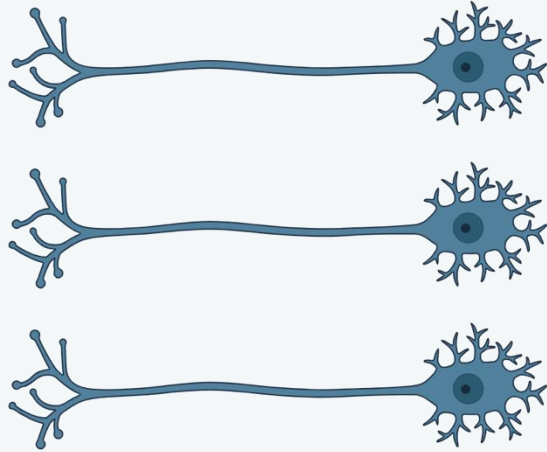
Genes provide the instructions to make proteins, and these proteins provide some function within the cell.



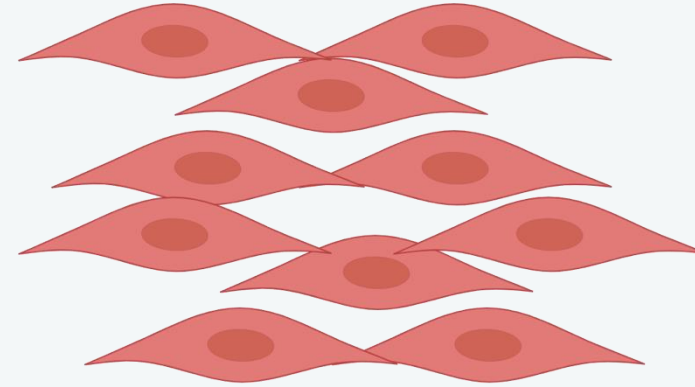
But how is it that all cells contain the same DNA sequence, yet we have different cell types?



Nerve Cell

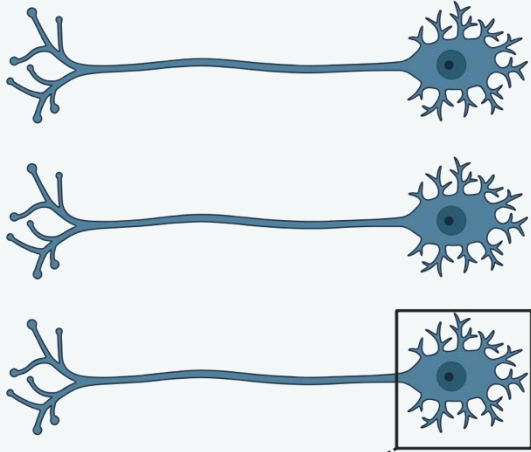


Muscle
Cell

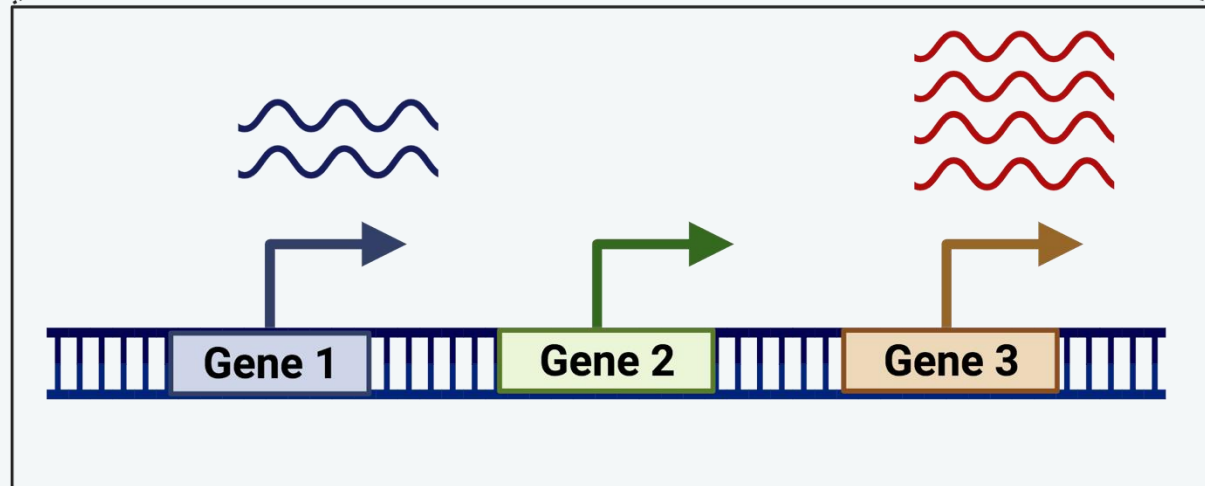
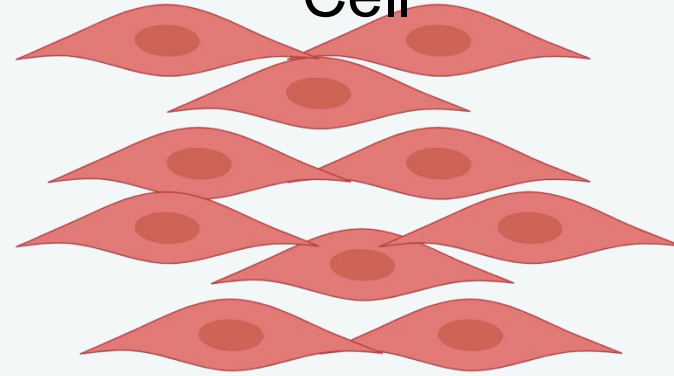


We would like to understand the *genetic mechanism* that drives the difference in cellular phenotype

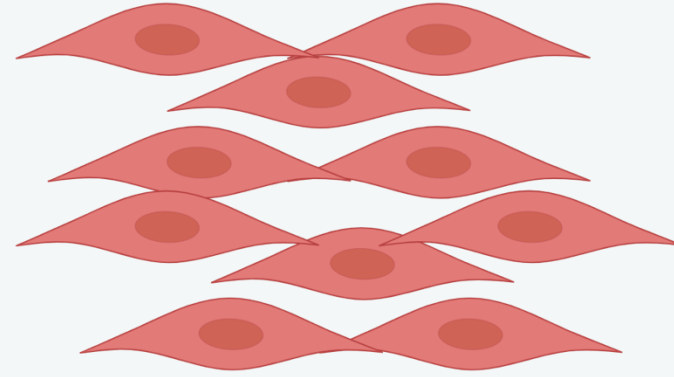
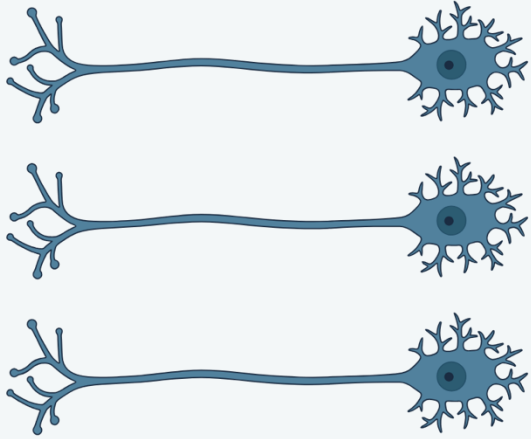
Nerve Cell



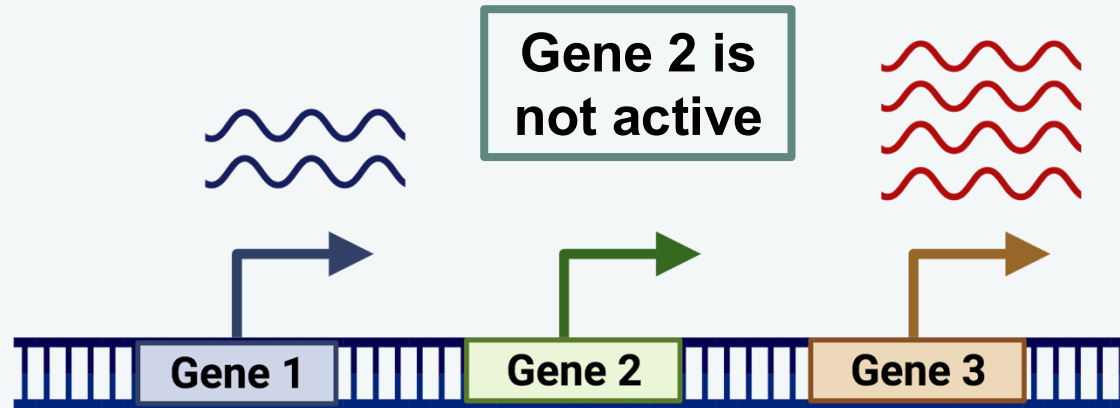
Muscle Cell

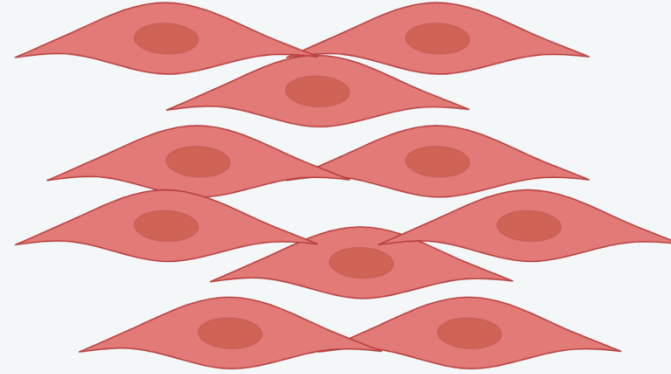
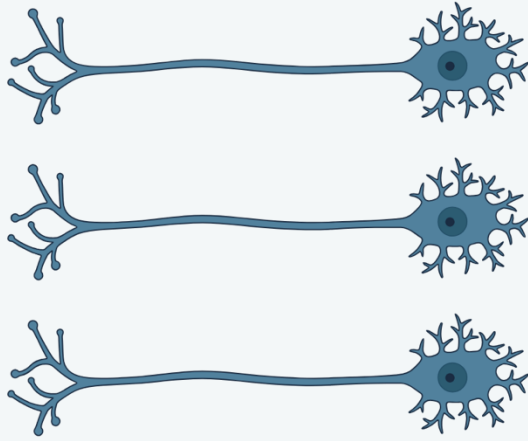


*Some genes
are more
active than
others*

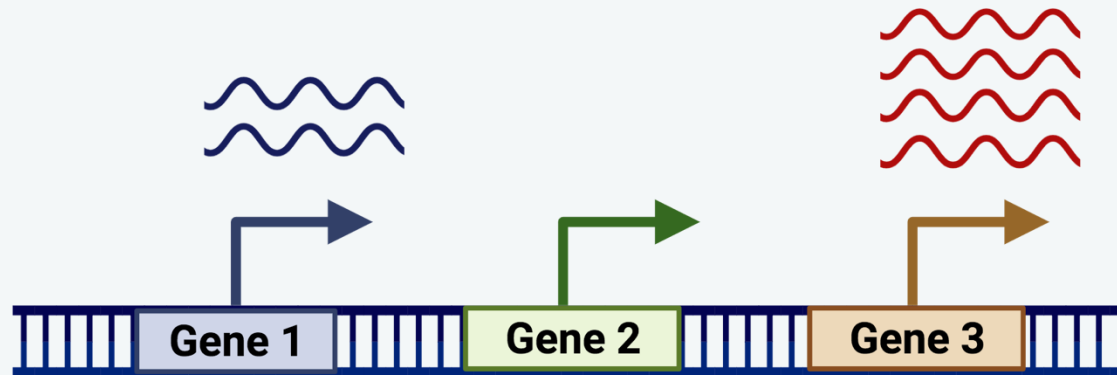


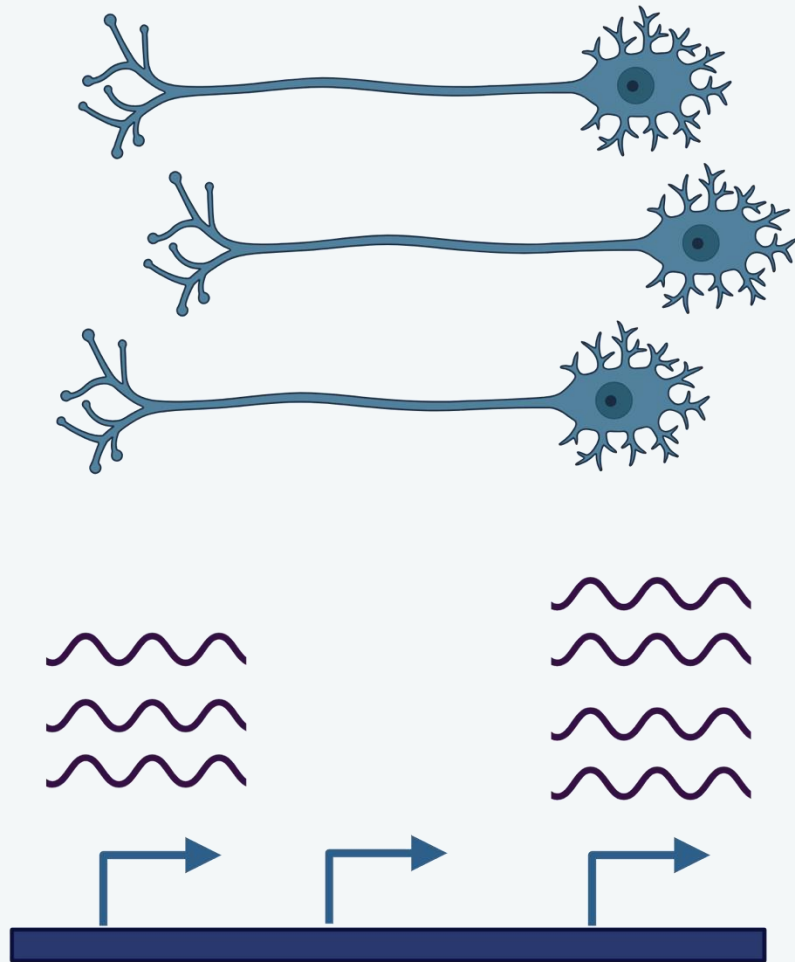
**Gene 3 is
most active**



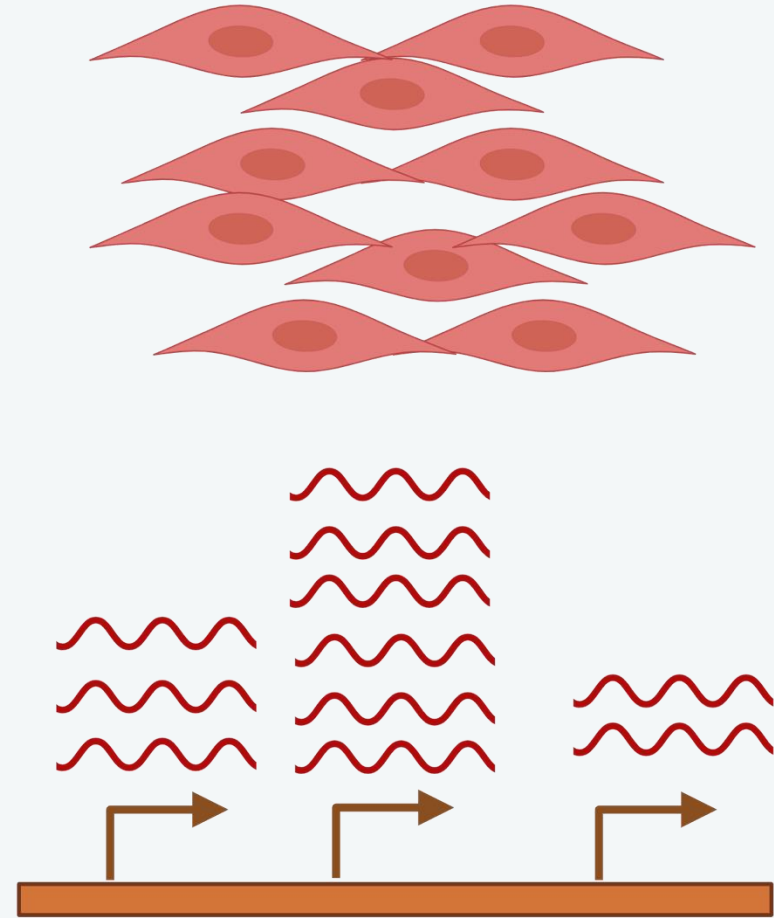


We need a technique to tell us which genes are active and how much they are being transcribed – RNA-Seq

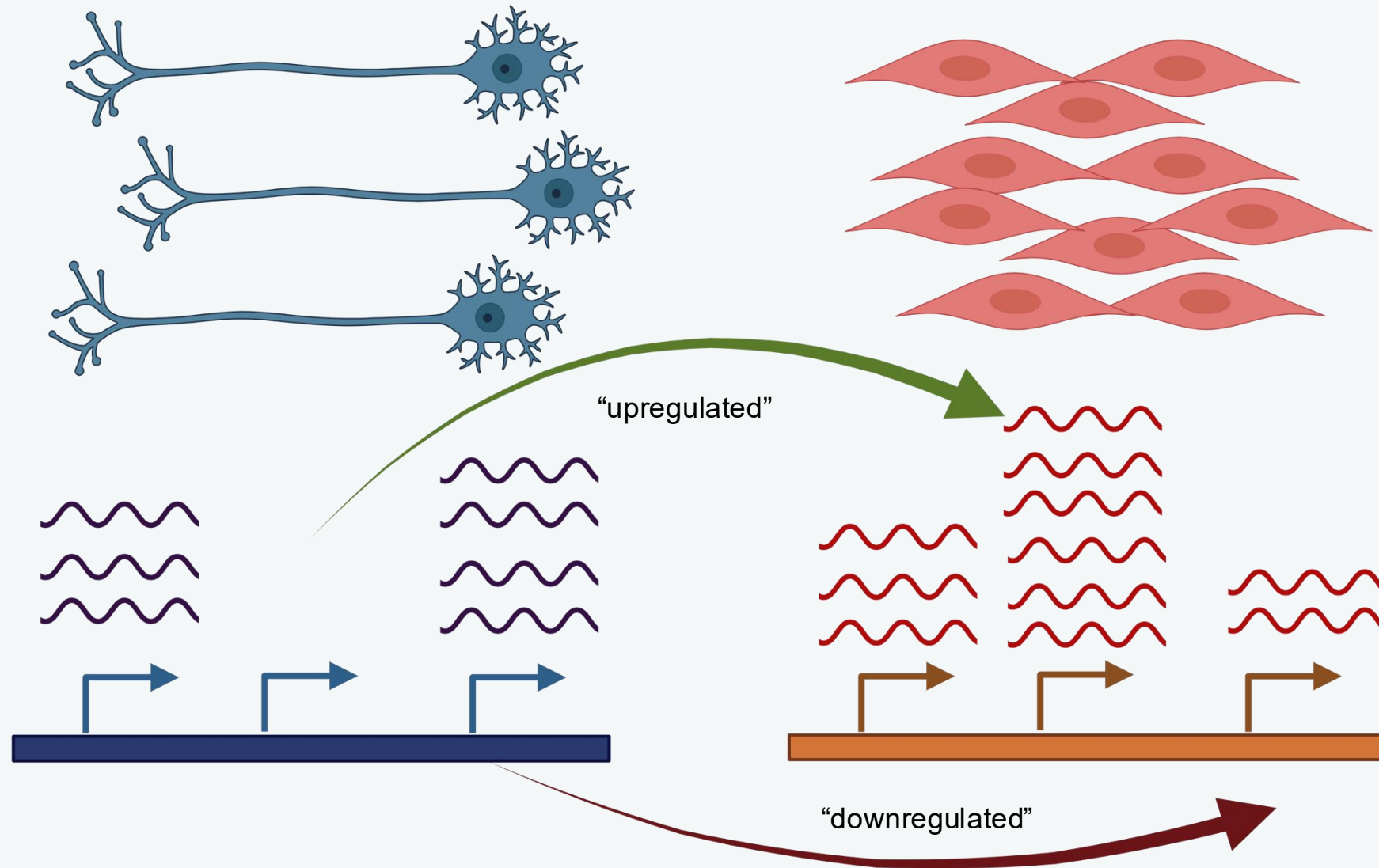




**We can use RNA-Seq to
measure gene expression
in nerve cells**

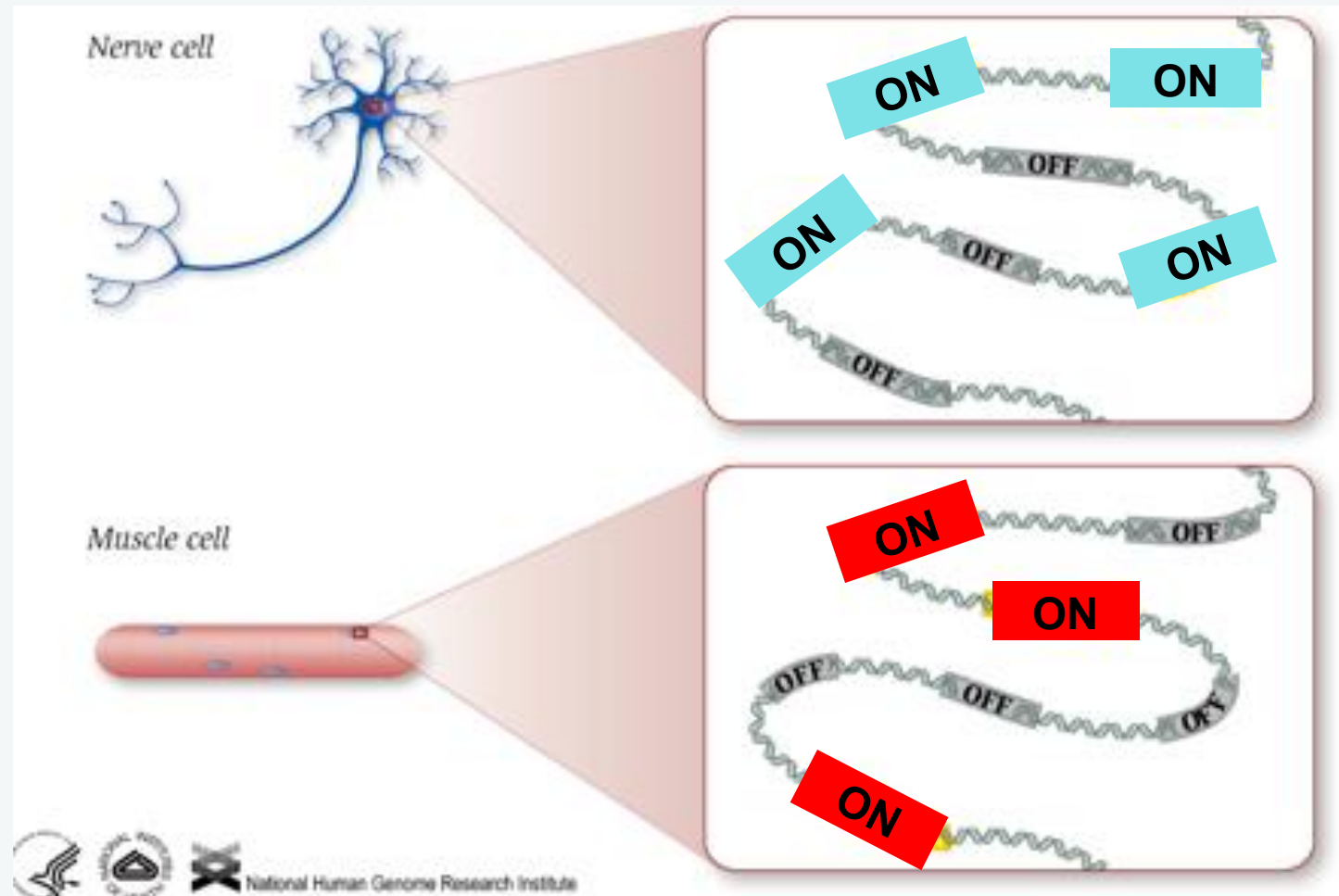


**....and then use it to
measure gene expression
in muscle cells**



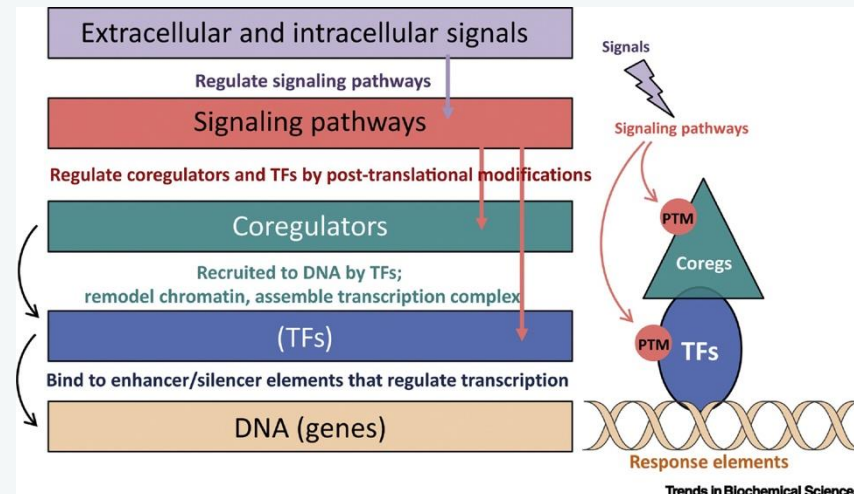
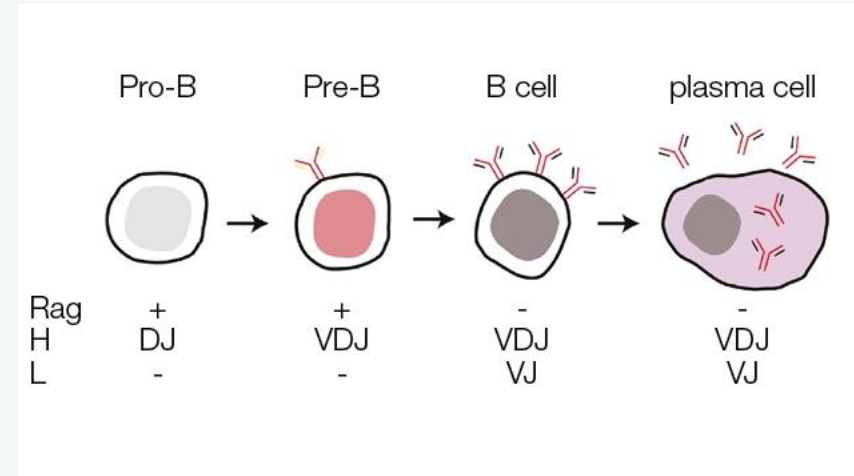
**Then we can compare the two cell types
bioinformatically to find what is different!**

Ultimately, what we would find is a different set of genes are turned ON for each cell type, resulting in different RNAs and proteins produced.

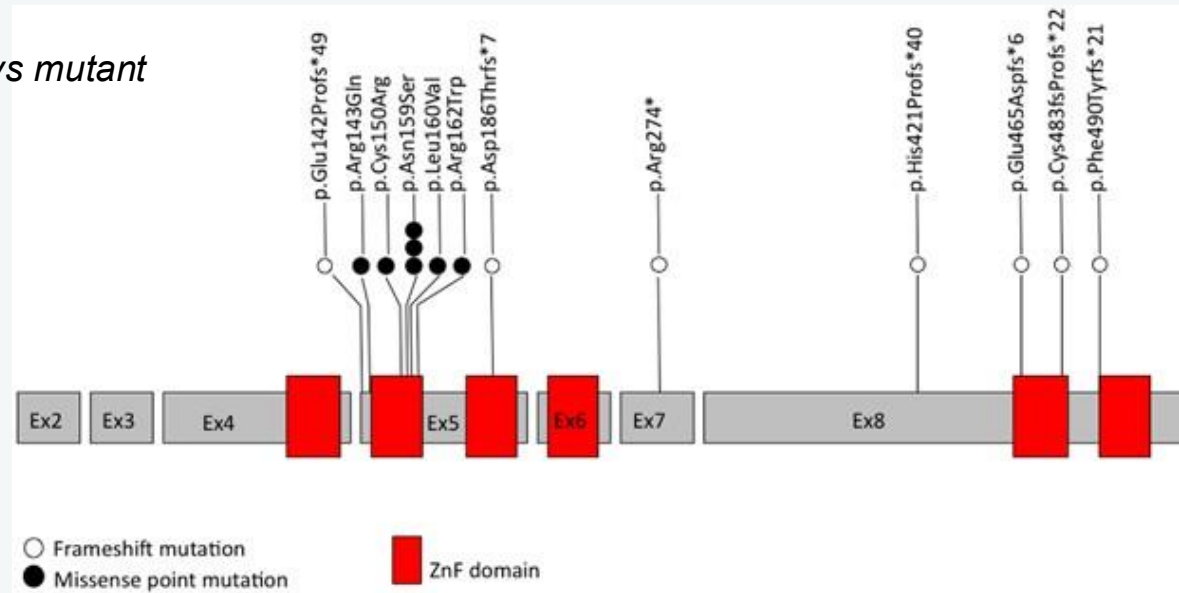


What affects which genes are being turned on/off ... and by how much?

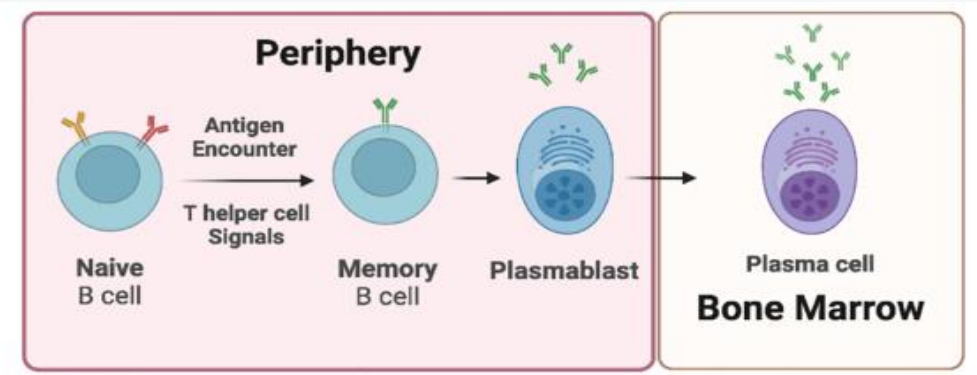
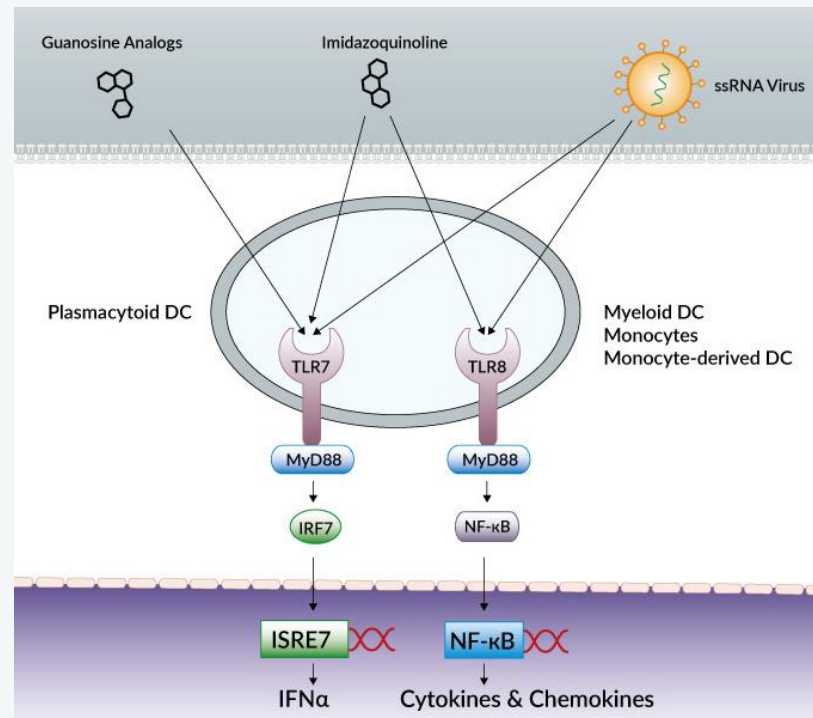
- Different biological processes (ex. apoptosis development)
- Presence of mutations
- Extrinsic factors (ex. TF, coregulators)



Control vs mutant



*Cells treated with guanosine
vs ssRNA virus*



Naïve B cell vs plasma cell

Transcriptomics is powerful; led to the identification of new subtypes in ALL



Gu et al., 2019

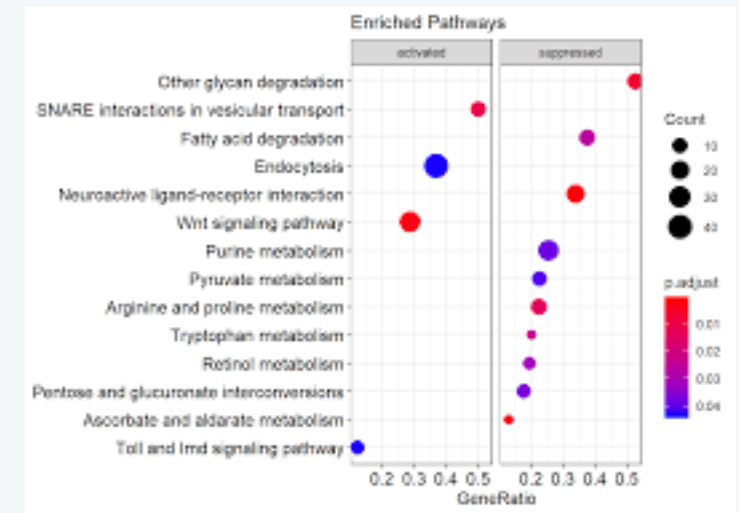
Bulk RNA-Seq: When to Use it?

- Measures average gene expression across a tissue or cell population

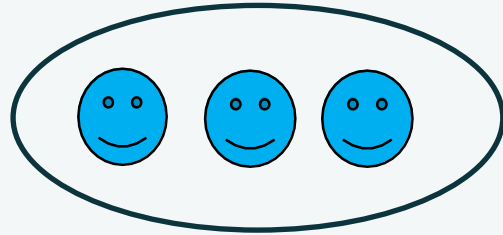
Best for:

- ❖ Comparing disease vs control
- ❖ Treatment response studies

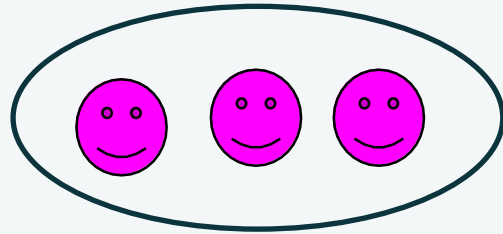
- Pros: cost effective & robust
- Limitations: obscures cell-type composition
- Useful when the research question is about global expression changes or pathways, but not cell-type resolution.



Biological Replicates



Condition 1



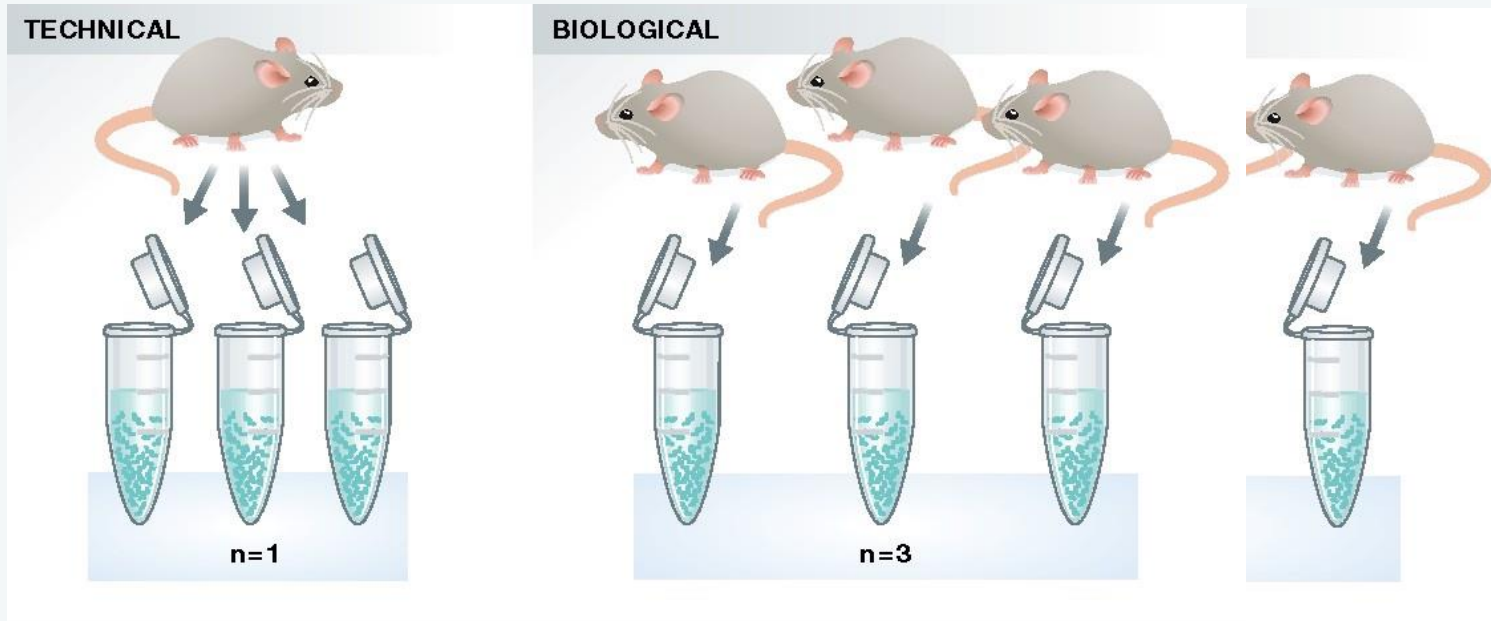
Condition 2

❖ To detect Differentially Expressed Genes (DEGs) between groups we should have several samples, which are also known as replicates

*Differential Gene Expression:
quantitative evaluation and
comparison of transcript levels
across treatments or groups*

Biological Replicates

- We are not talking about technical replicates
- Assessing biological variation requires biological replicates
- Three is the standard minimum
- Yet I would recommend **four** (More is always better!)

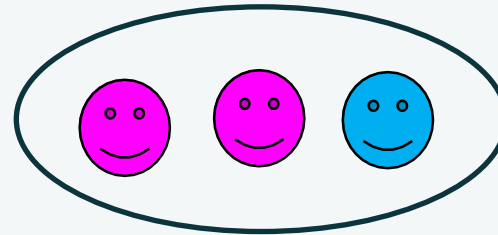
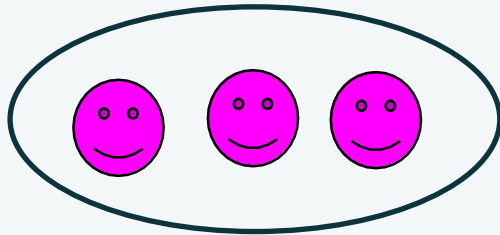
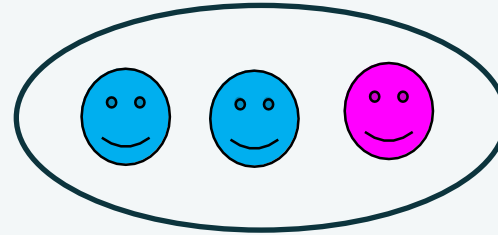
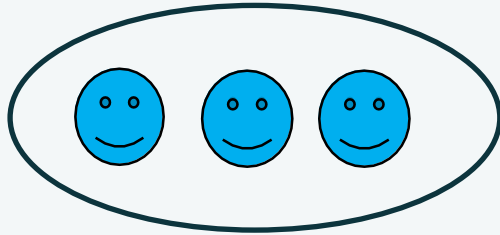


Probability of detecting DEGs

Replicates per group			
	3	5	10
Fold change			
2	87%	98%	100%



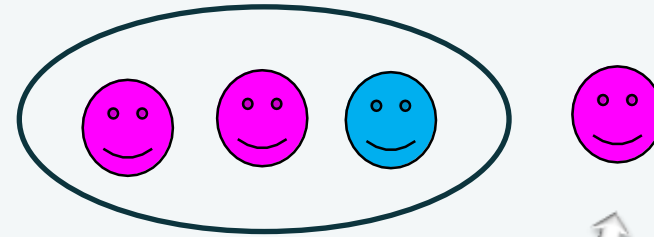
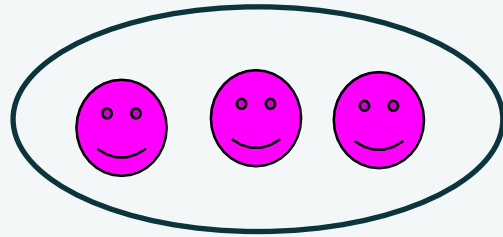
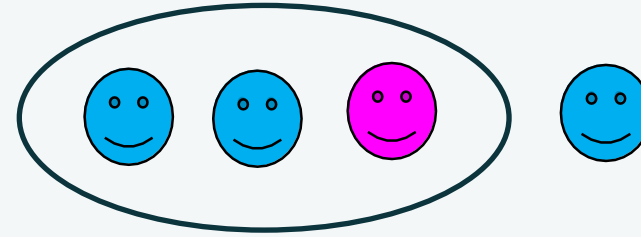
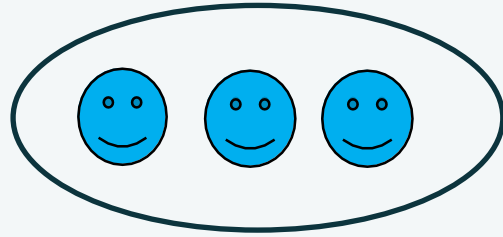
Grouping of Replicates



What you want

What you get

Grouping of Replicates



What you want

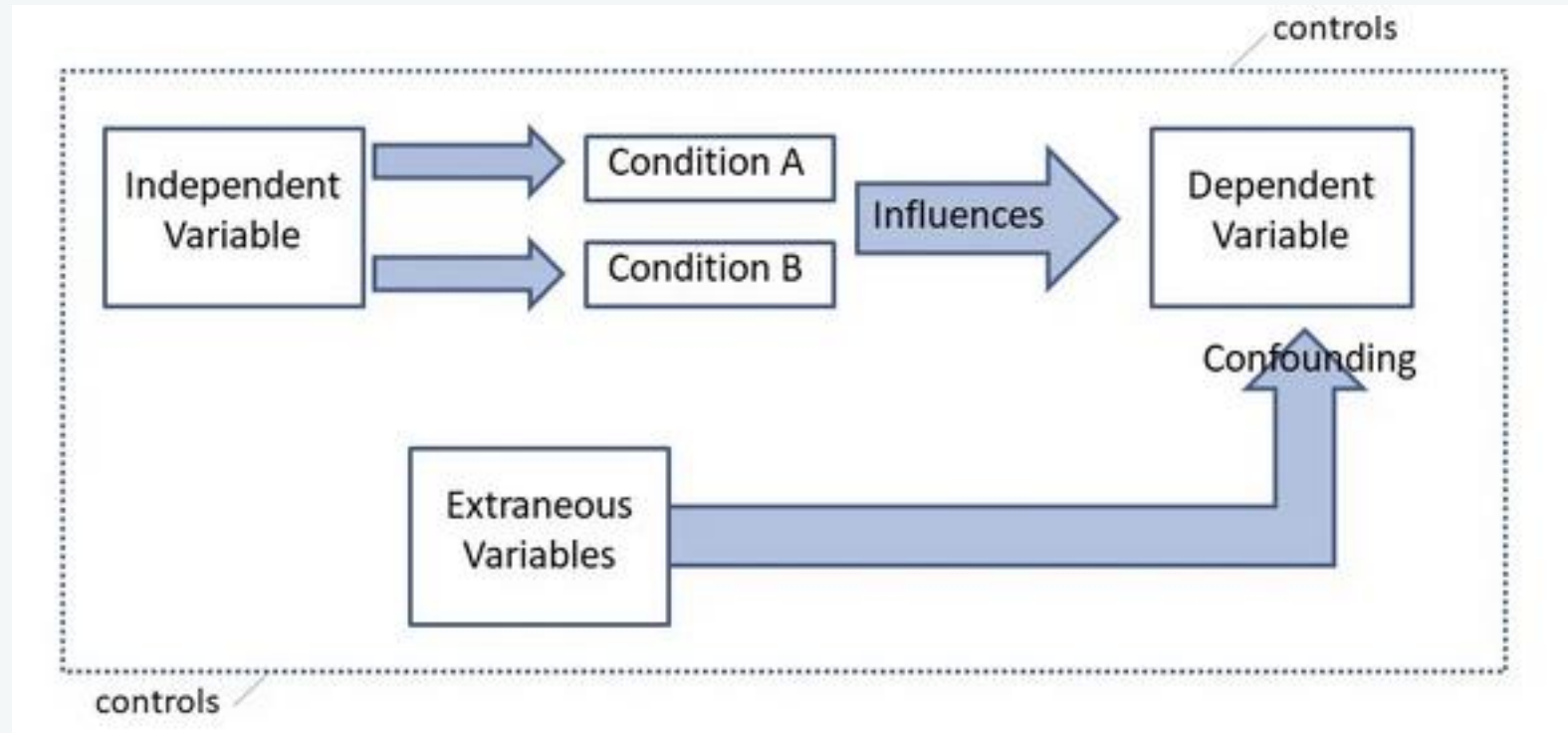
What you get

That spare comes in handy
Highly recommend especially
with mice!

What causes this?

Confounding variables

- A variable that influences or *confounds* the relationship between an independent and dependent variable

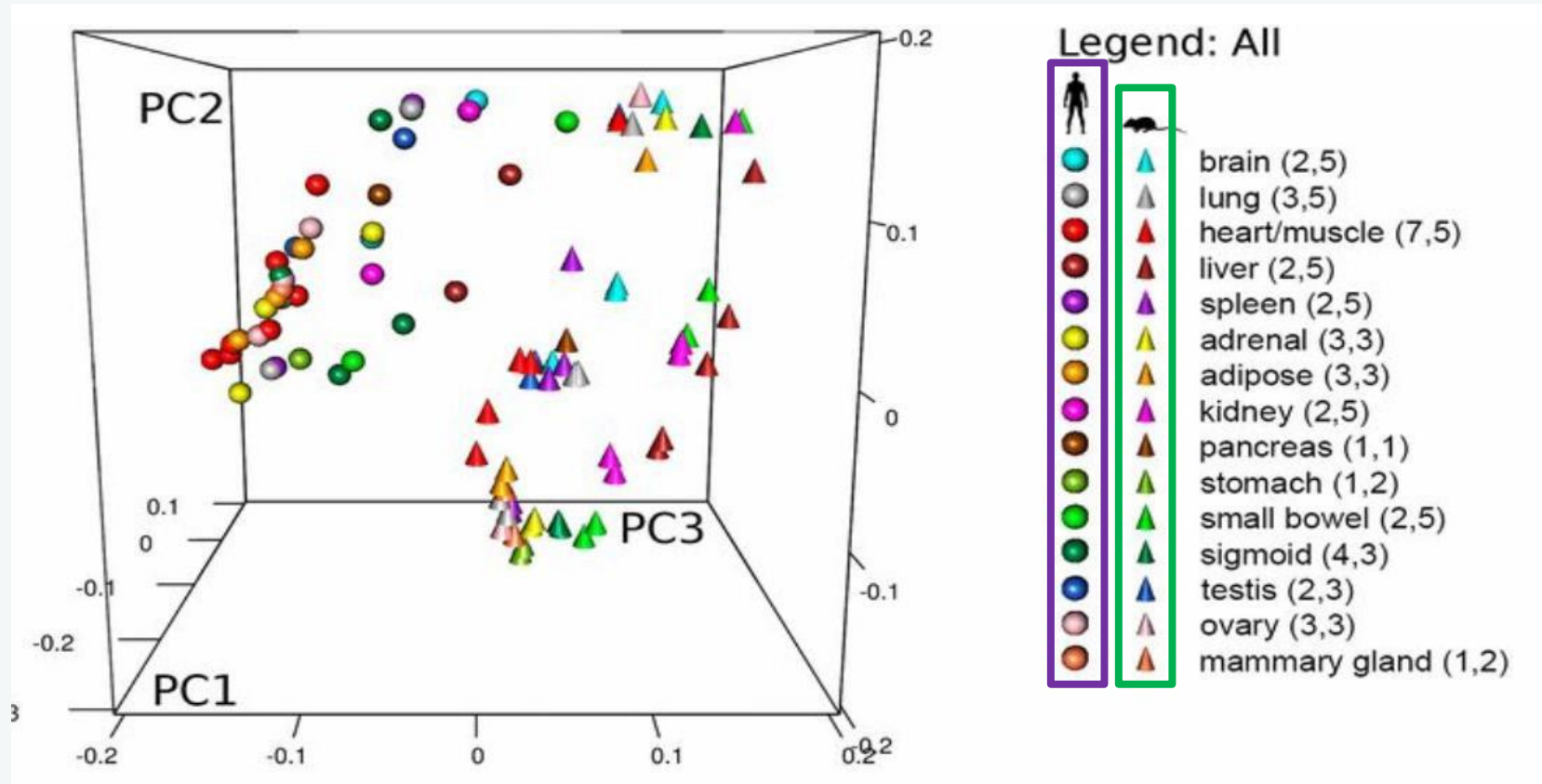


Examples of common confounding variables

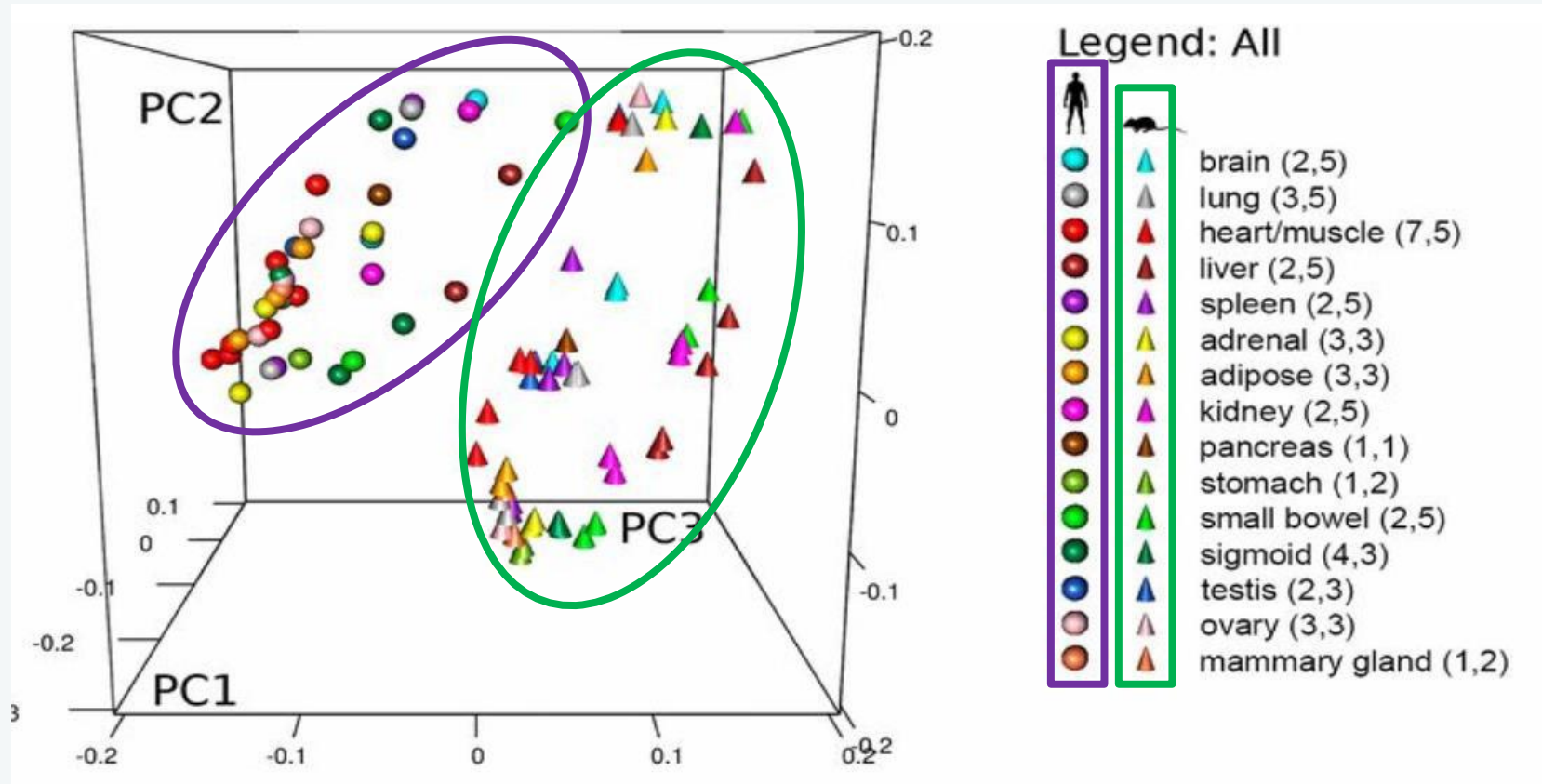
- New technician is processing your samples and/or running the sequencer
- Extracting RNA with different kits
- Inappropriate multiplexing strategy
- Sequencing on multiple different types of platforms

A well-planned experiment with an additional sample, does end up saving you time and money down the road

ENCODE reported that gene expression was likely to follow a species-specific rather tissue-specific pattern



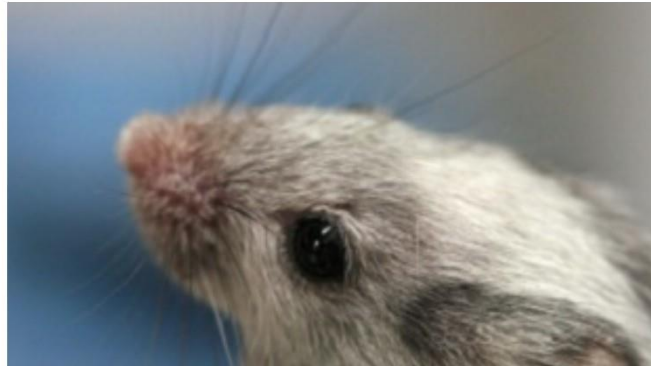
ENCODE reported that gene expression was likely to follow a species-specific rather tissue-specific pattern



Reanalysis of Mouse ENCODE data suggests mouse and human genes are expressed in tissue-specific, rather than species-specific, patterns.

May 19, 2015

JYOTI MADHUSOODANAN



WIKIMEDIA, RAMA

Late last year, members of the Mouse ENCODE consortium [reported](#) in *PNAS* that, across a wide range of tissues, gene expression was more likely to follow a [species-specific](#) rather than tissue-specific pattern. For example, genes in the mouse heart were expressed in a pattern more similar to that of other mouse tissues, such as the brain or liver, than the human heart.

But earlier this month, [Yoav Gilad](#) of the University of Chicago called these results into question [on Twitter](#). With a dozen or so 140-character dispatches (including three heat maps), Gilad suggested the results published in *PNAS* were an anomaly—a result of how the tissue samples were sequenced in different batches. If this “batch effect” was eliminated, he proposed, mouse and human tissues clustered in a tissue-specific manner, confirming previous results rather than supporting the conclusions reported by the Mouse ENCODE team.

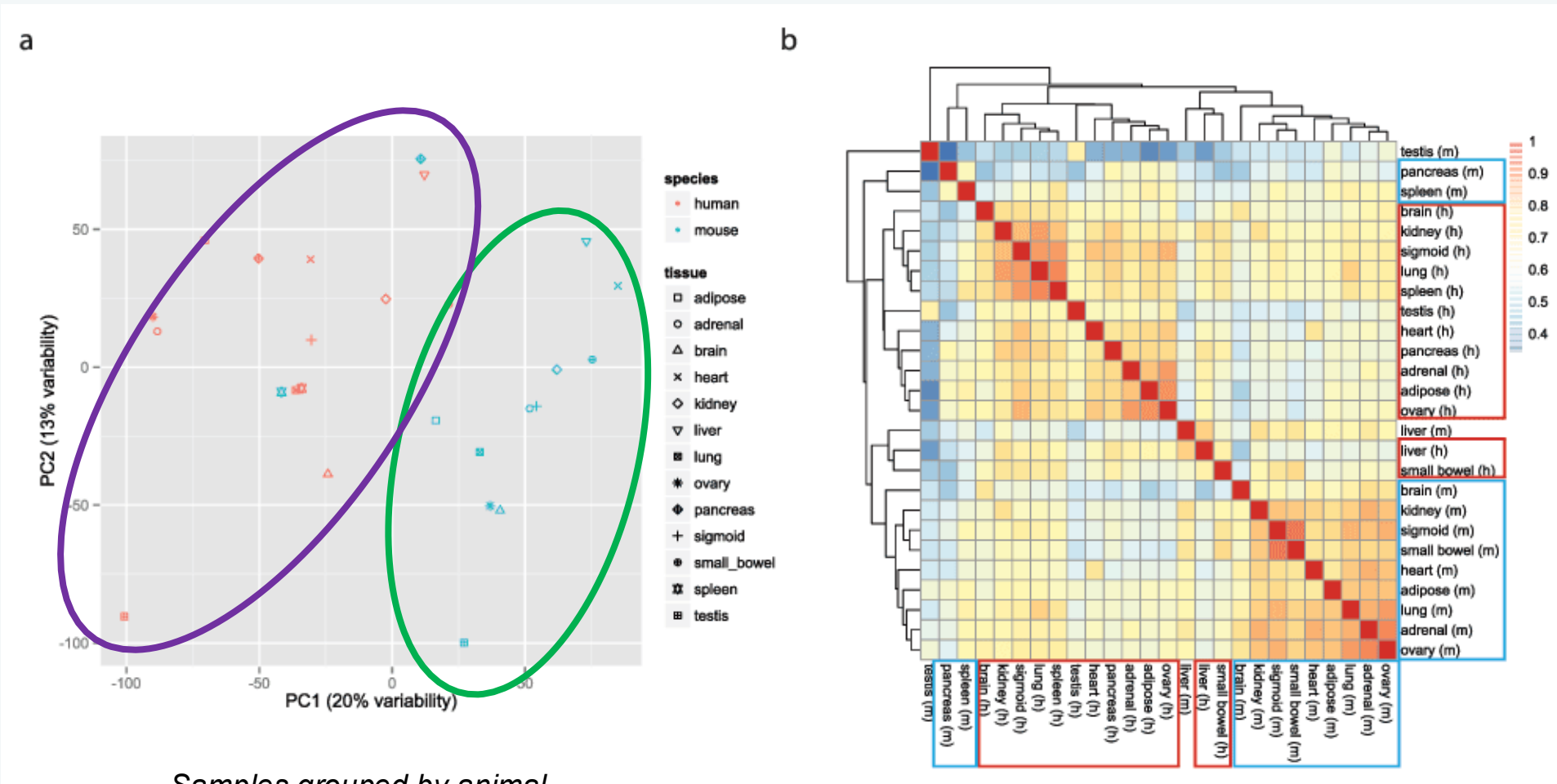
Sequence study design (sequencer ID, run ID, lane number):

D87PMJN1 (run 253, lane 7)	D87PMJN1 (run 253, lane 8)	D4LHBFN1 (run 276, lane 4)	MONK (run 312, lane 6)	HWI- ST373 (run 375, lane 7)
heart	adipose	adipose	heart	brain
kidney	adrenal	adrenal	kidney	pancreas
liver	sigmoid colon	sigmoid colon	liver	brain
small bowel	lung	lung	small bowel	spleen
spleen	ovary	ovary	testis	● human
testis		pancreas		● mouse

Sequencing lane (a batch effect) was almost completely confounded with species in the PNAS study. From

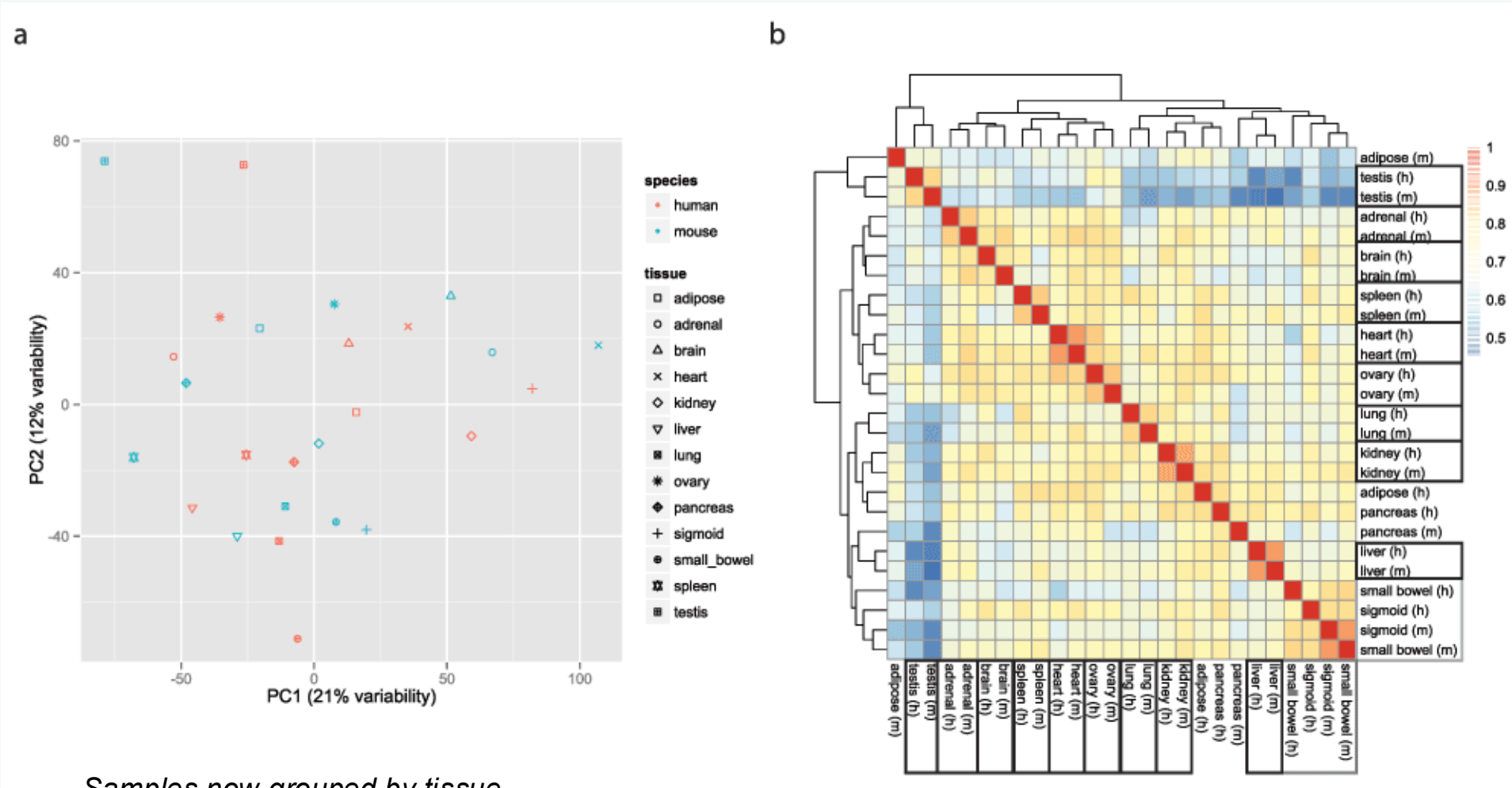
@Y_Gilad

Before accounting for batch effect



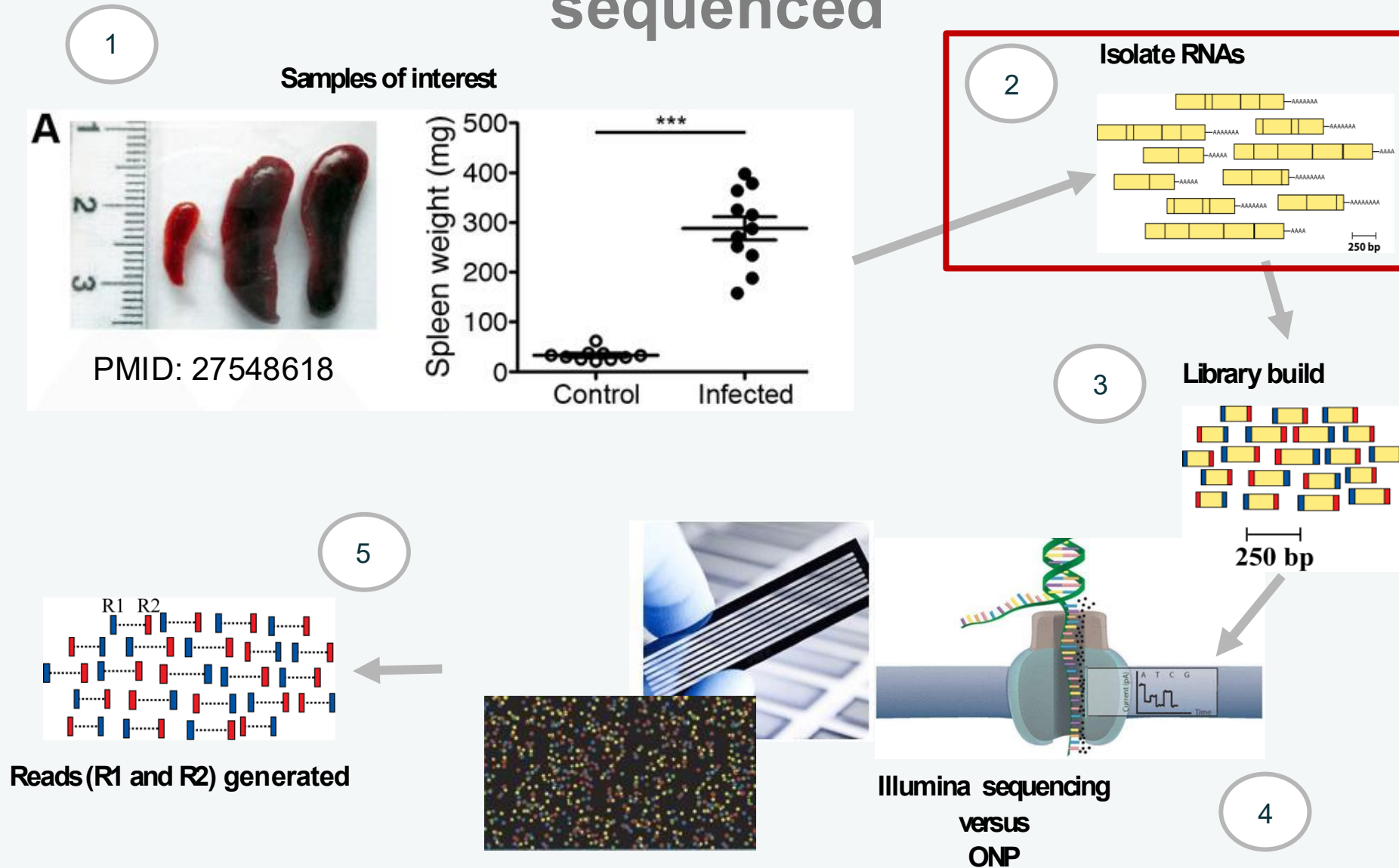
Samples grouped by animal

After accounting for batch effect



Samples now grouped by tissue

Experimental workflow before it gets sequenced



Working with RNA

- RNA-Seq is dependent on the isolation of pure RNA
- RNA is more labile than DNA so extra precautions should be taken

Isolation of RNA



- ✓ 24/7 automatic processing of online orders
- ✓ Knowledgeable and professional Product & Technical Support
- ✓ Fast and reliable (re)-ordering

RNeasy Plus Mini Kit (50)

Cat. No. / ID: 74134

For 50 minipreps: RNeasy Mini Spin Columns, gDNA Eliminator Spin Columns, Collection Tubes, RNase-Free Water and Buffers

\$435.00 ⓘ [Log in](#) to see your account pricing.

1. Column type / Plate type

Micro

Mini

96 well

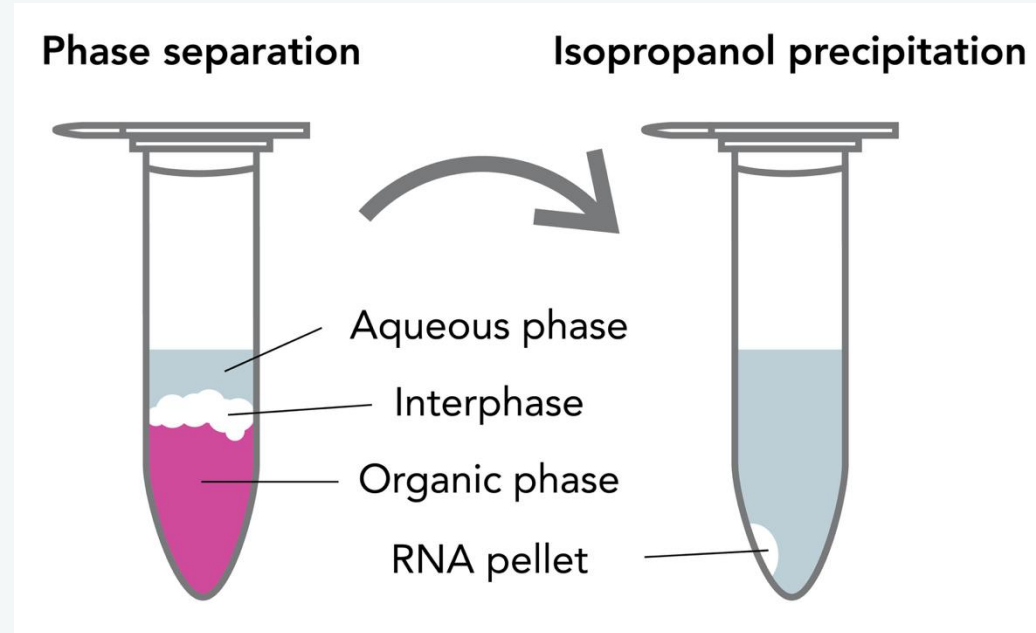
2. Preparations

50

250

- This kit comes everything you need for total RNA isolation including the gDNA eliminator spin columns

The cheaper way to isolate RNA



- The UVM VIGR core **does not** recommend as residual phenol will inhibit the library build and you cannot get rid of it. They will not sequence your samples if you use this mode of isolation!
- Pro-tip: maintain separate reagents and consumables.

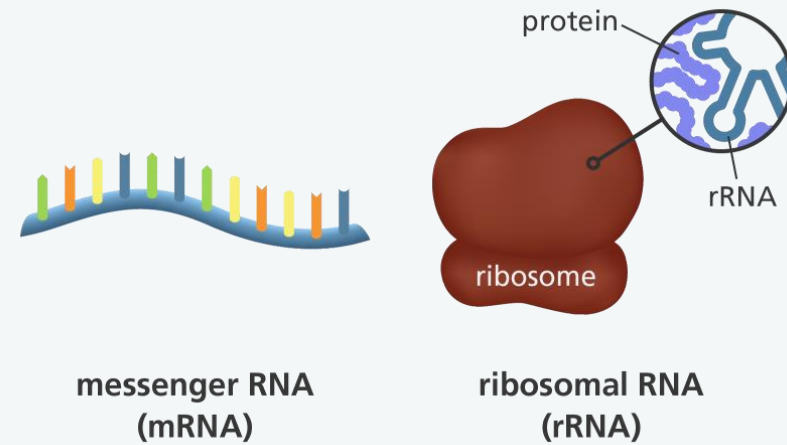
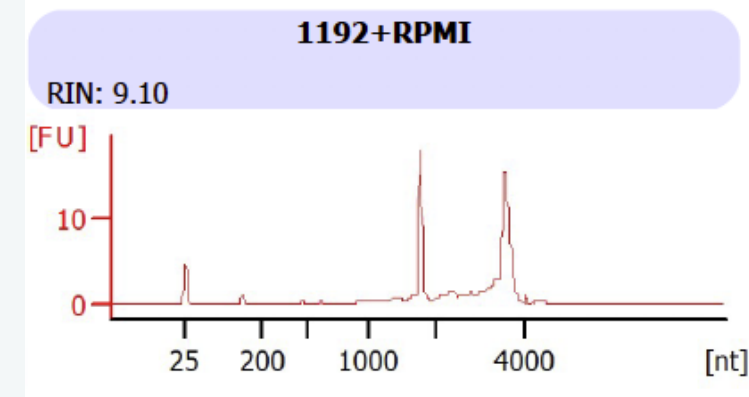
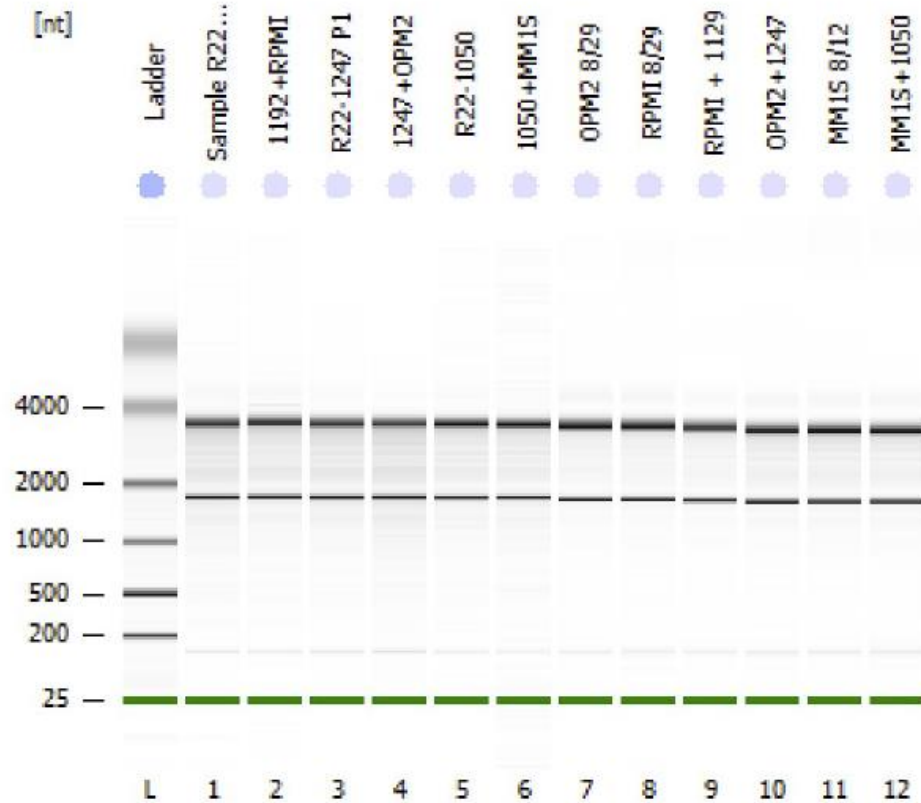
Other recommendations

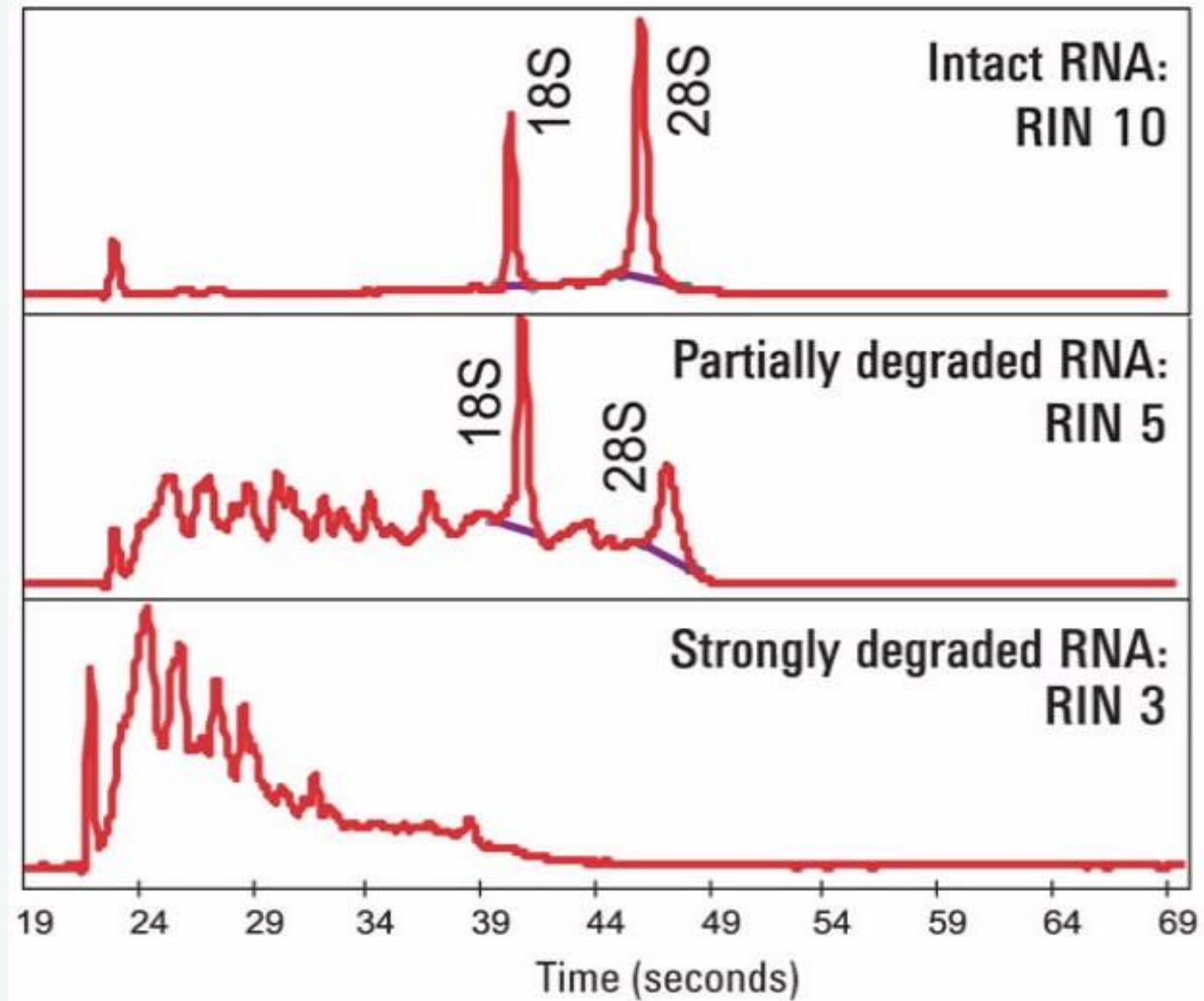
- Always keep your RNA samples on ice
- Spray bench, pipettes, centrifuge, and tip boxes with RNase Zap
- Maintain RNA integrity with **Ribolock** (ThermoFisher: EO0381)
- Once isolated, RNA is stored in the -80C



Check purity with RNA bioanalyzer

Electrophoresis File Run Summary



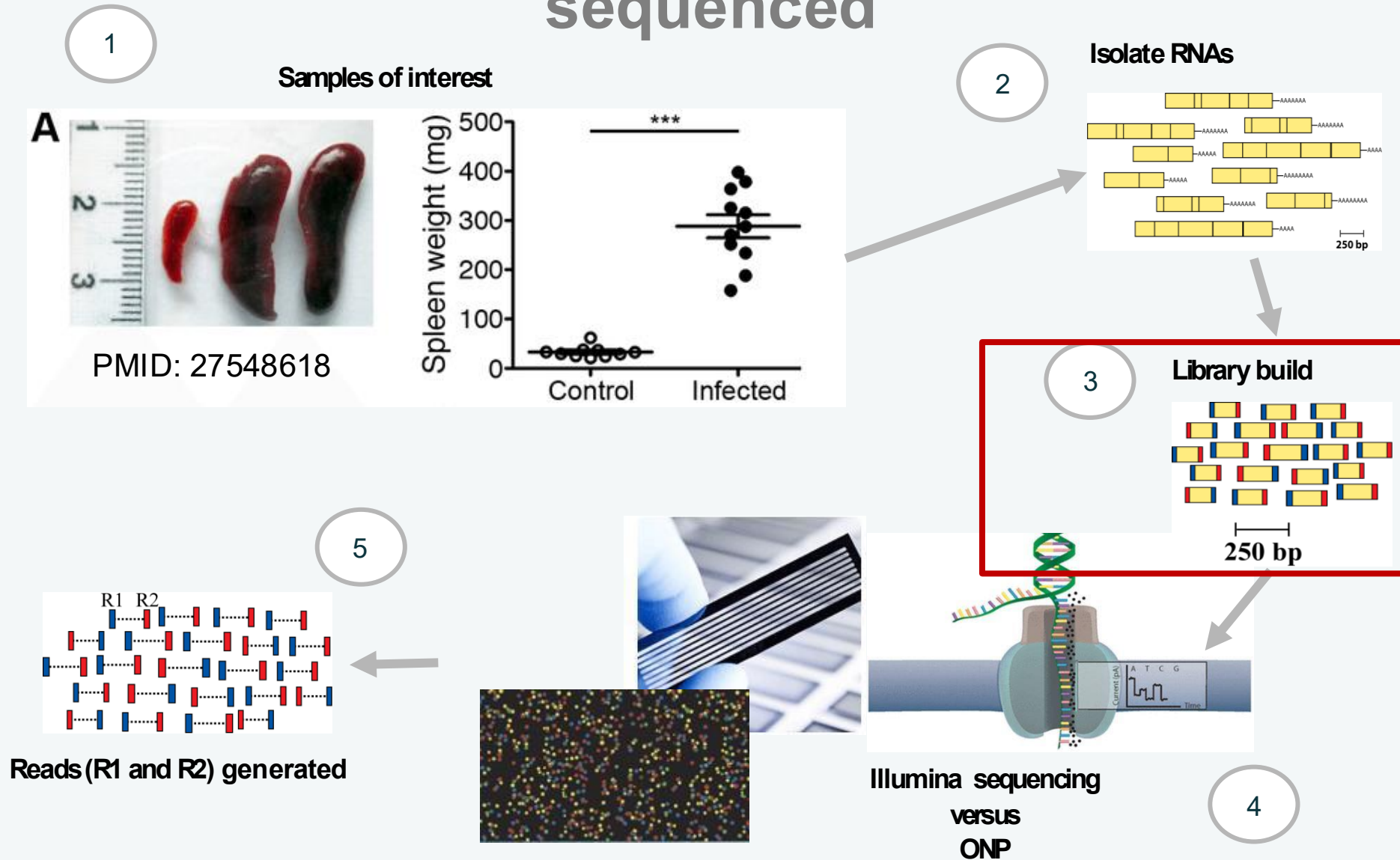


RNA Bioanalyzer submission at UVM

- Need to take a Qubit reading first not nanodrop
- Then determine whether a *pico* or *nano* chip is required based on this
- Submit 2 ul of sample to core in 0.6mL tube
- Fill out form on iLabs



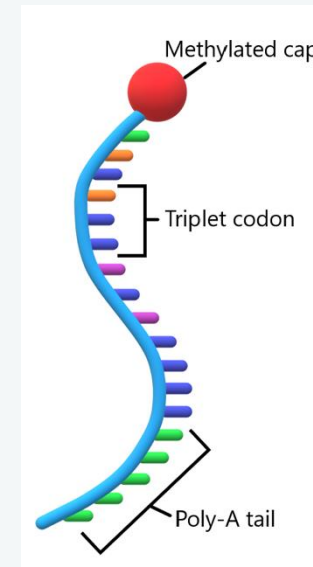
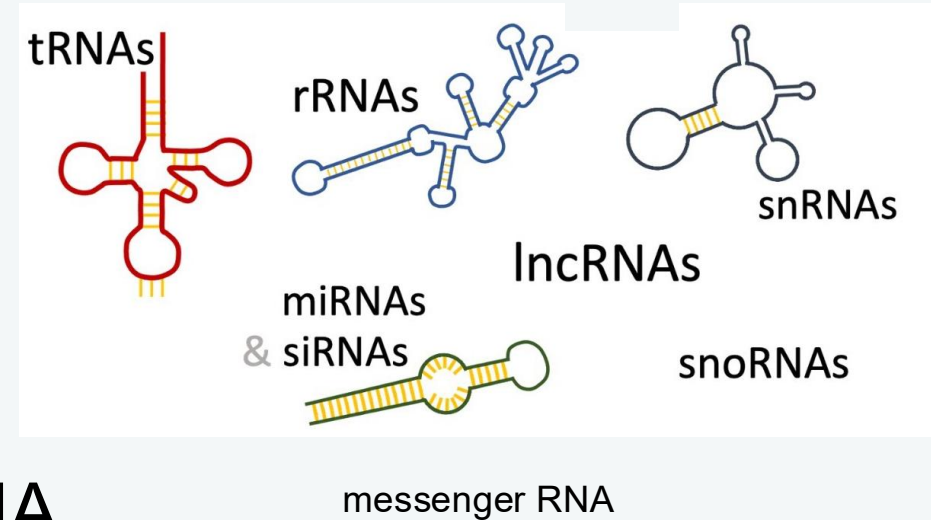
Experimental workflow before it gets sequenced



RNA composition

RNA comes in many different flavors

- Ribosomal-related RNAs:
 - rRNA, tRNA, snoRNA (up to 90% of RNAs)
- Protein-coding RNAs:
 - mRNA
- Regulatory RNAs:
 - microRNAs, lncRNAs

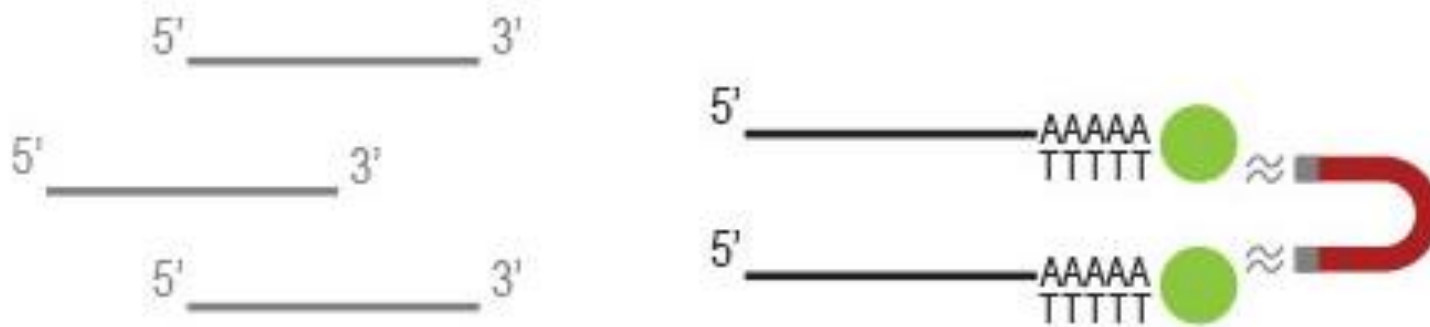


Before building the libraries need to perform target enrichment

- If rRNA is not removed, the majority of the final sequencing reads would be from rRNA and not mRNA
- Therefore, it necessary to enrich for mRNA
- Two common strategies:
 1. Poly A+ selection = mRNA only
 2. rRNA depletion = mRNA + *other species*

Poly-A versus rRNA depletion?

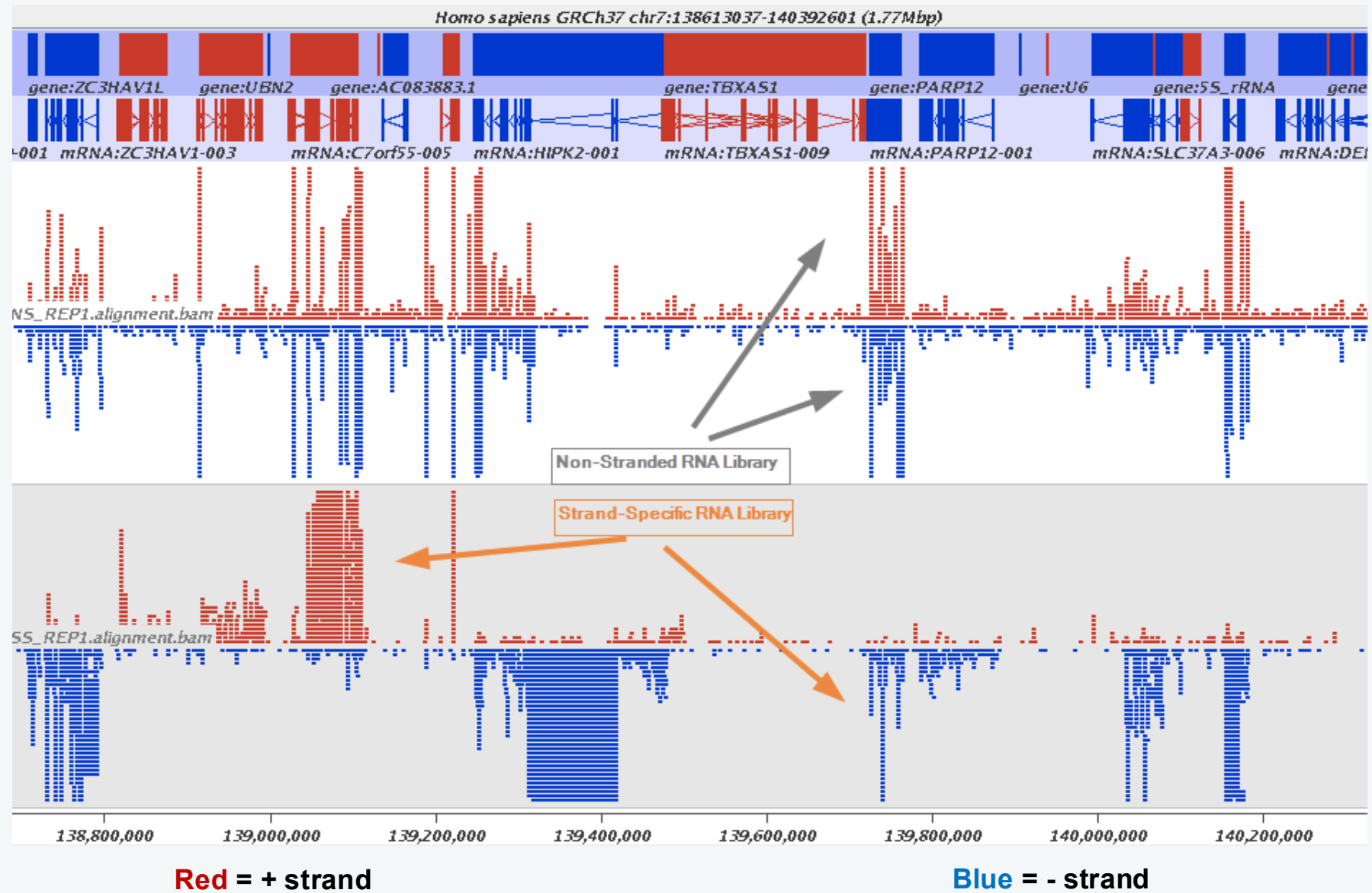
- For differential gene expression, most enrich using Poly(A)+
- However:
 - If you are aiming to obtain information about long non-coding RNA's perform ribosomal RNA depletion
 - Bacterial mRNAs are also not poly-adenylated



Strandedness

- Another consideration is whether to generate strand-preserving libraries
 - ❖ Libraries can be stranded or unstranded
- The implication of **stranded** libraries is that you could distinguish whether the reads are derived from forward or reverse-encoded transcripts

Strandedness



Motivation - Single cell level insights

Bulk RNA-seq

Taste the **average** of all fruits

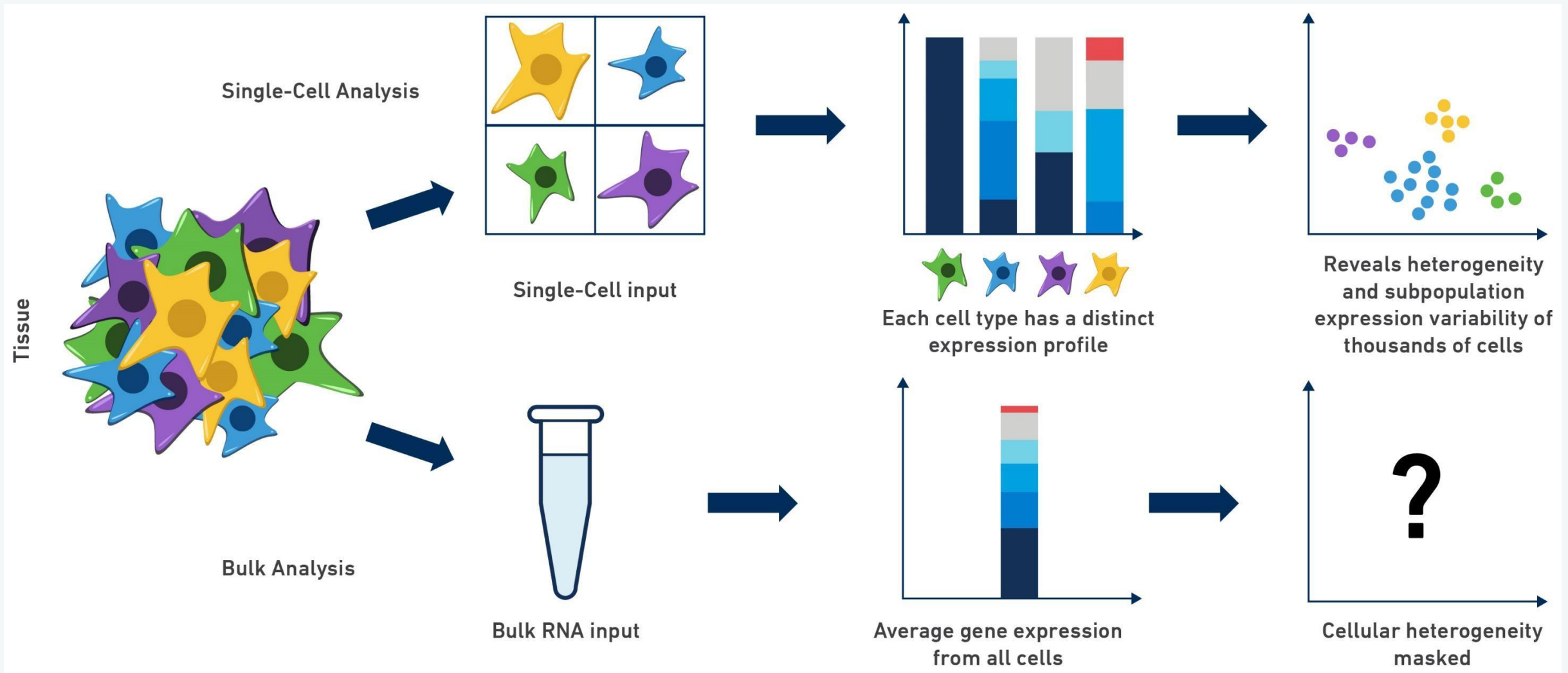


Single cell RNA-seq

Taste each fruit **individually**



Bulk RNA- vs. single cell RNA-seq



Single-Cell RNA-Seq

- Measures expression **cell-by-cell**
- Best for:
 - ❖ Identifying novel cell types/states
 - ❖ Uncovering tumor heterogeneity
 - ❖ Immune profiling
 - ❖ Understanding treatment-resistant populations
- Pros: High-resolution
- Limitations: Expensive, technological challenges as the data is noisy
- Select scRNA-seq when you suspect heterogeneity matters



Single cell level SPATIAL insights

Bulk RNA-seq



Taste the
average of
all fruits

Single cell RNA-seq

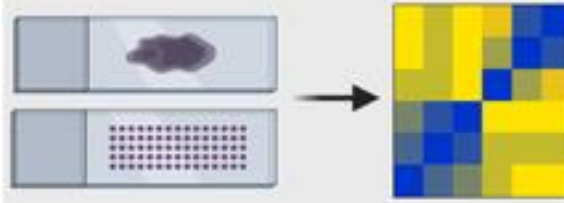


Taste each fruit **individually**

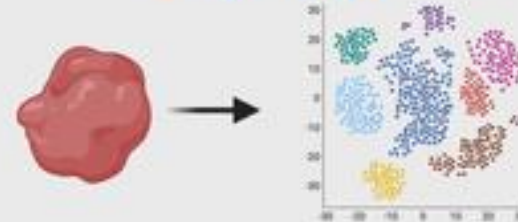
Spatial transcriptomics



Bulk RNA sequencing



Single-cell RNA Sequencing



High Throughput Spatial Transcriptomics



Resolution

Patient-level gene expression

Single-cell resolution

Spatial and single-cell resolution

Data Format

Aggregated gene expression from whole tissue

Gene expression at the individual level

Gene expression mapped to 2D tissue coordinates

Biological Insights

Identifies differentially expressed genes at the tissue level

Identifies cell subpopulations and heterogeneity

Reveals spatially variable genes, tissue structure, and cell-cell interactions

Advantages/
Disadvantages

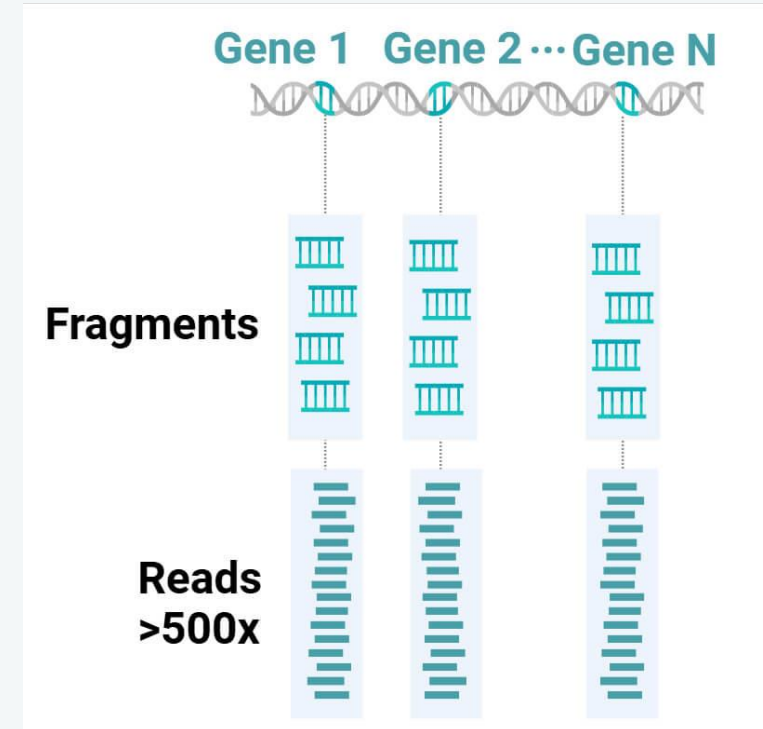
- ✓ High sensitivity for overall gene expression
- ✗ No spatial or single-cell resolution

- ✓ High resolution for individual cell types
- ✗ Loses spatial context and cell-cell interactions

- ✓ Preserves tissue architecture and spatial relationships
- ✗ Lower sensitivity for minimally-expressed genes

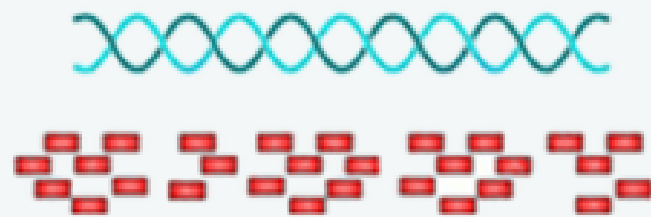
Whole-Exome & Targeted Panel Sequencing

- WES: captures all protein-coding regions
- Panels: focus on clinically relevant genes
- Best for:
 - ❖ Variant identification
 - ❖ Cancer mutation profiling
- Pros: Strong databases available for identification
- Limitations: Hard to capture structural variation
- Panels are ideal for well-defined clinical questions; while WES is used to survey the dataset



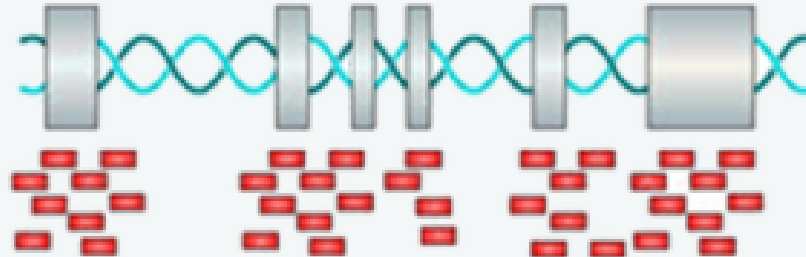
One step above this is WGS...

Whole genome sequencing



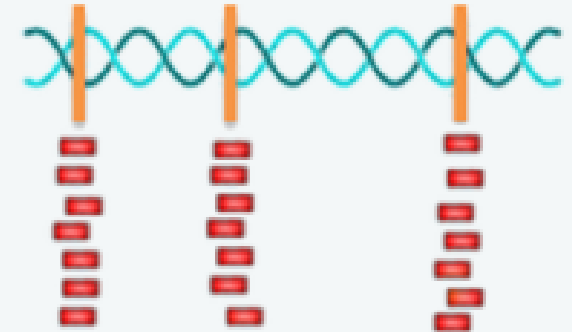
- Sequencing region : whole genome
- Sequencing Depth: >30X
- Covers everything – can identify all kinds of variants including SNPs, INDELs and SV.

Whole exome sequencing



- Sequencing region: whole exome
- Sequencing Depth : >50X ~ 100X
- Identify all kinds of variants including SNPs, INDELs and SV in coding region.
- Cost effective

Targeted sequencing

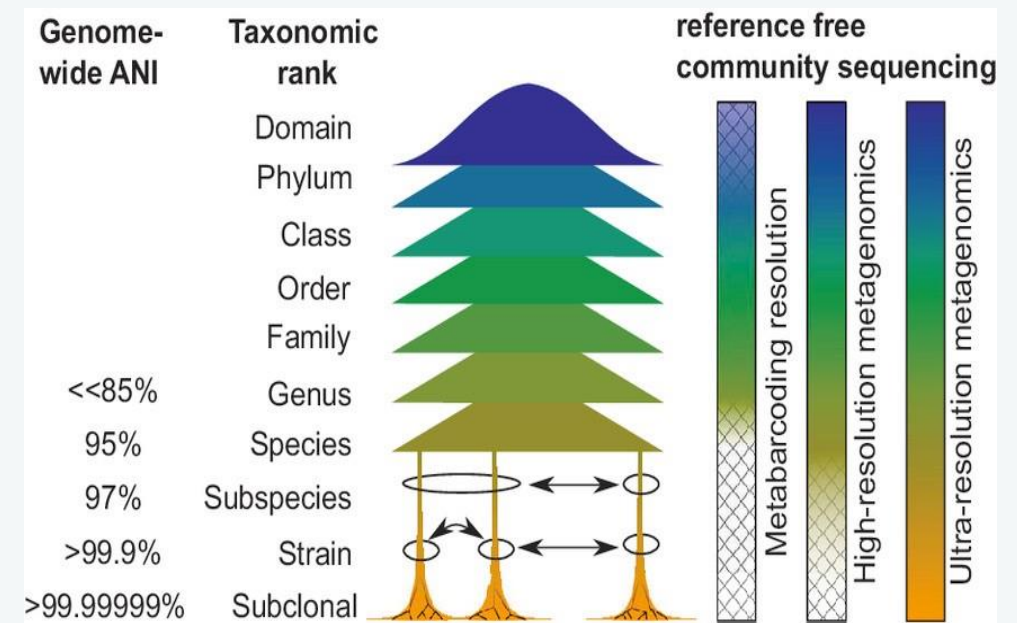


- Sequencing region: specific regions (could be customized)
- Sequencing Depth : >500X
- Identify all kinds of variants including SNPs, INDELs in specific regions
- Most Cost effective



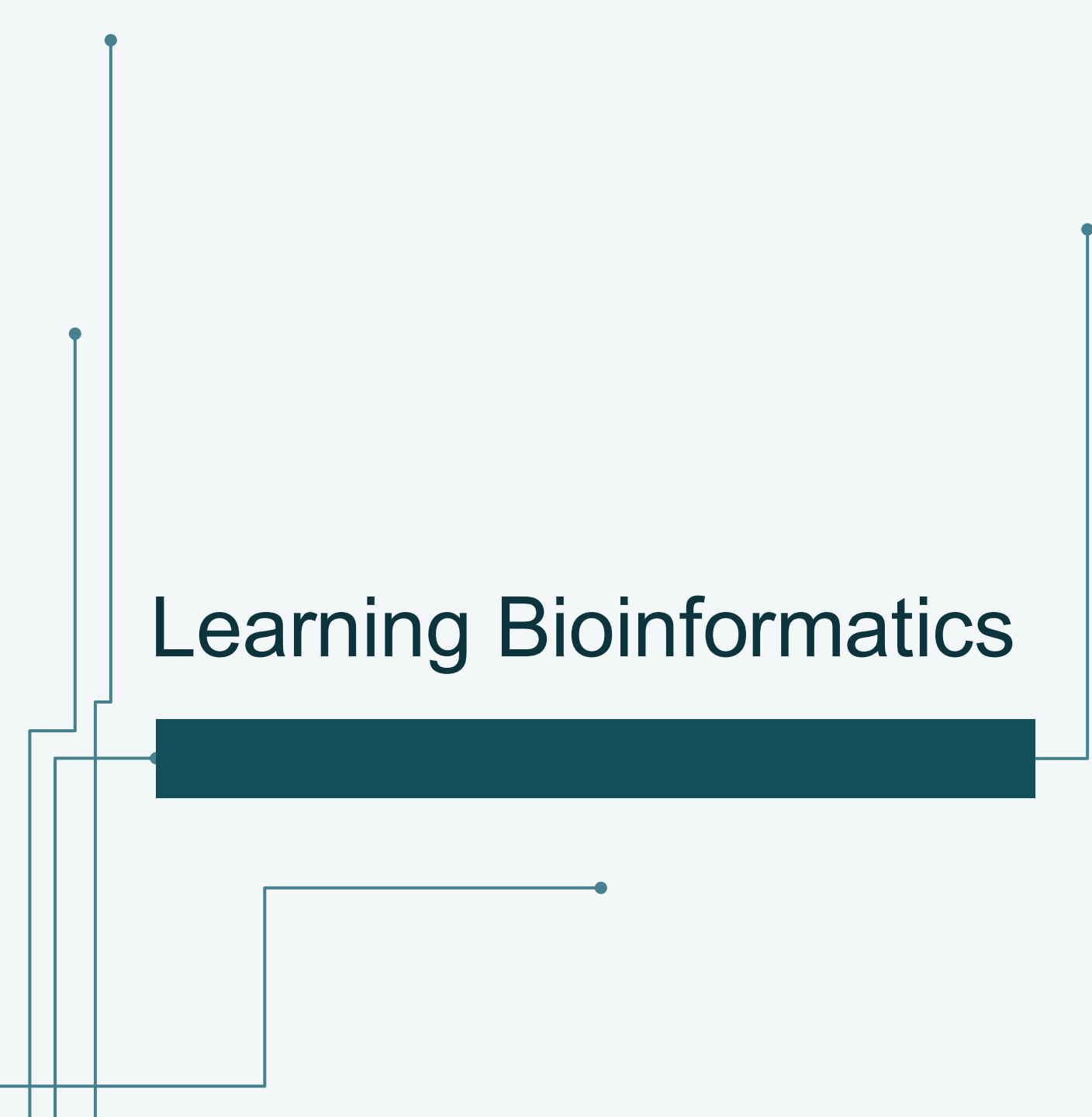
16S & Metagenomic Sequencing

- **A major reason why researchers perform sequencing is to differentiate between multiple bacterial species within a sample.**
The ability to identify novel, rare, or unexpected pathogens at a **species and strain level** is of high priority.
- 16S: bacterial composition at genus level
- Shotgun metagenomics: species level & functional pathways
- 16S is inexpensive; shotgun is comprehensive



Choosing the Right Design

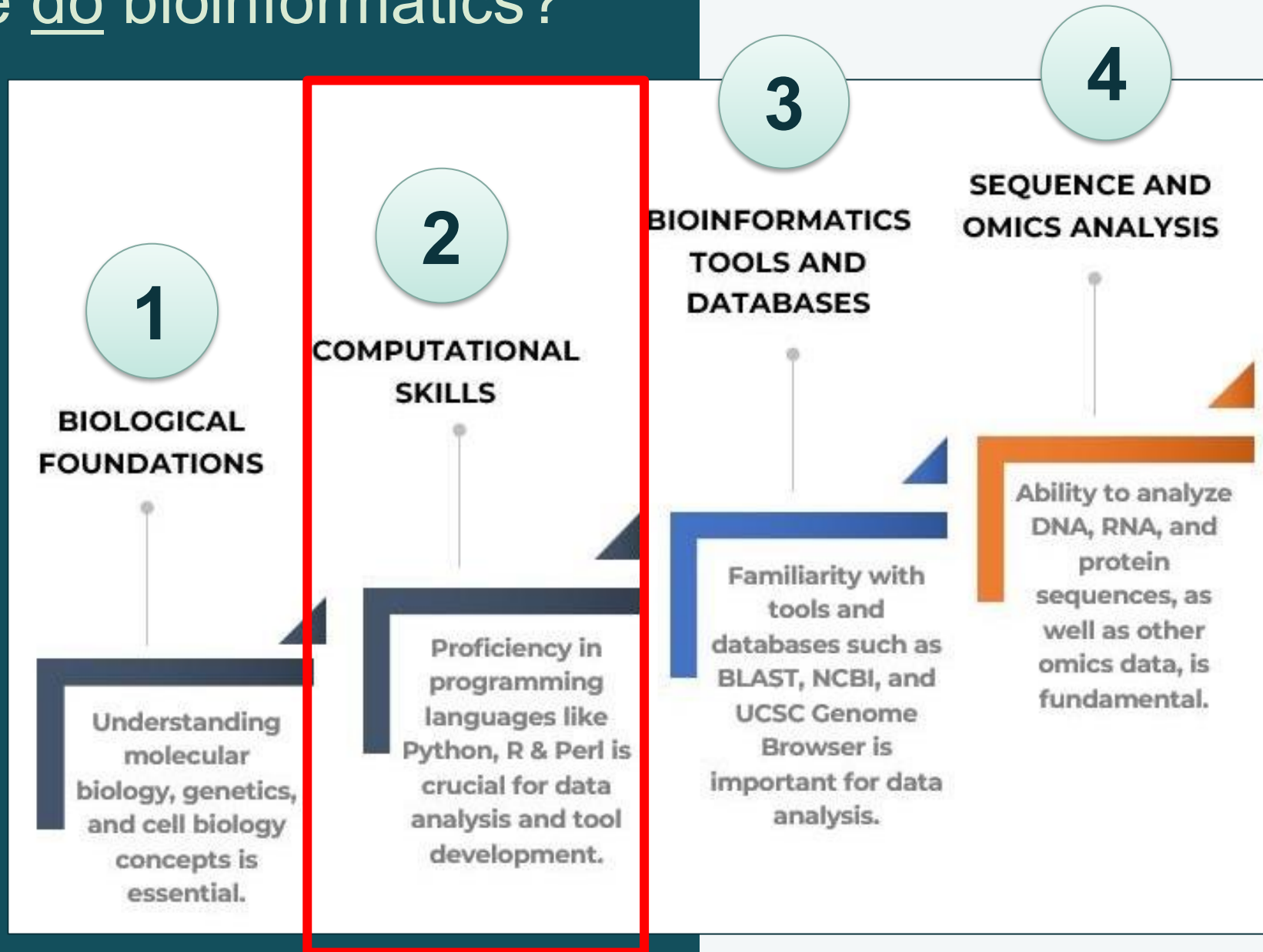
- Need for matching your research question with the correct modality
- “Which pathways change after treatment?” **rnaseq**
- “What cell type drives disease?” **scrnaseq**
- “What mutations define relapse?” **WES**
- “How does the microbiome differ between groups?” **metagenomics**



Learning Bioinformatics

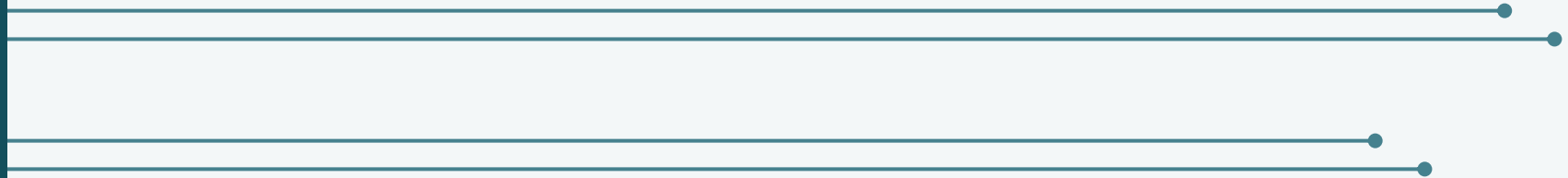


How do we do bioinformatics?

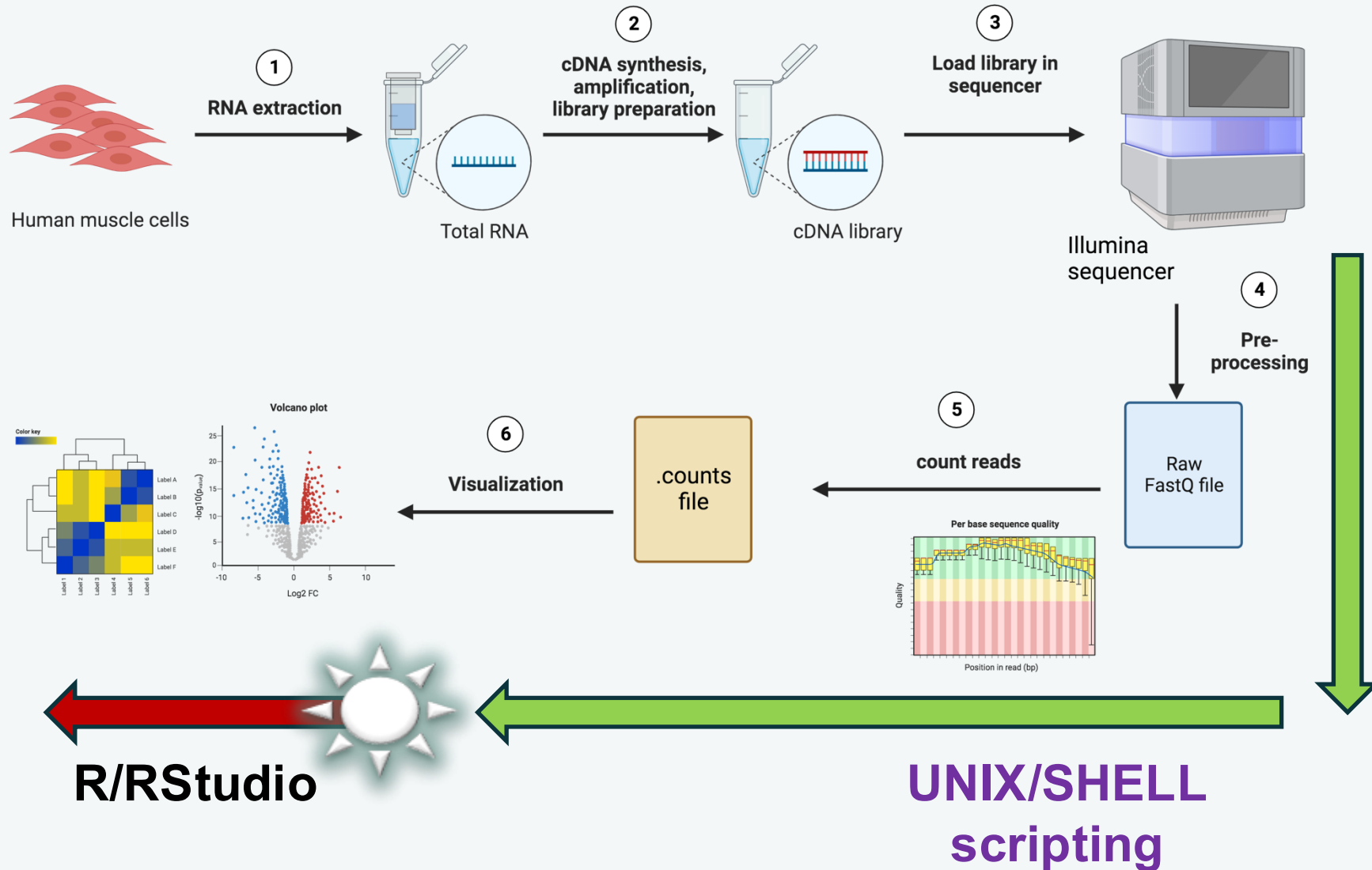


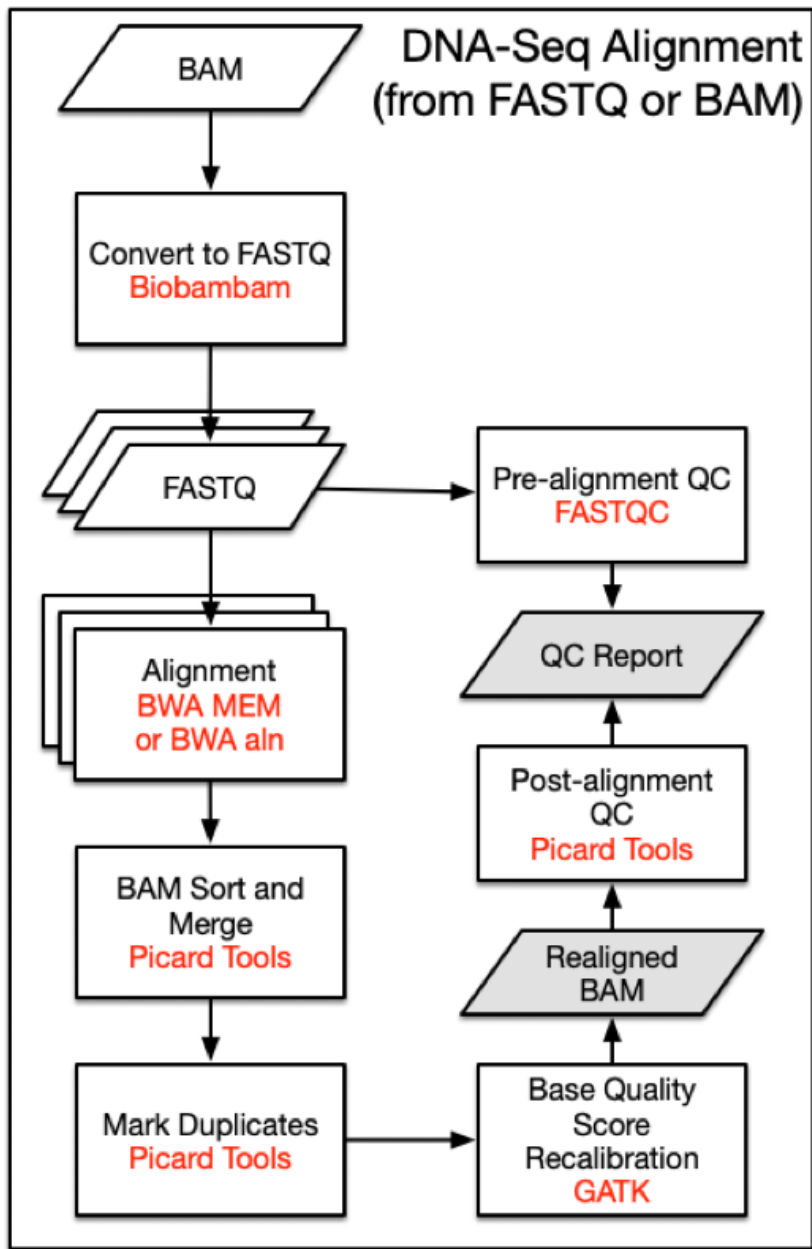
What are programming languages?

- Programming languages are tools
- They make it possible to instruct computers
- In biomedical research, the prevailing languages are R, Python, and command line



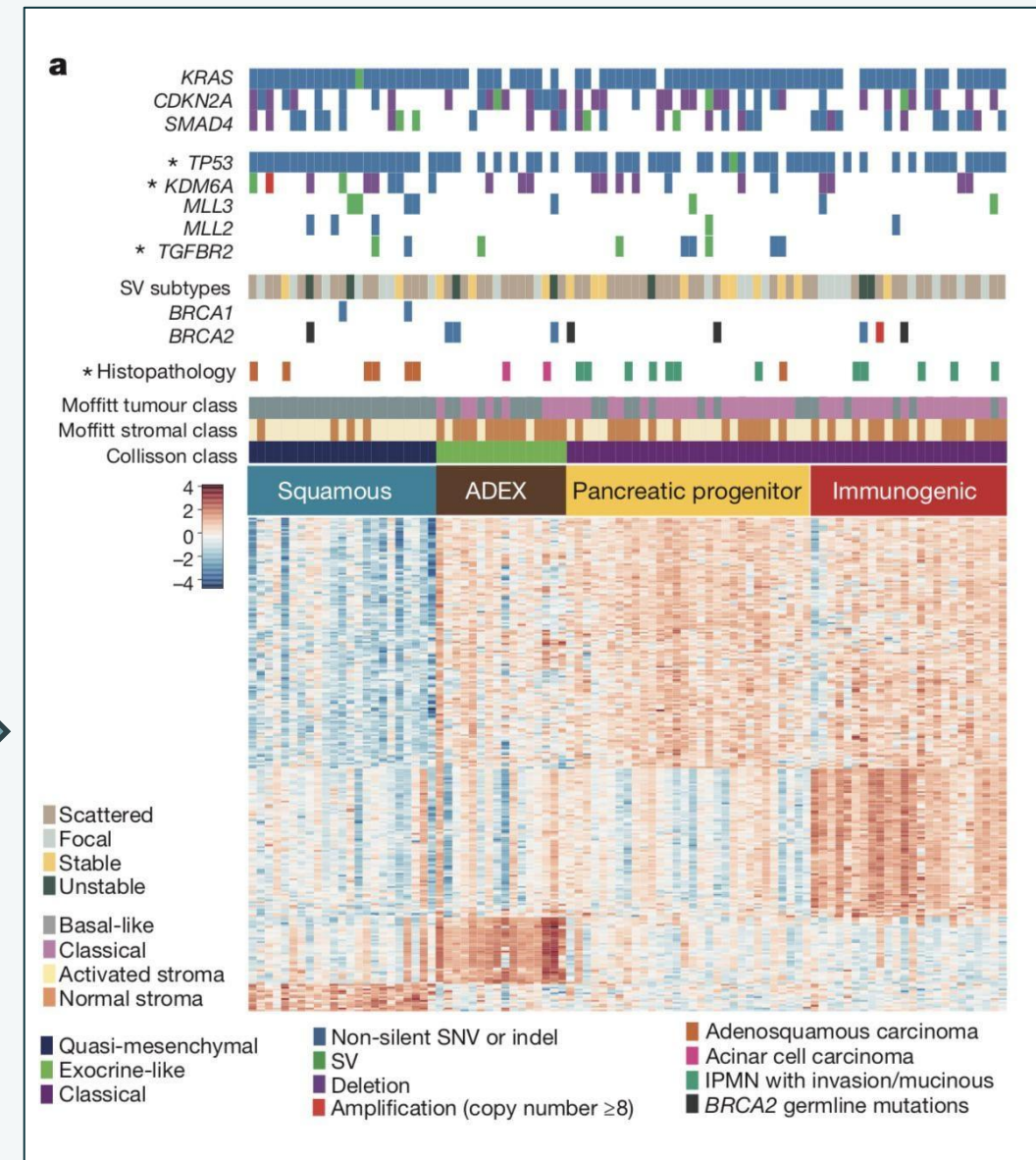
In order to a single RNA-Seq analysis it requires:





Step 1: Command-line
Programming language 1

Two Language Problem



Step 2: R
Programming language 2

Phases of FASTQ file “lifecycle”

Phase 1: Processing

Step 1:
Command-line
*Programming
language 1*

Processing Step	Tools
Quality control of raw sequences	FastQC, cutadapt, trimGalore
Alignment/Mapping to the genome	STAR
Generating gene counts	HT-seq, Salmon

Phase 2: Analysis

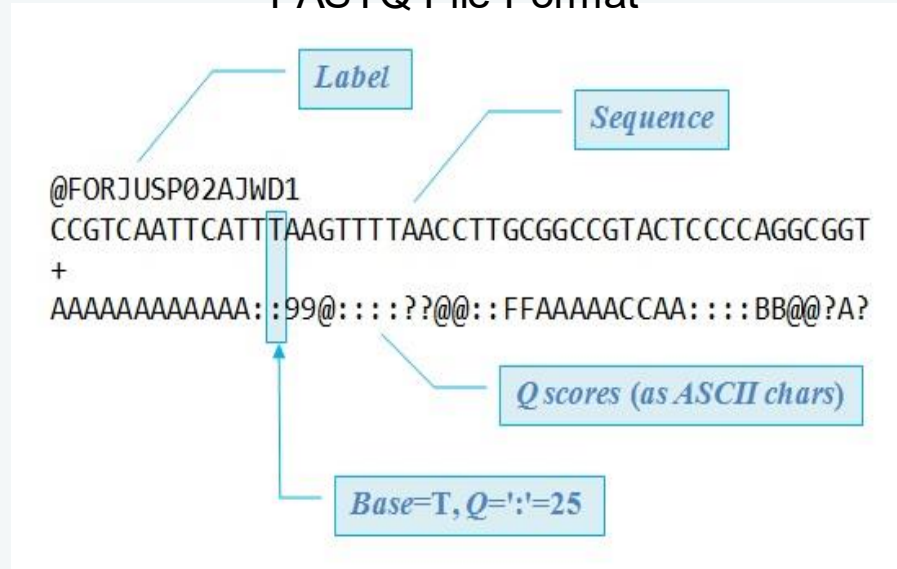
Step 2: R
Programming language 2

Analysis Step	Tools
Quality control of replicates	DESeq2, edgeR
Differential expression analysis	DESeq2, edgeR
Pathway analysis	GSEA, IPA, clusterProfiler



Quality Control of Raw Sequences

FASTQ File Format



Quality Control Steps

1. Generate FASTQC reports.
2. Check the quality of base-calls.
3. Check for “over-represented sequences.”
4. Trim low-quality bases.
5. Remove adapter sequences.

Example Commands:

```
fastqc 2D_WT_shGFP_E2_R50_001_S10_R1_001.fastq.gz
```

```
trim_galore --paired --illumina 2D_WT_shGFP_E2_R50_001_S10_R2_001.fastq.gz
2D_WT_shGFP_E2_R50_001_S10_R2_001.fastq.gz
```

Alignment/Mapping to the Genome

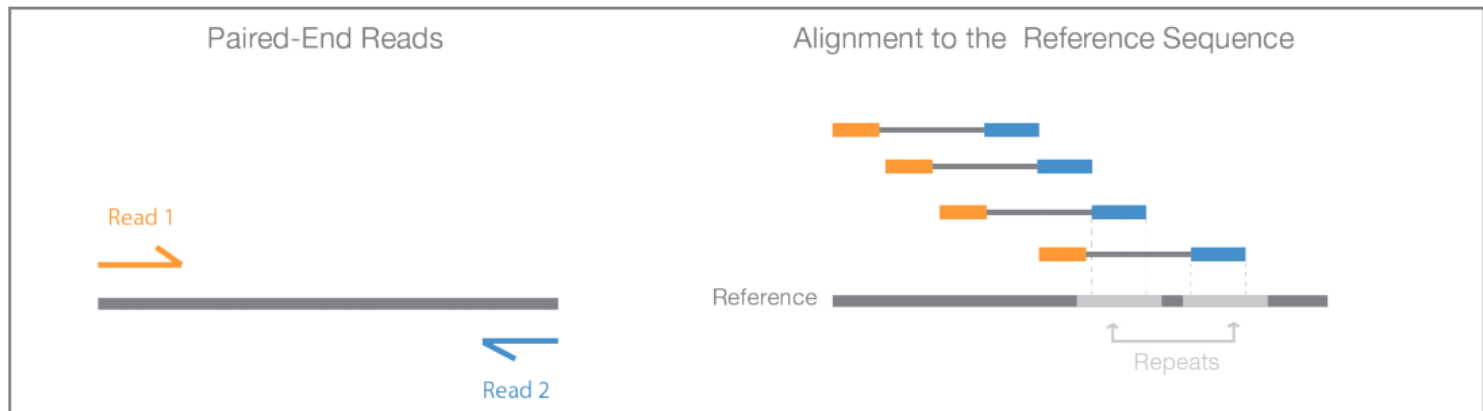


Figure 4: Paired-End Sequencing and Alignment—Paired-end sequencing enables both ends of the DNA fragment to be sequenced. Because the distance between each paired read is known, alignment algorithms can use this information to map the reads over repetitive regions more precisely. This results in better alignment of reads, especially across difficult-to-sequence, repetitive regions of the genome.

Alignment Steps

1. Align FASTQ files to a reference genome.
2. Check the alignment statistics.
3. Adjust alignment settings as needed.
4. Convert to BAM format.

Where do you get the reference genome files? https://www.ncbi.nlm.nih.gov/assembly/GCF_000001405.26/

Example Command:

```
STAR --runThreadN 12 \  
--genomeDir /home/langeca/gilli431/software/STAR_hg38 \  
--readFilesCommand zcat \  
--readFilesIn 2D_WT_shGFP_E2_R50_001_S10_R1_001.fastq.gz  
2D_WT_shGFP_E2_R50_001_S10_R2_001.fastq.gz \  
--outFileNamePrefix 2D_WT_shGFP_E2_R50_001
```

STAR Alignment Summary:

```
Number of input reads | 35202511  
Average input read length | 299  
UNIQUE READS:  
Uniquely mapped reads number | 33777023  
Uniquely mapped reads % | 95.95%  
Average mapped length | 298.68
```

Generating Gene Counts

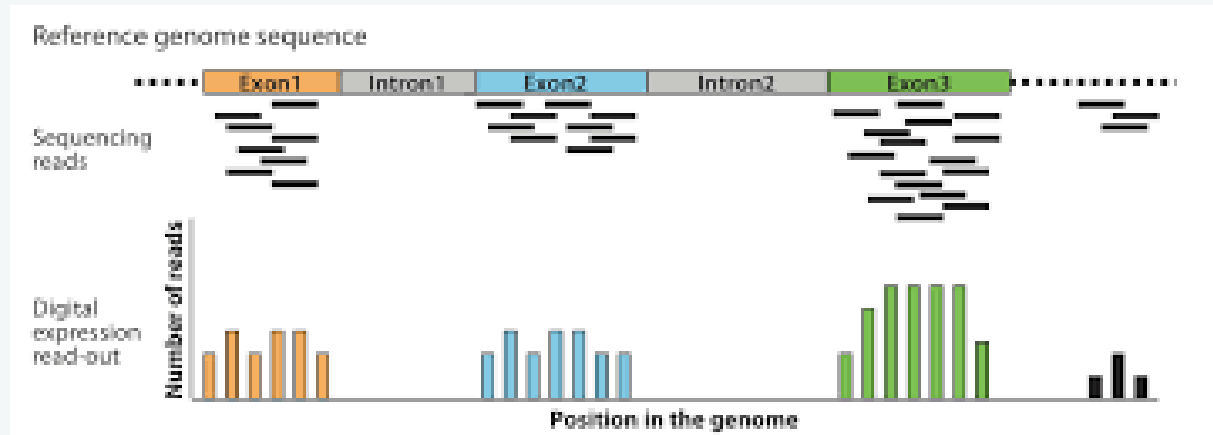


Example Command:

```
htseq-count -f bam -r name -s reverse -m union -i gene_id bam_files/${base}_sorted.bam  
~/software/gencode.v38.primary_assembly.annotation.gtf > count_files/${base}.counts
```

Gene Counting Steps

1. Check that bam files are sorted and indexed
2. Run HT-seq on bam files.
3. Save .cnts files for analysis.



All these steps can be strung together into a “pipeline.”



```
RNA-seq_pipeline_slurm.sh functional_RNA-seq_pipeline_slurm.sh
16
17 # Load necessary modules - check what you have installed on your server vs. what is in your bash
18 # module load fastqc
19 # module load cutadapt
20 # module load star
21 # module load samtools
22 # module load htseq
23 # module load multiqc
24
25 # Create directory for FastQC reports
26 mkdir fastqc_reports
27
28 # Create directory for trimmed fastq files
29 mkdir trimmed_fastq
30
31 # Run FastQC on all input FASTQ files and perform adapter trimming with TrimGalore
32 for f in *_R1_001.fastq.gz; do #check the base name of your files and modify as needed
33   base=$(basename $f _R1_001.fastq.gz)
34   fastqc -o fastqc_reports ${base}_R1_001.fastq.gz ${base}_R2_001.fastq.gz; # Generate FastQC report
35   trim_galore --paired --illumina --fastqc --output_dir trimmed_fastq ${base}_R1_001.fastq.gz ${base}_R2_001.fastq.gz; # Trim
36   done
37
38 # Make a directory for the bam files
39 mkdir bam_files
40
41 # Run STAR on the trimmed FASTQ files
42 for f in trimmed_fastq/*_R1_001_val_1.fq.gz; do
43   base=$(basename $f _R1_001_val_1.fq.gz); # Extract basename of input file
44   r2="${base}_R2_001_val_2.fq.gz"
45   STAR --runThreadN 12 \
46     --genomeDir /home/langeca/gilli431/software/STAR_hg38 \
47     --readFilesCommand zcat \
48     --readFilesIn $f trimmed_fastq/${base}_R2_001_val_2.fq.gz \
49     --outFileNamePrefix bam_files/${base}_ \
50     --outSAMtype BAM \
51
52 # Sort and index the BAM files
53 samtools sort -o bam_files/${base}_sorted.bam bam_files/${base}_Aligned.sortedByCoord.out.bam
54 samtools index bam_files/${base}_sorted.bam
55
56 # Make a directory for the count files
57 mkdir count_files
```

1. Put the raw data for the experiment you are analyzing into a single folder (on MSI).
2. Modify the SLURM header in the pipeline script
3. Send the job out for analysis.
4. Wait 12-24 hours.
5. Check your QC metrics and retrieve the count files.

What can be done with the processed data?

